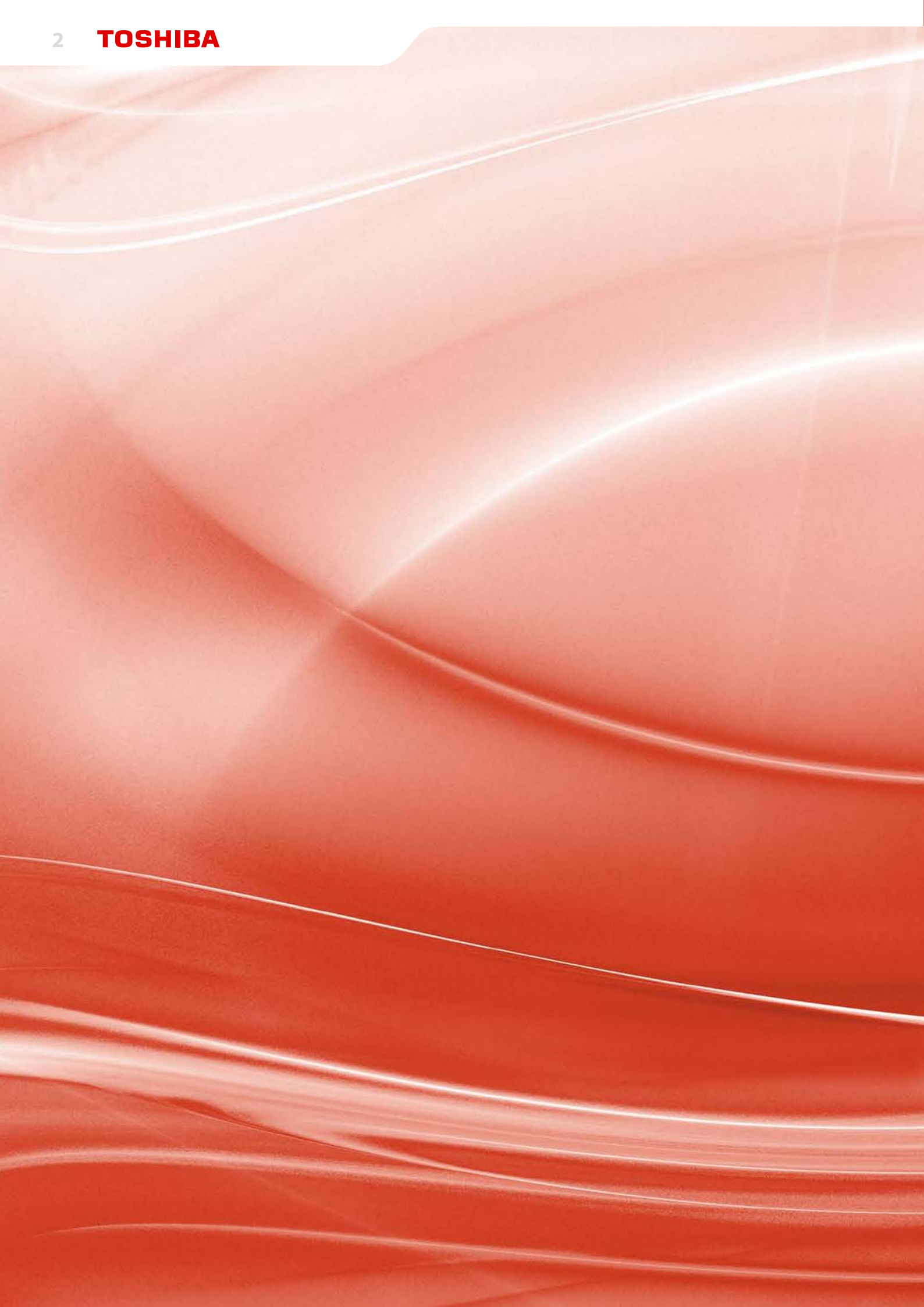




# General catalogue 2009

TOSHIBA AIRCONDITIONING  
Advancing the **eco**-evolution





# Contents

## General catalogue 2009

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## Toshiba Group

Toshiba's origins go back to 1875 and two great inventors.

Hisashige Tanaka, the Thomas Edison of Asia, was renowned for his inventiveness. The sign that welcomed visitors to Tanaka Engineering Works "Improving life with things people need," expressed his deep commitment to improve people's quality of life.

Ichisuke Fujioka was Japan's father of electronics. He manufactured the country's first incandescent lamps and pioneered the development of Japan's electronics industry.

Today, more than 130 years after, Toshiba Group provides a wide range

of products and services that feature the best in innovative technologies and the finest quality.

By continually developing innovative technologies, we strive to create products and services that enhance human life, and which lead to a thriving healthy society.

In 2008, Toshiba Group enjoyed consolidated annual sales of over US\$66 billion and employed approximately 198,000 people worldwide.

## A wide innovative range

**Digital products:** high value-added mobile phones; advanced audio and visual products; Personal Computers and business communication systems.

**Electronic devices and components:** advanced semiconductors and display technologies.

**Social infrastructure systems:** critical components that support power generating, broadcast, transportation, financial and distribution equipment and systems.

**Consumer appliances:** from refrigerators, and residential and commercial air conditioning systems, to automotive electronics and network services.

## Our brand statement

**TOSHIBA**  
Leading Innovation >>>

Toshiba delivers technology and products remarkable for their innovation and artistry – contributing to a safer, more comfortable, more productive life.

We bring together the spirit of innovation with our passion and conviction to shape the future and help protect the global environment – our shared heritage.

We foster close relationships, rooted in trust and respect, with our customers, business partners and communities around the world.

TOSHIBA AIRCONDITIONING

Advancing the **eco**-evolution





Toshiba produced its first air conditioning units in the **1950's**, and immediately worked on introducing improvements.

Its role as an innovator continued with the introduction of the rotary compressor and electronic controls.

By the **1980's** with a broad product offering, Toshiba was the first to introduce the inverter driven unit (1981) and the twin rotary compressor (in 1988).

In **1999**, Toshiba again led the industry with the launch of its product range operating with non-ozone depleting refrigerants (R-410A and R407C).

Toshiba's spirit of innovation continues with its relentless drive for product and system improvements.

**2000:** revolutionary DC hybrid inverter.

**2003:** new Digital Inverter.

**2004:** new SMMS (the VRF system with the industry's highest COP).

**2006:** new MiNi-SMMS.

**2007:** Super Daiseikai (with Ag Plasma air purifier).



# Today Toshiba

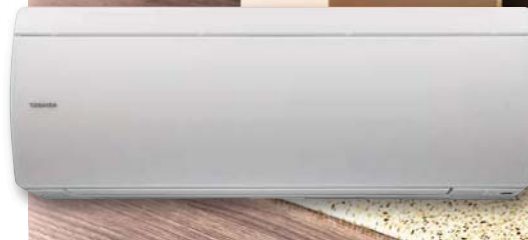
Toshiba offers a solution for all applications: residential, light commercial and larger commercial buildings.

Residential indoor units are designed to blend perfectly with all interiors and most high-wall units incorporate advanced filtration systems to deliver optimum indoor air quality.

For small commercial premises, products are designed to deliver optimum performance combined with energy efficiency.

Digital and Super Digital inverter systems offer a wide range of indoor units for maximum flexibility.

For larger applications, VRF systems combine flexibility, energy efficiency and respect for the environmental. These reliable systems offer a wide choice of stylish indoor units.







## Our vision – “Advancing the Eco-Evolution”

As relative newcomers in the evolution of the Earth, it's undeniable that humankind has had a greater impact on our planet's eco-system than any other life form.

All organisms to some extent affect change on our global environment, but none quite as fast and drastic as humans. Many scientists agree that human activity has accelerated pollution and climatic changes beyond the natural evolutionary process.

Globally, levels of carbon dioxide and average regional temperatures rise at alarming rates, impacting nature and civilization. And the quality of air we breathe continues to deteriorate in the cities we live in.

Whatever the causes, the message is clear: the future is up to us all, and we can do more to make things better.

The core of Toshiba Air Conditioning's philosophy is a profound respect for our global environment and the passionate pursuit toward improving the quality of life for our customers worldwide.

As part of that global commitment, we develop cutting-edge technologies and advancements that serve the benefit of people everywhere, by offering an ideal balance of comfort and ecologically superior products.

We advance Research & Development in the field of super energy-efficient,

cleaner technologies and innovative products that not only use significantly less energy but help maintain air quality through state-of-the-art air purification systems for the home and business.

We endeavor to lead by example by delivering the highest quality environmental systems that give added value, and contribute responsibly to the advancement of humankind.

We call this vision, **“Advancing the Eco-Evolution”**.



## Technology & Environment

Toshiba constantly works to ensure that our air conditioners help protect the environment while delivering reassurance and reliability to users around the world.

Toshiba has earned international praise for building energy-saving products, and for its environmental-oriented production facilities. The company's environmental management system has earned ISO 14001 certification.

Our efforts also extend to actively addressing the issue of global warming, as exemplified by our early adoption of R410A, the new HFC refrigerant with zero ozone depletion co-efficient, in our Daiseikai and Super-Multi Module series air conditioners.

Toshiba is the first manufacturer in Japan to adopt R410A refrigerant for all its products.

The products resulting from Toshiba's research and development dedicated to protecting our environment have earned international recognition and an industry-leading 14 environmental awards, including the Ministry of International Trade and Industry Prize in Japan.



## Quality and savings

Toshiba has been studying, designing and creating innovative air conditioning systems for 30 years and as a result has always offered the highest performance on the market. Quality has always been Toshiba's strength and will remain the trademark that will differentiate Toshiba air conditioners from the competition. This is the philosophy behind every Toshiba product.

### A/A Class

Since energy labeling was introduced to highlight actual energy consumption, Toshiba systems have always shown the precious green A.

This means energy savings: good news for the environment and for everyone's wallet!

### Eurovent

In order to certify the technical data and to guarantee it, Toshiba participates in the Eurovent program of product testing to verify conformance between test results and actual product performance.



| Energy                                                                               |    | Air-conditioner |
|--------------------------------------------------------------------------------------|----|-----------------|
| Manufactured                                                                         |    | <b>TOSHIBA</b>  |
| Outside unit                                                                         |    |                 |
| Inside unit                                                                          |    |                 |
| More efficient                                                                       |    | <b>A</b>        |
| A                                                                                    |    |                 |
| B                                                                                    |    |                 |
| C                                                                                    |    |                 |
| D                                                                                    |    |                 |
| E                                                                                    |    |                 |
| F                                                                                    |    |                 |
| G                                                                                    |    |                 |
| Less efficient                                                                       |    |                 |
| Annual energy consumption, kWh in cooling mode                                       |    | <b>607</b>      |
| <small>(Actual consumption depends on how the appliance is used and climate)</small> |    |                 |
| Cooling output                                                                       | kW | <b>3.9</b>      |
| Energy efficiency ratio                                                              |    | <b>3.21</b>     |
| <small>Full load (the higher the better)</small>                                     |    |                 |
| Type                                                                                 |    |                 |
| Cooling only                                                                         | ←  |                 |
| Cooling + Heating                                                                    | ←  |                 |
| Air cooled                                                                           | ←  |                 |
| Water cooled                                                                         | ←  |                 |
| Noise                                                                                |    |                 |
| <small>(dB(A) re 1 pW)</small>                                                       |    |                 |
| Further information is contained on product brochure                                 |    |                 |
| Norm EN 814<br>Air-conditioner<br>Energy Label Directive 2002/31/EC                  |    |                 |





## No compromise, no concessions. Only quality, pure and simple

For the past 30 years, Toshiba has invested heavily and worked hard to maintain its position as a leader in the air conditioning market.

For Toshiba quality has always been a priority and today and into the future,

the quality of Toshiba products will continue to differentiate us from other manufacturers.

## Tomorrow's technology today

Toshiba is the world leader in microelectronics. From processors to the most advanced chips, from inverter systems to energy efficiency,

every detail is optimised to achieve unrivalled performance.

## The study of air flow

In our laboratories we attach high importance to the study and analyses of airflows within our air conditioners.

The results are apparent in the comfort and quietness of our units.

## The sum of perfection

We know that a perfect unit is the sum of perfection in every detail. Solutions that are studied and verified in every tiny element are recognised

universally by air conditioning professionals for their total reliability.

## Respect for environment

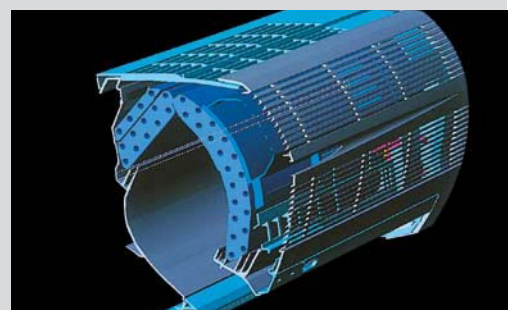
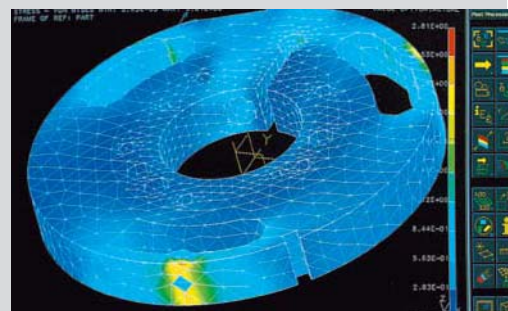
Continuous research to improve on our own energy efficiency targets, already the highest in the industry, are a clear demonstration of Toshiba's

commitment to respect of the environment.

## Absolute comfort

We care deeply about every detail: indoor air quality, low sound levels, energy efficiency... each element is

studied and developed to reach the ultimate goal – absolute comfort.



## The residential range. The applications

Air conditioning plays a fundamental role in your well-being at home as well as at work.

In fact, it is not only important to grant the desired temperature; it is also the right tool to get the best indoor air quality.

Selecting the most suitable system is key to maximizing performance and optimizing comfort.

Air treatment, low sound levels, guaranteed energy savings and easy installation of the indoor units make

the Toshiba residential range the top choice for any home.



## Well-being and beyond

The new R-410A residential products have been designed to offer the best possible comfort to users.

The air filtration systems provide cleaned indoor air at all times.

In fact, the new high-walls grant high energy savings and unmatched indoor air quality.

Thanks to a new design, the indoor units also look elegant and blend beautifully with any room interior.

They have all the features a user can require: silent, discreet and unobtrusive, they deliver high energy efficiency and complete air purification.



# Residential

The residential range  
Ultimate inverter technology, ultimate comfort.



## When technology meets comfort

Toshiba was the first company to incorporate inverter technology into air conditioning systems in 1981 and since then it has always maintained a technological advantage over its competitors.

The development of the new and exclusive **DC hybrid inverter** system has reaffirmed this ability to innovate and maintain technological leadership in a fast-growing market. But for Toshiba, innovation also means a

strong commitment to international institutions that carefully evaluate the impact of new technologies on our environment.

Toshiba combines technological development with care for future generations: the result is a range of extremely **energy-efficient air conditioners**, reducing greenhouse gas emissions at source.

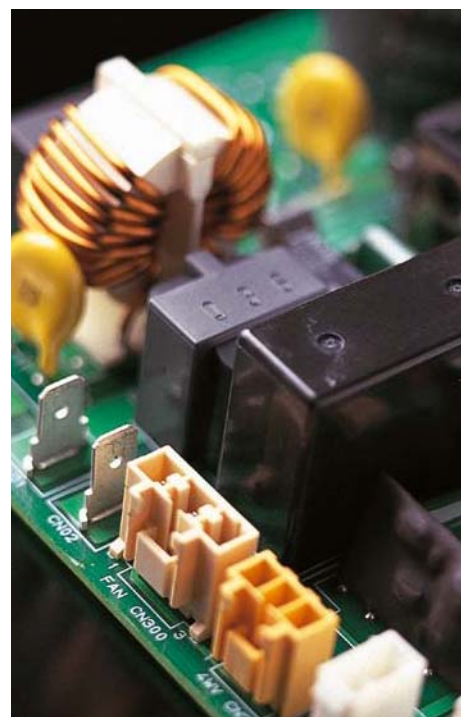
Toshiba continuous research developed PWM technology, which is used together with the traditional PAM control. The application of these two distinct technologies allows total control of performance and energy use.

## The ultimate expression of inverter technology

Toshiba DC hybrid inverter technology controls the capacity supplied by the air conditioner. By modifying the supply current frequency or intensity, it ensures smooth linear variation of the rotation speed and capacity of the compressor – the heart of the air conditioner.

This allows the cooling and heating capacities to be matched to the actual operating conditions required.

When the room temperature is different from the set point, the air conditioner operates at maximum capacity, ensuring that a comfortable temperature will be reached quickly. Once the desired temperature is reached, the inverter precisely adjusts the capacity to maintain a temperature close to the set point.







Superior COP

Advanced filtration systems

Optimised compatibility with the environment

Unequalled capacity reserves



## Filtration, purification, innovation

True quality in the residential environment goes beyond control of the air filtration.

With Toshiba residential air conditioners, air quality is guaranteed by many stages of filtration and additional functions such as pre-treatment, removal of fine particles, viruses, bacteria and allergens or total purification.

Electrostatic purification also guarantees a basic cost advantage as there are no ongoing replacement part costs.

## Care for the environment

Toshiba has anticipated legislation on the control of refrigerant emissions to the atmosphere, and pioneers solutions that our technological leadership enables us to offer.



UP TO 5,36 COP

R-410A

DUAL STAGE COMPRESSOR

SINGLE-SPLIT

FIVE SIZES RANGE



## Super Daiseikai PKVP/PAVP Inverter high-wall

### Features

The new SuperDaiseikai has been designed and created with the objective to provide excellence, respecting the latest eco-evolution trends and maintaining the ultimate comfort.

World's best class energy efficiency.

Pure air.

Premium design.

### Key features

Improved class A efficiency with COP value above 5 (5,36 for size 10).

New technology and advanced electronic reduced the Annual Energy Consumption of 30%\*.

Dual stage compressor improve the load efficiency in a wide range of conditions.

Fast filtration: impurities are ionized by the ion charger and absorbed by the new heat exchanger.

Self cleaning to prevent the growth of mold inside the unit.

Nordic version with base plate heat exchanger and winter operation mode.

New air purifier

Plasma ion charger

Self cleaning

Stylish new design

Wide air flow area



\* in cooling; compared to SKV

Technical specifications **heat pump**

| Outdoor unit                      |                         |   | RAS-07PAVP-E       | RAS-10PAVP-E       | RAS-13PAVP-E       | RAS-16PAVP-E       | RAS-18PAVP-E       |
|-----------------------------------|-------------------------|---|--------------------|--------------------|--------------------|--------------------|--------------------|
| Indoor unit                       |                         |   | RAS-07PKVP-E       | RAS-10PKVP-E       | RAS-13PKVP-E       | RAS-16PKVP-E       | RAS-18PKVP-E       |
| Cooling capacity                  | kW                      |   | 2,0                | 2,5                | 3,5                | 4,5                | 5,0                |
| Cooling range (min. – max.)       | kW                      |   | 0,3 – 3,0          | 0,3 – 3,5          | 0,3 – 4,5          | 0,3 – 5,0          | 0,3 – 5,5          |
| Power input (min. – rated – max.) | kW                      | C | 0,07 – 0,35 – 0,68 | 0,07 – 0,47 – 0,88 | 0,07 – 0,77 – 1,25 | 0,07 – 1,22 – 1,49 | 0,07 – 1,49 – 1,75 |
| EER                               | W/W                     |   | 5,63               | 5,26               | 4,55               | 3,69               | 3,36               |
| Energy efficiency class           |                         | C | A                  | A                  | A                  | A                  | A                  |
| Annual energy consumption         | kWh                     |   | 177                | 237                | 385                | 610                | 745                |
| Heating capacity                  | kW                      |   | 2,5                | 3,0                | 4,0                | 5,5                | 6,0                |
| Heating range (min. – max.)       | kW                      |   | 0,3 – 5,0          | 0,3 – 5,8          | 0,3 – 6,1          | 0,3 – 6,5          | 0,3 – 6,7          |
| Power input (min. – rated – max.) | kW                      | H | 0,07 – 0,44 – 1,30 | 0,07 – 0,56 – 1,60 | 0,07 – 0,84 – 1,60 | 0,07 – 1,34 – 1,70 | 0,07 – 1,54 – 1,75 |
| COP                               | W/W                     |   | 5,68               | 5,36               | 4,76               | 4,1                | 3,9                |
| Energy efficiency class           |                         | H | A                  | A                  | A                  | A                  | A                  |
| Indoor unit                       |                         |   | RAS-07PKVP-E       | RAS-10PKVP-E       | RAS-13PKVP-E       | RAS-16PKVP-E       | RAS-18PKVP-E       |
| Air flow (h/l)                    | m <sup>3</sup> /h – l/s | C | 612/288 – 170/80   | 624/306 – 173/85   | 696/318 – 193/88   | 744/372 – 207/103  | 804/408 – 223/113  |
| Sound pressure level (h/l)        | dB(A)                   | C | 42/26              | 43/27              | 45/27              | 47/30              | 49/31              |
| Sound power level (h/l)           | dB(A)                   | C | 57/41              | 58/42              | 60/42              | 62/45              | 64/46              |
| Air flow (h/l)                    | m <sup>3</sup> /h – l/s | H | 648/348 – 180/97   | 666/348 – 185/97   | 696/348 – 193/97   | 744/384 – 207/107  | 804/420 – 223/117  |
| Sound pressure level (h/l)        | dB(A)                   | H | 42/26              | 43/27              | 45/27              | 47/30              | 49/31              |
| Sound power level (h/l)           | dB(A)                   | H | 57/41              | 58/42              | 60/42              | 62/45              | 64/46              |
| Dimensions (h x w x d)            | mm                      |   | 295 x 790 x 242    | 295 x 790 x 242    | 295 x 790 x 242    | 295 x 790 x 242    | 295 x 790 x 242    |
| Weight                            | kg                      |   | 12                 | 12                 | 12                 | 12                 | 12                 |
| Outdoor unit                      |                         |   | RAS-07PAVP-E       | RAS-10PAVP-E       | RAS-13PAVP-E       | RAS-16PAVP-E       | RAS-18PAVP-E       |
| Air flow                          | m <sup>3</sup> /h – l/s | C | 1662 – 462         | 1800 – 500         | 2232 – 620         | 2232 – 620         | 2370 – 658         |
| Sound pressure level              | dB(A)                   | C | 46                 | 48                 | 50                 | 50                 | 52                 |
| Sound power level                 | dB(A)                   | C | 61                 | 63                 | 65                 | 65                 | 67                 |
| Operating range                   | °C                      | C | –10 ÷ 46           | –10 ÷ 46           | –10 ÷ 46           | –10 ÷ 46           | –10 ÷ 46           |
| Air flow                          | m <sup>3</sup> /h – l/s | H | 1530 – 425         | 1662 – 462         | 2088 – 580         | 2088 – 580         | 2232 – 620         |
| Sound pressure level              | dB(A)                   | H | 46                 | 48                 | 50                 | 50                 | 52                 |
| Sound power level                 | dB(A)                   | H | 61                 | 63                 | 65                 | 65                 | 67                 |
| Operating range                   | °C                      | H | –15 ÷ 24           | –15 ÷ 24           | –15 ÷ 24           | –15 ÷ 24           | –15 ÷ 24           |
| Dimensions (h x w x d)            | mm                      |   | 550 x 780 x 290    | 550 x 780 x 290    | 550 x 780 x 290    | 550 x 780 x 290    | 550 x 780 x 290    |
| Weight                            | kg                      |   | 39                 | 39                 | 40                 | 40                 | 40                 |
| Compressor type                   |                         |   | Twin Rotary        | Twin Rotary        | Twin Rotary        | Twin Rotary        | Twin Rotary        |
| Flare connections (gas – liquid)  |                         |   | 3/8" – 1/4"        | 3/8" – 2/8"        | 3/8" – 2/8"        | 4/8" – 2/8"        | 1/2" – 1/4"        |
| Minimum pipe length               | m                       |   | 2                  | 2                  | 2                  | 2                  | 2                  |
| Maximum pipe length               | m                       |   | 20                 | 20                 | 20                 | 20                 | 20                 |
| Maximum height difference         | m                       |   | 10                 | 10                 | 10                 | 10                 | 10                 |
| Chargeless pipe length            | m                       |   | 15                 | 15                 | 15                 | 15                 | 15                 |
| Power supply                      | V-ph-Hz                 |   | 220/240-1-50       | 220/240-1-50       | 220/240-1-50       | 220/240-1-50       | 220/240-1-50       |

C = cooling mode  
H = heating mode

**heat pump**

| Outdoor unit | RAS-07PAVP-ND | RAS-10PAVP-ND | RAS-13PAVP-ND | RAS-16PAVP-ND | RAS-18PAVP-ND |
|--------------|---------------|---------------|---------------|---------------|---------------|
| Indoor unit* | RAS-07PKVP-ND | RAS-10PKVP-ND | RAS-13PKVP-ND | RAS-16PKVP-ND | RAS-18PKVP-ND |



**R-410A****DC HYBRID INVERTER****SINGLE/MULTI-SPLIT****THE HIGHEST PERFORMANCE**

## Super Daiseikai SKVP Inverter high-wall

### Features

The third generation of Daiseikai provides the highest energy saving and unmatched IAQ with Ag + Plasma Purifier air filtration. The new modern design integrates perfectly with any environment.

### Key features

Lowest energy consumption with 5,1 EER in cooling (size 10).

Ag + Plasma air purifier: doesn't need cleaning since its deodorising effect gets constantly regenerated.

New, modern aesthetic.

Self-cleaning with low density Ozone will eliminate all bacteria normally remaining after operation.

DC Hybrid Inverter technology with Twin Rotary compressor.

Special version for very cold climate: electric heater in CDU base pan and possibility of setting down to 8 °C in heating.

**Ag + Plasma air Purifier****Dust collection****Bacteria elimination****Deodorising****Modern design**

Technical specifications **heat pump**

| Outdoor unit                      |                         |   | RAS-10SAVP-E       | RAS-13SAVP-E       | RAS-16SAVP-E       |
|-----------------------------------|-------------------------|---|--------------------|--------------------|--------------------|
| Indoor unit                       |                         |   | RAS-B10SKVP-E      | RAS-B13SKVP-E      | RAS-B16SKVP-E      |
| Cooling capacity                  | kW                      |   | 2,5                | 3,5                | 4,5                |
| Cooling range (min. – max.)       | kW                      |   | 0,5 – 3,5          | 0,6 – 4,5          | 0,8 – 5,0          |
| Power input (min. – rated – max.) | kW                      | C | 0,10 – 0,49 – 0,87 | 0,11 – 0,85 – 1,37 | 0,15 – 1,35 – 1,82 |
| EER                               | W/W                     |   | 5,10               | 4,12               | 3,33               |
| Energy efficiency class           |                         | C | A                  | A                  | A                  |
| Annual energy consumption         | kWh                     |   | 245                | 425                | 675                |
| Heating capacity                  | kW                      |   | 3,2                | 4,2                | 5,5                |
| Heating range (min. – max.)       | kW                      |   | 0,6 – 6,1          | 0,6 – 6,9          | 0,8 – 8,0          |
| Power input (min. – rated – max.) | kW                      | H | 0,12 – 0,63 – 1,71 | 0,12 – 0,95 – 2,09 | 0,15 – 1,49 – 2,51 |
| COP                               | W/W                     |   | 5,08               | 4,42               | 3,69               |
| Energy efficiency class           |                         | H | A                  | A                  | A                  |
| Indoor unit                       |                         |   | RAS-B10SKVP-E      | RAS-B13SKVP-E      | RAS-B16SKVP-E      |
| Air flow (h/l)                    | m <sup>3</sup> /h – l/s | C | 546/276 – 152/77   | 564/276 – 157/77   | 606/318 – 168/88   |
| Sound pressure level (h/l)        | dB(A)                   | C | 42/27              | 43/27              | 45/29              |
| Sound power level (h/l)           | dB(A)                   | C | 57/42              | 58/42              | 60/44              |
| Air flow (h/l)                    | m <sup>3</sup> /h – l/s | H | 612/282 – 170/78   | 636/300 – 177/83   | 678/342 – 188/95   |
| Sound pressure level (h/l)        | dB(A)                   | H | 43/27              | 44/27              | 45/29              |
| Sound power level (h/l)           | dB(A)                   | H | 58/42              | 59/42              | 60/45              |
| Dimensions (h x w x d)            | mm                      |   | 250 x 790 x 208    | 250 x 790 x 208    | 250 x 790 x 208    |
| Weight                            | kg                      |   | 9                  | 9                  | 9                  |
| Outdoor unit                      |                         |   | RAS-10SAVP-E       | RAS-13SAVP-E       | RAS-16SAVP-E       |
| Air flow                          | m <sup>3</sup> /h – l/s |   | 2148 – 597         | 2406 – 668         | 2406 – 668         |
| Sound pressure level              | dB(A)                   | C | 46                 | 48                 | 49                 |
| Sound power level                 | dB(A)                   | C | 59                 | 61                 | 62                 |
| Operating range                   | °C                      | C | –10 ÷ 46           | –10 ÷ 46           | –10 ÷ 46           |
| Sound pressure level              | dB(A)                   | H | 47                 | 50                 | 50                 |
| Sound power level                 | dB(A)                   | H | 60                 | 63                 | 63                 |
| Operating range                   | °C                      | H | –15 ÷ 24           | –15 ÷ 24           | –15 ÷ 24           |
| Dimensions (h x w x d)            | mm                      |   | 550 x 780 x 290    | 550 x 780 x 290    | 550 x 780 x 290    |
| Weight                            | kg                      |   | 38                 | 38                 | 38                 |
| Compressor type                   |                         |   | DC Twin Rotary     | DC Twin Rotary     | DC Twin Rotary     |
| Flare connections (gas – liquid)  |                         |   | 3/8" – 2/8"        | 3/8" – 2/8"        | 4/8" – 2/8"        |
| Minimum pipe length               | m                       |   | 2                  | 2                  | 2                  |
| Maximum pipe length               | m                       |   | 25                 | 25                 | 25                 |
| Maximum height difference         | m                       |   | 10                 | 10                 | 10                 |
| Chargeless pipe length            | m                       |   | 15                 | 15                 | 15                 |
| Power supply                      | V-ph-Hz                 |   | 220/240-1-50       | 220/240-1-50       | 220/240-1-50       |

\* also available in Multisplit (Cooling only and Heat pump) - see Matching table page 32

C = cooling mode  
H = heating mode

| heat pump                          |           |           |           | cooling only |            |            |
|------------------------------------|-----------|-----------|-----------|--------------|------------|------------|
| Indoor units                       | B10SKVP-E | B13SKVP-E | B16SKVP-E | -            | -          | -          |
| MONOSPLIT Outdoor units            | 10SAVP-E  | 13SAVP-E  | 16SAVP-E  | -            | -          | -          |
| Indoor units                       | B10SKVP-E | B13SKVP-E | B16SKVP-E | M10SKCVP-E   | M13SKCVP-E | M16SKCVP-E |
| MULTISPLIT Outdoor units (2 rooms) | M14GAV-E  | M14GAV-E  | -         | M14GACV-E    | M14GACV-E  | -          |
| (2 rooms)                          | M18GAV-E  | M18GAV-E  | M18GAV-E  | M18GACV-E    | M18GACV-E  | M18GACV-E  |
| (3 rooms)                          | 3M18SAV-E | 3M18SAV-E | 3M18SAV-E | 3M18SACV-E   | 3M18SACV-E | 3M18SACV-E |
| (3 rooms)                          | 3M26GAV-E | 3M26GAV-E | 3M26GAV-E | 3M23GACV-E   | 3M23GACV-E | 3M23GACV-E |
| (4 rooms)                          | 4M23SAV-E | 4M23SAV-E | 4M23SAV-E | 4M23SACV-E   | 4M23SACV-E | 4M23SACV-E |
| (4 rooms)                          | 4M27GAV-E | 4M27GAV-E | 4M27GAV-E | 4M27GACV-E   | 4M27GACV-E | 4M27GACV-E |

**R-410A****DC HYBRID INVERTER****AIR PURIFICATION****DAISEIKAI**

## Daiseikai SKVR Inverter high-wall

### Features

This elegant member of the Daiseikai family provides top performance in energy saving, active air purification and comfortable indoor climate creation.

Now, with the new 5 kW and 6 kW sizes, the range is complete and can meet any requirement.

### Key features

Very high efficiency in cooling and heating to guarantee the lowest energy cost, 4,18 EER for size 10.

PAP: 10 times Active Filtration and Air Purification with Double Freshness: purifies air 10 times faster, odours are removed in 30 mins; it doesn't need to be replaced.

The ioniser delivers an energising current of air and gives the ambient the same freshness as a forest.

Extremely low noise for both IDU and CDU.

One-touch Personal Comfort to enable users to programme their preferred settings.

One-touch Preset: for immediate comfort setting based on European user preference.

DC Hybrid Inverter technology with PAM and PWM.

Low temperature operation -15 °C in heating and -10 °C in cooling.

**Active filtration and air purification**

**Air ionizer and zeolite plus**

**Self-cleaning automatic operation**

**Auto-diagnosis system**

**Modern design with aluminium flat panel**





Technical specifications **heat pump**

| Outdoor unit                      |                         |   | RAS-10SAVR-E       | RAS-13SAVR-E/E2    | RAS-16SAVR-E       | RAS-18SAV-E        | RAS-22SAV-E        |
|-----------------------------------|-------------------------|---|--------------------|--------------------|--------------------|--------------------|--------------------|
| Indoor unit                       |                         |   | RAS-10SKVR-E       | RAS-13SKVR-E/E2    | RAS-16SKVR-E       | RAS-18SKVR-E       | RAS-22SKVR-E       |
| Cooling capacity                  | kW                      |   | 2,5                | 3,5                | 4,5                | 5,0                | 6,0                |
| Cooling range (min. – max.)       | kW                      |   | 1,1 – 3,1          | 0,8 – 4,1          | 0,8 – 5,0          | 1,1 – 6,0          | 1,2 – 6,7          |
| Power input (min. – rated – max.) | kW                      | C | 0,25 – 0,60 – 0,82 | 0,15 – 1,00 – 1,25 | 0,15 – 1,39 – 1,72 | 0,18 – 1,42 – 2,00 | 0,20 – 1,99 – 2,65 |
| EER                               | W/W                     |   | 4,18               | 3,50               | 3,23               | 3,52               | 3,01               |
| Energy efficiency class           |                         | C | A                  | A                  | A                  | A                  | B                  |
| Annual energy consumption         | kWh                     |   | 299                | 500                | 698                | 710                | 998                |
| Heating capacity                  | kW                      |   | 3,2                | 4,2                | 5,5                | 5,8                | 7                  |
| Heating range (min. – max.)       | kW                      |   | 0,9 – 4,8          | 0,9 – 5,8          | 0,9 – 6,9          | 0,8 – 6,3          | 1,0 – 7,5          |
| Power input (min. – rated – max.) | kW                      | H | 0,17 – 0,75 – 1,40 | 0,15 – 1,08 – 1,64 | 0,15 – 1,52 – 1,98 | 0,14 – 1,56 – 1,70 | 0,18 – 2,05 – 2,21 |
| COP                               | W/W                     |   | 4,27               | 3,89               | 3,62               | 3,72               | 3,41               |
| Energy efficiency class           |                         | H | A                  | A                  | A                  | A                  | B                  |
| Indoor unit                       |                         |   | RAS-10SKVR-E       | RAS-13SKVR-E       | RAS-16SKVR-E       | RAS-18SKVR-E       | RAS-22SKVR-E       |
| Air flow                          | m <sup>3</sup> /h – l/s | C | 515 – 143          | 540 – 150          | 686 – 191          | 942 – 262          | 1062 – 295         |
| Sound pressure level (h/l)        | dB(A)                   | C | 38/26              | 39/26              | 45/30              | 44/32              | 47/35              |
| Sound power level (h/l)           | dB(A)                   | C | 51/39              | 52/39              | 58/43              | 59/47              | 62/50              |
| Air flow                          | m <sup>3</sup> /h – l/s | H | 571 – 159          | 614 – 171          | 735 – 204          | 972 – 270          | 1080 – 300         |
| Sound pressure level (h/l)        | dB(A)                   | H | 39/28              | 40/28              | 45/31              | 44/32              | 47/35              |
| Sound power level (h/l)           | dB(A)                   | H | 52/41              | 53/41              | 58/44              | 59/47              | 62/50              |
| Dimensions (h x w x d)            | mm                      |   | 275 x 790 x 205    | 275 x 790 x 205    | 275 x 790 x 205    | 320 x 1050 x 228   | 320 x 1050 x 228   |
| Weight                            | kg                      |   | 9                  | 9                  | 9                  | 13                 | 13                 |
| Outdoor unit                      |                         |   | RAS-10SAVR-E       | RAS-13SAVR-E       | RAS-16SAVR-E       | RAS-18SAV-E        | RAS-22SAV-E        |
| Air flow                          | m <sup>3</sup> /h – l/s |   | 1800 – 500         | 2250 – 625         | 2160 – 600         | 1914 – 532         | 2232 – 620         |
| Sound pressure level (h/l)        | dB(A)                   | C | 46                 | 48                 | 49                 | 49                 | 52                 |
| Sound power level (h/l)           | dB(A)                   | C | 59                 | 61                 | 62                 | 64                 | 67                 |
| Operating range                   | °C                      | C | –10 ÷ 46           | –10 ÷ 46           | –10 ÷ 46           | –10 ÷ 46           | –10 ÷ 46           |
| Sound pressure level (h/l)        | dB(A)                   | H | 47                 | 50                 | 50                 | 50                 | 51                 |
| Sound power level (h/l)           | dB(A)                   | H | 60                 | 63                 | 63                 | 65                 | 66                 |
| Operating range                   | °C                      | H | –15 ÷ 24           | –15 ÷ 24           | –15 ÷ 24           | –15 ÷ 24           | –15 ÷ 24           |
| Dimensions (h x w x d)            | mm                      |   | 550 x 780 x 290    | 550 x 780 x 290    | 550 x 780 x 290    | 550 x 780 x 290    | 550 x 780 x 290    |
| Weight                            | kg                      |   | 35                 | 35                 | 39                 | 39                 | 40                 |
| Compressor type                   |                         |   | DC Rotary          | DC Twin Rotary     | DC Twin Rotary     | DC Twin Rotary     | DC Twin Rotary     |
| Flare connections (gas – liquid)  |                         |   | 3/8" – 1/4"        | 3/8" – 1/4"        | 1/2" – 1/4"        | 1/2" – 1/4"        | 1/2" – 1/4"        |
| Minimum pipe length               | m                       |   | 1                  | 1                  | 1                  | 2                  | 2                  |
| Maximum pipe length               | m                       |   | 20                 | 20                 | 20                 | 20                 | 20                 |
| Maximum height difference         | m                       |   | 10                 | 10                 | 10                 | 10                 | 10                 |
| Chargeless pipe length            | m                       |   | 15                 | 15                 | 15                 | 15                 | 15                 |
| Power supply                      | V-ph-Hz                 |   | 220/240-1-50       | 220/240-1-50       | 220/240-1-50       | 220/240-1-50       | 220/240-1-50       |

C = cooling mode  
H = heating mode

**R-410A****DC HYBRID INVERTER****SINGLE/MULTI-SPLIT****AIR PURIFICATION**

## Suzumi SKV Inverter high-wall

### Features

The most silent indoor unit in its category. Unobtrusively, it delivers high energy efficiency and complete air purification, thanks to the 7 in 1 filtering solution.

This hi-wall is now available in 2 additional sizes (5 kW and 6 kW): the complete range can satisfy the most demanding requirements.

### Key features

DC Hybrid Inverter technology with PAM and PWM. A energy class in cooling and heating.

Extremely low noise, for both indoor and outdoor units: no disturbance.

7 in 1 filtering system, with Double Anti Oxidant enhanced with Vitamin C.

One-touch Preset: preset ideal comfort parameters.

New modern design with easy to clean flat panel. Size 10 is the smallest in its category.

**A class in cooling and heating**

**7 in 1 filtering solution**

**Auto-diagnosis and self-cleaning functions**

**Modern design with flat panel**



Technical specifications **heat pump**

| Outdoor unit                      |                         |   | RAS-10SAV-E/E2     | RAS-13SAV-E        | RAS-16SAV-E        | RAS-18SAV-E        | RAS-22SAV-E        |
|-----------------------------------|-------------------------|---|--------------------|--------------------|--------------------|--------------------|--------------------|
| Indoor unit                       |                         |   | RAS-10SKV-E/E2     | RAS-13SKV-E        | RAS-16SKV-E        | RAS-18SKV-E        | RAS-22SKV-E        |
| Cooling capacity                  | kW                      |   | 2,5                | 3,5                | 4,5                | 5,0                | 6,0                |
| Cooling range (min. – max.)       | kW                      |   | 1,1 – 3,0          | 1,1 – 4,0          | 0,8 – 5,0          | 1,1 – 6,0          | 1,2 – 6,7          |
| Power input (min. – rated – max.) | kW                      | C | 0,25 – 0,75 – 0,96 | 0,25 – 1,07 – 1,33 | 0,15 – 1,59 – 1,90 | 0,18 – 1,42 – 2,00 | 0,20 – 1,99 – 2,65 |
| EER                               | W/W                     |   | 3,33               | 3,27               | 2,82               | 3,52               | 3,01               |
| Energy efficiency class           |                         | C | A                  | A                  | C                  | A                  | B                  |
| Annual energy consumption         | KWh                     |   | 375                | 535                | 798                | 710                | 998                |
| Heating capacity                  | kW                      |   | 3,2                | 4,2                | 5,3                | 5,8                | 7                  |
| Heating range (min. – max.)       | kW                      |   | 0,9 – 4,1          | 0,9 – 5,0          | 0,9 – 6,2          | 0,8 – 6,3          | 1,0 – 7,5          |
| Power input (min. – rated – max.) | kW                      | H | 0,20 – 0,86 – 1,20 | 0,17 – 1,13 – 1,48 | 0,15 – 1,55 – 1,81 | 0,14 – 1,56 – 1,70 | 0,18 – 2,05 – 2,21 |
| COP                               | W/W                     |   | 3,72               | 3,72               | 3,42               | 3,72               | 3,41               |
| Energy efficiency class           |                         | H | A                  | A                  | B                  | A                  | B                  |
| Indoor unit                       |                         |   | RAS-10SKV-E        | RAS-13SKV-E        | RAS-16SKV-E        | RAS-18SKV-E        | RAS-22SKV-E        |
| Air flow                          | m <sup>3</sup> /h – l/s | C | 522 – 145          | 563 – 156          | 691 – 192          | 954 – 265          | 1080 – 300         |
| Sound pressure level (h/l)        | dB(A)                   | C | 38/29              | 39/26              | 45/30              | 44/32              | 47/35              |
| Sound power level (h/l)           | dB(A)                   | C | 51/42              | 52/39              | 58/43              | 59/47              | 62/50              |
| Air flow                          | m <sup>3</sup> /h – l/s | H | 576 – 160          | 630 – 175          | 743 – 206          | 990 – 275          | 1098 – 305         |
| Sound pressure level (h/l)        | dB(A)                   | H | 40/30              | 40/28              | 45/31              | 44/32              | 47/35              |
| Sound power level (h/l)           | dB(A)                   | H | 53/43              | 53/41              | 58/44              | 50/47              | 62/50              |
| Dimensions (h x w x d)            | mm                      |   | 250 x 740 x 195    | 275 x 790 x 205    | 275 x 790 x 205    | 320 x 1050 x 228   | 320 x 1050 x 228   |
| Weight                            | kg                      |   | 8                  | 9                  | 9                  | 13                 | 13                 |
| Outdoor unit                      |                         |   | RAS-10SAV-E        | RAS-13SAV-E        | RAS-16SAV-E        | RAS-18SAV-E        | RAS-22SAV-E        |
| Air flow                          | m <sup>3</sup> /h – l/s |   | 1620 – 450         | 2100 – 583         | 2100 – 583         | 1914 – 532         | 2232 – 620         |
| Sound pressure level              | dB(A)                   | C | 48                 | 48                 | 49                 | 49                 | 52                 |
| Sound power level                 | dB(A)                   | C | 61                 | 61                 | 62                 | 64                 | 67                 |
| Operating range                   | °C                      | C | -15 ÷ 43           | -10 ÷ 46           | -10 ÷ 46           | -10 ÷ 46           | -10 ÷ 46           |
| Sound pressure level              | dB(A)                   | H | 50                 | 50                 | 50                 | 50                 | 51                 |
| Sound power level                 | dB(A)                   | H | 63                 | 63                 | 63                 | 65                 | 66                 |
| Operating range                   | °C                      | H | -10 ÷ 24           | -15 ÷ 24           | -15 ÷ 24           | -15 ÷ 24           | -15 ÷ 24           |
| Dimensions (h x w x d)            | mm                      |   | 550 x 660 x 240    | 550 x 780 x 290    | 550 x 780 x 290    | 550 x 780 x 290    | 550 x 780 x 290    |
| Weight                            | kg                      |   | 29                 | 35                 | 35                 | 39                 | 40                 |
| Compressor type                   |                         |   | DC Rotary          | DC Rotary          | DC Twin Rotary     | DC Twin Rotary     | DC Twin Rotary     |
| Flare connections (gas – liquid)  |                         |   | 3/8" – 1/4"        | 3/8" – 1/4"        | 1/2" – 1/4"        | 1/2" – 1/4"        | 1/2" – 1/4"        |
| Minimum pipe length               | m                       |   | 1                  | 1                  | 1                  | 2                  | 2                  |
| Maximum pipe length               | m                       |   | 10                 | 20                 | 20                 | 20                 | 20                 |
| Maximum height difference         | m                       |   | 8                  | 10                 | 10                 | 10                 | 10                 |
| Chargeless pipe length            | m                       |   | 10                 | 15                 | 15                 | 15                 | 15                 |
| Power supply                      | V-ph-Hz                 |   | 220/240-1-50       | 220/240-1-50       | 220/240-1-50       | 220/240-1-50       | 220/240-1-50       |

C = cooling mode  
H = heating mode

## heat pump

## cooling only

| Indoor units                       |  | B10SKV-E  | B13SKV-E  | B16SKV-E  | -          | -          | -          |
|------------------------------------|--|-----------|-----------|-----------|------------|------------|------------|
| MONOSPLIT Outdoor units            |  | 10SAV-E   | 13SAV-E   | 16SAV-E   | -          | -          | -          |
| Indoor units                       |  | B10SKV-E  | B13SKV-E  | B16SKV-E  | M10SKCV-E  | M13SKCV-E  | M16SKCV-E  |
| MULTISPLIT Outdoor units (2 rooms) |  | M14GAV-E  | M14GAV-E  | -         | M14GACV-E  | M14GACV-E  | -          |
| (2 rooms)                          |  | M18GAV-E  | M18GAV-E  | M18GAV-E  | M18GACV-E  | M18GACV-E  | M18GACV-E  |
| (3 rooms)                          |  | 3M18SAV-E | 3M18SAV-E | 3M18SAV-E | 3M18SACV-E | 3M18SACV-E | 3M18SACV-E |
| (3 rooms)                          |  | 3M26GAV-E | 3M26GAV-E | 3M26GAV-E | 3M23GACV-E | 3M23GACV-E | 3M23GACV-E |
| (4 rooms)                          |  | 4M23SAV-E | 4M23SAV-E | 4M23SAV-E | 4M23SACV-E | 4M23SACV-E | 4M23SACV-E |
| (4 rooms)                          |  | 4M27GAV-E | 4M27GAV-E | 4M27GAV-E | 4M27GACV-E | 4M27GACV-E | 4M27GACV-E |



**R-410A****DC HYBRID INVERTER****AIR PURIFICATION**

## AvAnt Inverter high-wall

### Features

AvAnt-garde Inverter model specifically conceived for residential users.

If your need is ideal temperature all year round, delivered with high energy saving and extremely low noise, Toshiba AvAnt Inverter is the solution.

### Key features

A energy class in cooling and heating (capacities 10 and 13): high energy saving.

The lowest indoor unit noise of the category: only 20 dB(A) at low fan speed.\*

3 in 1 filtration system: to reduce bacteria, prevent allergies and eliminate odours.

New modern design with reduced dimension to fit into every room.

Easy to use remote control.

**A class in cooling and heating**

**3 in 1 filtration system**

**Extremely low sound level**

**Modern and compact design**



\* model 137SKV, from 2,5 m distance

Technical specifications **heat pump**

| Outdoor unit                      |                         |   | RAS-107SAV-E/E2    | RAS-137SAV-E       | RAS-167SAV-E       |
|-----------------------------------|-------------------------|---|--------------------|--------------------|--------------------|
| Indoor unit                       |                         |   | RAS-107SKV-E/E2    | RAS-137SKV-E       | RAS-167SKV-E       |
| Cooling capacity                  | kW                      |   | 2,5                | 3,5                | 4,5                |
| Cooling range (min. – max.)       | kW                      |   | 1,1 – 3,0          | 1,1 – 4,0          | 0,8 – 5,0          |
| Power input (min. – rated – max.) | kW                      | C | 0,26 – 0,76 – 0,97 | 0,25 – 1,08 – 1,33 | 0,15 – 1,59 – 1,90 |
| EER                               | W/W                     |   | 3,29               | 3,24               | 2,82               |
| Energy efficiency class           |                         | C | A                  | A                  | C                  |
| Annual energy consumption         | KWh                     |   | 375                | 535                | 798                |
| Heating capacity                  | kW                      |   | 3,2                | 4,2                | 5,3                |
| Heating range (min. – max.)       | kW                      |   | 0,9 – 4,1          | 0,9 – 5,0          | 0,9 – 6,2          |
| Power input (min. – rated – max.) | kW                      | H | 0,20 – 0,87 – 1,20 | 0,17 – 1,14 – 1,48 | 1,15 – 1,55 – 1,81 |
| COP                               | W/W                     |   | 3,68               | 3,68               | 3,42               |
| Energy efficiency class           |                         | H | A                  | A                  | B                  |
| Indoor unit                       |                         |   | RAS-107SKV-E       | RAS-137SKV-E       | RAS-167SKV-E       |
| Air flow                          | m <sup>3</sup> /h – l/s | C | 520 – 144          | 560 – 155          | 690 – 192          |
| Sound pressure level (l/m/h)      | dB(A)                   | C | 29/33/38           | 26/33/39           | 30/40/45           |
| Sound power level (l/m/h)         | dB(A)                   | C | 42/46/51           | 39/46/52           | 43/53/58           |
| Air flow                          | m <sup>3</sup> /h – l/s | H | 570 – 158          | 630 – 175          | 740 – 205          |
| Sound pressure level (l/m/h)      | dB(A)                   | H | 30/35/40           | 28/34/40           | 31/40/45           |
| Sound power level (l/m/h)         | dB(A)                   | H | 43/48/53           | 41/47/53           | 44/53/58           |
| Dimensions (h x w x d)            | mm                      |   | 250 x 740 x 195    | 275 x 790 x 205    | 275 x 790 x 205    |
| Weight                            | kg                      |   | 8                  | 9                  | 9                  |
| Outdoor unit                      |                         |   | RAS-107SAV-E       | RAS-137SAV-E       | RAS-167SAV-E       |
| Air flow                          | m <sup>3</sup> /h – l/s |   | 1620 – 450         | 2100 – 583         | 2100 – 583         |
| Sound pressure level              | dB(A)                   | C | 48                 | 48                 | 49                 |
| Sound power level                 | dB(A)                   | C | 61                 | 61                 | 62                 |
| Operating range                   | °C                      | C | 15 ÷ 43            | –10 ÷ 46           | –10 ÷ 46           |
| Sound pressure level              | dB(A)                   | H | 50                 | 50                 | 50                 |
| Sound power level                 | dB(A)                   | H | 63                 | 63                 | 63                 |
| Operating range                   | °C                      | H | –10 ÷ 24           | –15 ÷ 24           | –15 ÷ 24           |
| Dimensions (h x w x d)            | mm                      |   | 530 x 660 x 240    | 550 x 780 x 290    | 550 x 780 x 290    |
| Weight                            | kg                      |   | 29                 | 35                 | 35                 |
| Compressor type                   |                         |   | DC Rotary          | DC Rotary          | DC Twin Rotary     |
| Flare connections (gas – liquid)  |                         |   | 3/8" – 1/4"        | 3/8" – 1/4"        | 1/2" – 1/4"        |
| Minimum pipe length               | m                       |   | 10                 | 15                 | 15                 |
| Maximum pipe length               | m                       |   | 10                 | 20                 | 20                 |
| Maximum height difference         | m                       |   | 8                  | 10                 | 10                 |
| Chargeless pipe length            | m                       |   | 10                 | 15                 | 15                 |
| Power supply                      | V-ph-Hz                 |   | 220/240-1-50       | 220/24-1-50        | 220/240-1-50       |

C = cooling mode  
H = heating mode

**R-410A****DC HYBRID INVERTER****MULTI-SPLIT****IR REMOTE CONTROL**

## GDV Inverter ducted

### Features

These ducted multi-split inverter units offer reliable and controllable comfort year-round.

The compact and quiet units are suitable for a wide range of residential and light commercial applications, with one outdoor unit serving up to four indoor units.

### Key features

Easy-to-use infrared remote control or wired remote control as an option.

Low noise level: the unit operates very quietly.

Very slim design and only 230 mm high, for easier and more flexible installation.

Flexible air inlet: rear or below the unit.

Drain pump kit available as an option.

Up to 63,7 Pa static pressure.

**Only 230 mm high**

**Flexible air return**

**High static pressure**



Technical specifications **cooling only**

| Indoor unit      |    | RAS-M10GDCV-E | RAS-M13GDCV-E | RAS-M16GDCV-E |
|------------------|----|---------------|---------------|---------------|
| Cooling capacity | kW | 2,7           | 3,7           | 4,5           |

Technical specifications **heat pump**

| Indoor unit      |    | RAS-M10GDV-E | RAS-M13GDV-E | RAS-M16GDV-E |
|------------------|----|--------------|--------------|--------------|
| Cooling capacity | kW | 2,7          | 3,7          | 4,5          |
| Heating capacity | kW | 4,0          | 5,0          | 5,5          |

| Indoor unit                                   |                         | CO | RAS-M10GDCV-E     | RAS-M13GDCV-E     | RAS-M16GDCV-E     |
|-----------------------------------------------|-------------------------|----|-------------------|-------------------|-------------------|
|                                               |                         | HP | RAS-M10GDV-E      | RAS-M13GDV-E      | RAS-M16GDV-E      |
| Air flow (h/l)                                | m <sup>3</sup> /h – l/s | C  | 720/400 – 200/111 | 780/428 – 217/119 | 780/428 – 217/119 |
| Sound pressure level (h/l)                    | dB(A)                   | C  | 31/23             | 32/24             | 33/25             |
| Sound power level (h/l)                       | dB(A)                   | C  | 44/36             | 45/37             | 46/38             |
| Operating range                               | °C                      | C  | 21 ÷ 32           | 21 ÷ 32           | 21 ÷ 32           |
| Air flow (h/l)                                | m <sup>3</sup> /h – l/s | H  | 720/450 – 200/111 | 780/428 – 217/119 | 780/428 – 217/119 |
| Sound pressure level (h/l)                    | dB(A)                   | H  | 32/24             | 33/25             | 34/26             |
| Sound power level (h/l)                       | dB(A)                   | H  | 45/37             | 46/33             | 47/34             |
| Operating range                               | °C                      | H  | 0 ÷ 28            | 0 ÷ 28            | 0 ÷ 28            |
| Dimensions (h × w × d)                        | mm                      |    | 230 × 750 × 440   | 230 × 750 × 440   | 230 × 750 × 440   |
| Weight                                        | kg                      |    | 19                | 19                | 19                |
| Flare connections (gas – liquid)              |                         |    | 3/8" – 1/4"       | 3/8" – 1/4"       | 1/2" – 1/4"       |
| External static pressure* (stand/upper limit) | Pa                      |    | 35,3/54,9         | 41,2/63,7         | 41,2/63,7         |

\* static pressure at nominal condition

C = cooling mode  
H = heating mode

## heat pump

## cooling only

|                                    |           |           |           |            |            |            |
|------------------------------------|-----------|-----------|-----------|------------|------------|------------|
| Indoor units                       | -         | -         | -         | -          | -          | -          |
| MONOSPLIT Outdoor units            | -         | -         | -         | -          | -          | -          |
| Indoor units                       | M10GDV-E  | M13GDV-E  | M16GDV-E  | M10GDCV-E  | M13GDCV-E  | M16GDCV-E  |
| MULTISPLIT Outdoor units (2 rooms) | M14GAV-E  | M14GAV-E  | -         | M14GACV-E  | M14GACV-E  | -          |
| (2 rooms)                          | M18GAV-E  | M18GAV-E  | M18GAV-E  | M18GACV-E  | M18GACV-E  | M18GACV-E  |
| (3 rooms)                          | 3M26GAV-E | 3M26GAV-E | 3M26GAV-E | 3M23GACV-E | 3M23GACV-E | 3M23GACV-E |
| (4 rooms)                          | 4M27GAV-E | 4M27GAV-E | 4M27GAV-E | 4M27GACV-E | 4M27GACV-E | 4M27GACV-E |



R-410A HEAT PUMPS

DC HYBRID INVERTER

MULTI-SPLIT



**Compact and stylish indoor unit**

**Standard 600 × 600 mm grid**

**Easy installation**

**Easy control access via corner pockets**

## Inverter 4-way cassette

### Features

This 4-way cassette has been designed to suit all the standard 600 × 600 mm grid ceiling, for easy installation and maintenance.

Its compact and stylish panel design makes it discreet and unobtrusive. Both the installation and the maintenance are really easy, thanks to the simple access to electrical box by simply removing the suction grill.

### Key features

Stylish design and compact dimensions to suit all standard 600 × 600 mm grid ceilings.

Complete range from 10 to 16 k. All cassettes will fit standard 600 × 600 mm tiles.

**Easy maintenance:** easy access to electrical box by simply removing the suction grill.

Easy installation by panel adjust pocket.

All units equipped with IR remotes as standard.



Technical specifications **heat pump**

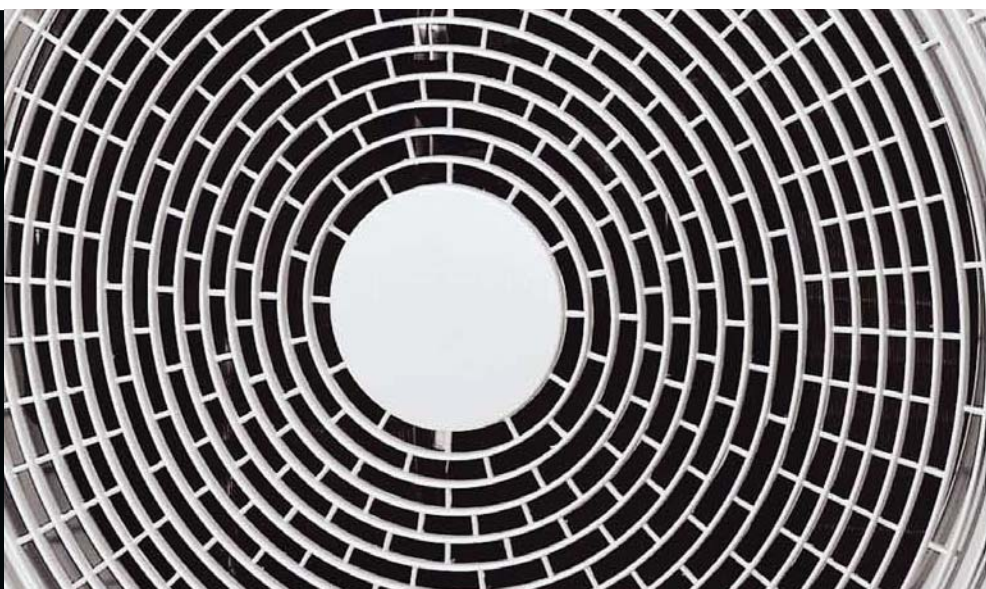
| Indoor unit                |                         | CO | RAS-M10SMUCV-E  | RAS-M13SMUCV-E  | RAS-M16SMUCV-E  |
|----------------------------|-------------------------|----|-----------------|-----------------|-----------------|
|                            |                         | HP | RAS-M10SMUV-E   | RAS-M13SMUV-E   | RAS-M16SMUV-E   |
| Air flow (h/l)             | m <sup>3</sup> /h – l/s | C  | 588 – 163       | 618 – 172       | 660 – 183       |
| Sound pressure level (h/l) | dB(A)                   | C  | 37/30           | 38/30           | 40/31           |
| Sound power level          | dB(A)                   | C  | 52              | 53              | 55              |
| Air flow (h/l)             | m <sup>3</sup> /h       | H  | 558 – 432       | 618 – 432       | 660 – 450       |
| Sound pressure level (h/l) | dB(A)                   | H  | 37/30           | 38/30           | 40/31           |
| Sound power level          | dB(A)                   | H  | 52              | 53              | 55              |
| Dimensions (h × w × d)     | mm                      |    | 268 × 575 × 575 | 268 × 575 × 575 | 268 × 575 × 575 |
| Weight                     | kg                      |    | 15              | 15              | 15              |

C = cooling mode

H = heating mode

**heat pump****cooling only**

| Indoor units                       |           | –         | –         | –          | –          | –          |
|------------------------------------|-----------|-----------|-----------|------------|------------|------------|
| MONOSPLIT Outdoor units            |           | –         | –         | –          | –          | –          |
| Indoor units                       | M10SMUV-E | M13SMUV-E | M16SMUV-E | M10SMUCV-E | M13SMUCV-E | M16GMUCV-E |
| MULTISPLIT Outdoor units (2 rooms) | –         | –         | –         | –          | –          | –          |
| (3 rooms)                          | 3M18SAV-E | 3M18SAV-E | 3M18SAV-E | 3M18SACV-E | 3M18SACV-E | 3M18SACV-E |
| (3 rooms)                          | 3M26GAV-E | 3M26GAV-E | 3M26GAV-E | –          | –          | –          |
| (4 rooms)                          | 4M23SAV-E | 4M23SAV-E | 4M23SAV-E | 4M23SACV-E | 4M23SACV-E | 4M23SACV-E |
| (4 rooms)                          | 4M27GAV-E | 4M27GAV-E | 4M27GAV-E | 4M27GACV-E | 4M27GACV-E | 4M27GACV-E |

**R-410A****DC HYBRID INVERTER****MULTI-SPLIT**

## GAV/SAV Multi-split Outdoor unit

**Wide range of both heat pump and cooling only outdoor units**

**Up to 70 m pipe run**

### Features

Toshiba multi-splits are equipped with Toshiba DC hybrid inverter, an enhanced feature that ensures improved performance and reliability.

Thanks to the efficient DC compressors, these units provide fast and precise temperature management, with energy savings of 40 – 50% compared to fixed-speed systems.

### Key features

Wide range of indoor units available: Daiseikai, high-wall, ducted, cassette 600 × 600.

A perfect combination of DC twin rotary compressor, DC hybrid inverter and R-410A refrigerant.

Superior reliability, due to the reduction of the compressor ON/ OFF cycles.

Low noise levels.

Flexibility: this system allows up to 25 m piping length for one room ensuring the overall length is not exceeded.

Technical specifications **cooling only**

| Outdoor unit                         |                         | 2-room Multisplit |                 | 3-room Multisplit      |                 | 4-room Multisplit      |                 |
|--------------------------------------|-------------------------|-------------------|-----------------|------------------------|-----------------|------------------------|-----------------|
|                                      |                         | RAS-M14GACV-E     | RAS-M18GACV-E   | RAS-3M18SACV-E         | RAS-3M23GACV-E  | RAS-4M23SACV-E         | RAS-4M27GACV-E  |
| Cooling capacity                     | kW                      | 4,0               | 5,2             | 5,2                    | 6,7             | 6,8                    | 8,0             |
| Cooling capacity (min. – max.)       | kW                      | 1,1 – 4,5         | 1,1 – 6,2       | 2,2 – 6,5              | 2,2 – 7,0       | 3,0 – 7,5              | 1,4 – 9,2       |
| Power input                          | kW                      | 1,08              | 1,60            | 1,34                   | 2,15            | 1,95                   | 2,50            |
| EER                                  | W/W                     | 3,70              | 3,25            | 3,88                   | 3,12            | 3,49                   | 3,20            |
| Energy efficiency class              |                         | A                 | A               | A                      | B               | A                      | A               |
| Air flow                             | m <sup>3</sup> /h – l/s | 1820 – 505        | 2100 – 583      | 2100 – 583             | 3000 – 833      | 3000 – 883             | 3000 – 833      |
| Sound pressure level                 | dB(A)                   | 46                | 48              | 47                     | 48              | 47                     | 48              |
| Sound power level                    | dB(A)                   | 59                | 61              | 62                     | 61              | 62                     | 61              |
| Operating range                      | °C                      | 5 ÷ 43            | 5 ÷ 43          | 5 ÷ 43                 | 10 ÷ 43         | 5 ÷ 43                 | 10 ÷ 43         |
| Dimensions (h × w × d)               | mm                      | 550 × 780 × 290   | 550 × 780 × 290 | 695 × 780 × 270        | 695 × 780 × 270 | 795 × 900 × 320        | 795 × 900 × 320 |
| Weight                               | kg                      | 36                | 40              | 47                     | 48              | 55                     | 63              |
| Compressor type                      |                         | DC Twin Rotary    | DC Twin Rotary  | DC Twin Rotary         | DC Twin Rotary  | DC Twin Rotary         | DC Twin Rotary  |
| Flare connections – gas              |                         | 3/8" × 2          | 3/8" × 2        | 3/8" × 3               | 3/8" × 2 + 1/2  | 3/8" × 4               | 3/8" × 3 + 1/2  |
| Flare connections – liquid           |                         | 1/4" × 2          | 1/4" × 2        | 1/4" × 3               | 1/4" × 3        | 1/4" × 4               | 1/4" × 4        |
| Maximum pipe length (per unit/total) | m                       | 20/30             | 20/30           | 20/50                  | 20/40           | 25/60                  | 25/70           |
| Maximum height difference            | m                       | 10                | 10              | 10                     | 10              | 15                     | 15              |
| Chargeless pipe length               | m                       | 20                | 20              | 50                     | 40              | 40                     | 70              |
| Power supply                         | V-ph-Hz                 | 220/240-1-50      | 220/240-1-50    | 220/240-1-50, 220-1-60 | 220/240-1-50    | 220/240-1-50, 220-1-60 | 220/240-1-50    |

Technical specifications **heat pump**

| Outdoor unit                         |                         | 2-room Multisplit |                 | 3-room Multisplit |                 | 4-room Multisplit |                 |
|--------------------------------------|-------------------------|-------------------|-----------------|-------------------|-----------------|-------------------|-----------------|
|                                      |                         | RAS-M14GAV-E      | RAS-M18GAV-E    | RAS-3M18SAV-E     | RAS-3M26GAV-E   | RAS-4M23SAV-E     | RAS-4M27GAV-E   |
| Cooling capacity                     | kW                      | 4,0               | 5,2             | 5,2               | 7,5             | 6,8               | 8,0             |
| Cooling capacity (min. – max.)       | kW                      | 1,4 – 4,5         | 1,4 – 6,2       | 2,2 – 6,5         | 1,4 – 8,9       | 3,0 – 7,5         | 1,4 – 9,2       |
| Power input                          | kW                      | C 1,08            | 1,60            | 1,34              | 2,25            | 1,95              | 2,50            |
| EER                                  | W/W                     | 3,70              | 3,25            | 3,88              | 3,33            | 3,49              | 3,20            |
| Energy efficiency class              | C                       | A                 | A               | A                 | A               | A                 | B               |
| Heating capacity                     | kW                      | 4,4               | 6,7             | 6,8               | 9,0             | 7,2               | 9,0             |
| Heating capacity (min. – max.)       | kW                      | 0,9 – 5,2         | 0,9 – 8,5       | 2,2 – 7,7         | 0,8 – 10,8      | 2,0 – 8,4         | 0,8 – 11,0      |
| Power input                          | kW                      | H 1,01            | 1,85            | 1,60              | 2,55            | 1,63              | 2,25            |
| COP                                  | W/W                     | 4,36              | 3,62            | 4,25              | 3,53            | 4,42              | 4,00            |
| Energy efficiency class              | H                       | A                 | A               | A                 | B               | A                 | A               |
| Air flow                             | m <sup>3</sup> /h – l/s | 1820 – 505        | 2100 – 583      | 2100 – 583        | 3000 – 833      | 3000 – 833        | 3000 – 833      |
| Sound pressure level                 | dB(A)                   | C 46              | 48              | 47                | 48              | 47                | 48              |
| Sound power level                    | dB(A)                   | C 59              | 61              | 62                | 61              | 62                | 61              |
| Operating range                      | °C                      | C 5 ÷ 43          | 5 ÷ 43          | 5 ÷ 43            | 10 ÷ 43         | 5 ÷ 43            | 10 ÷ 43         |
| Sound pressure level                 | dB(A)                   | H 48              | 50              | 49                | 48              | 48                | 48              |
| Sound power level                    | dB(A)                   | H 61              | 63              | 64                | 61              | 63                | 61              |
| Operating range                      | °C                      | H –10 ÷ 24        | –10 ÷ 24        | –15 ÷ 24          | –10 ÷ 22        | –15 ÷ 24          | –10 ÷ 22        |
| Dimensions (h × w × d)               | mm                      | 550 × 780 × 290   | 550 × 780 × 290 | 695 × 780 × 270   | 795 × 900 × 320 | 795 × 900 × 320   | 795 × 900 × 320 |
| Weight                               | kg                      | 36                | 40              | 47                | 64              | 55                | 65              |
| Compressor type                      |                         | DC Twin Rotary    | DC Twin Rotary  | DC Twin Rotary    | DC Twin Rotary  | DC Twin Rotary    | DC Twin Rotary  |
| Flare connections – gas              |                         | 3/8" × 2          | 3/8" × 2        | 3/8" × 3          | 3/8" × 2 + 1/2  | 3/8" × 4          | 3/8" × 3 + 1/2  |
| Flare connections – liquid           |                         | 1/4" × 2          | 1/4" × 2        | 1/4" × 3          | 1/4" × 3        | 1/4" × 4          | 1/4" × 4        |
| Maximum pipe length (per unit/total) | m                       | 20/30             | 20/30           | 20/50             | 25/50           | 25/60             | 25/70           |
| Maximum height difference            | m                       | 10                | 10              | 10                | 15              | 15                | 15              |
| Chargeless pipe length               | m                       | 20                | 20              | 50                | 50              | 40                | 70              |
| Power supply                         | V-ph-Hz                 | 220/240-1-50      | 220/240-1-50    | 220/240-1-50      | 220/240-1-50    | 220/240-1-50      | 220/240-1-50    |

C = cooling mode  
H = heating mode



Combination ratings (size 14) **cooling only**

Cooling, 230V

| Operating status | Combination |        | Total | Unit capacity (kW) |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |      | EER  | label |
|------------------|-------------|--------|-------|--------------------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|------|------|-------|
|                  | Unit A      | Unit B |       | Unit A             | Unit B | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max. |      |       |
| 1 unit operation | 10          | -      | 10    | 2,50               | -      | 1,1                   | 2,5   | 3,2  | 170             | 570   | 880  | 1,06                  | 2,92  | 4,11 | 4,39 | A     |
|                  | 13          | -      | 13    | 3,50               | -      | 1,1                   | 3,5   | 3,8  | 170             | 1100  | 1280 | 1,06                  | 5,03  | 5,80 | 3,18 | B     |
| 2 unit operation | 10          | 10     | 20    | 1,95               | 1,95   | 1,4                   | 3,9   | 4,4  | 230             | 1070  | 1290 | 1,43                  | 4,90  | 5,84 | 3,64 | A     |
|                  | 13          | 10     | 23    | 2,33               | 1,67   | 1,4                   | 4,0   | 4,5  | 230             | 1080  | 1300 | 1,43                  | 4,94  | 5,89 | 3,70 | A     |

Outdoor unit:

RAS-M14GACV-E

Indoor unit:

CO  
CO**SKVCP**RAS-M10SKCVP-E  
RAS-M13SKCVP-E**SKV**RAS-M10SKCV-E  
RAS-M13SKCV-E**GDV**RAS-M10GDCV-E  
RAS-M13GDCV-ECombination ratings (size 18) **cooling only**

Cooling, 230V

| Operating status | Combination |        | Total | Unit capacity (kW) |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |      | EER  | label |
|------------------|-------------|--------|-------|--------------------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|------|------|-------|
|                  | Unit A      | Unit B |       | Unit A             | Unit B | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max. |      |       |
| 1 unit operation | 10          | -      | 10    | 2,70               | -      | 1,1                   | 2,7   | 3,2  | 220             | 600   | 800  | 1,37                  | 2,90  | 3,66 | 4,50 | A     |
|                  | 13          | -      | 13    | 3,70               | -      | 1,1                   | 3,7   | 4,2  | 220             | 1100  | 1400 | 1,37                  | 4,88  | 6,21 | 3,36 | A     |
|                  | 16          | -      | 16    | 4,50               | -      | 1,1                   | 4,5   | 4,9  | 220             | 1500  | 1700 | 1,37                  | 6,65  | 7,54 | 3,00 | C     |
| 2 unit operation | 10          | 10     | 20    | 2,55               | 2,55   | 1,4                   | 5,1   | 6,1  | 260             | 1550  | 2150 | 1,61                  | 6,88  | 9,54 | 3,29 | A     |
|                  | 13          | 10     | 23    | 2,85               | 2,35   | 1,4                   | 5,2   | 6,2  | 260             | 1600  | 2170 | 1,61                  | 7,10  | 9,63 | 3,25 | A     |
|                  | 13          | 13     | 26    | 2,60               | 2,60   | 1,4                   | 5,2   | 6,2  | 260             | 1600  | 2170 | 1,61                  | 7,10  | 9,63 | 3,25 | A     |
|                  | 16          | 10     | 26    | 3,25               | 1,95   | 1,4                   | 5,2   | 6,2  | 260             | 1600  | 2170 | 1,61                  | 7,10  | 9,63 | 3,25 | A     |

Outdoor unit:

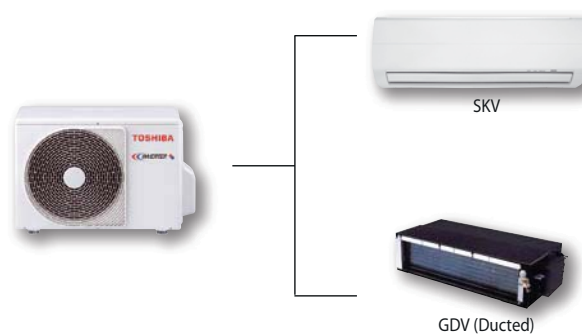
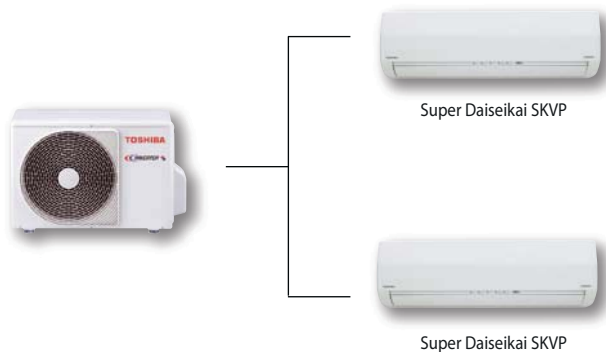
RAS-M18GACV-E

Indoor unit:

CO  
CO  
CO**SKVCP**RAS-M10SKCVP-E  
RAS-M13SKCVP-E  
RAS-M16SKCVP-E**SKV**RAS-M10SKCV-E  
RAS-M13SKCV-E  
RAS-M16SKCV-E**GDV**RAS-M10GDCV-E  
RAS-M13GDCV-E  
RAS-M16GDCV-E**SMUCV**RAS-M10SMUCV-E  
RAS-M13SMUCV-E  
RAS-M16SMUCV-E

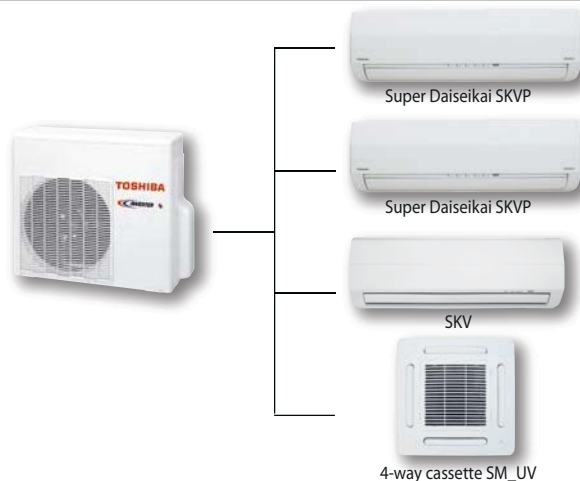
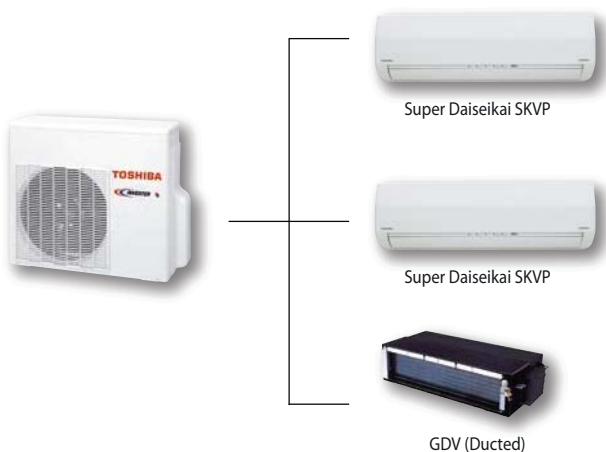
## 2 Rooms

## 2 Rooms



## 3 Rooms

## 4 Rooms



Combination ratings (size 18) **cooling only**

Cooling, 230V

| Operating status | Combination |        |        | Total | Unit capacity (kW) |        |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |      | EER  | label |
|------------------|-------------|--------|--------|-------|--------------------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|------|------|-------|
|                  | Unit A      | Unit B | Unit C |       | Unit A             | Unit B | Unit C | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max. |      |       |
| 1 unit operation | 10          | -      | -      | 10    | 2,50               | -      | -      | 1,4                   | 2,5   | 3,2  | 320             | 600   | 850  | 1,86                  | 3,39  | 4,02 | 4,17 | A     |
|                  | 13          | -      | -      | 13    | 3,50               | -      | -      | 1,4                   | 3,5   | 4,2  | 320             | 1070  | 1520 | 1,86                  | 4,90  | 6,81 | 3,27 | A     |
|                  | 16          | -      | -      | 16    | 4,50               | -      | -      | 1,4                   | 4,5   | 5,0  | 320             | 1670  | 1780 | 1,86                  | 7,49  | 7,98 | 2,69 | D     |
| 2 unit operation | 10          | 10     | -      | 20    | 2,40               | 2,40   | -      | 1,8                   | 4,8   | 5,9  | 360             | 1320  | 1800 | 2,06                  | 6,04  | 8,07 | 3,64 | A     |
|                  | 13          | 10     | -      | 23    | 2,80               | 2,00   | -      | 1,8                   | 4,8   | 5,9  | 360             | 1320  | 1800 | 2,06                  | 6,04  | 8,07 | 3,64 | A     |
|                  | 16          | 10     | -      | 26    | 3,21               | 1,79   | -      | 1,8                   | 5,0   | 6,2  | 360             | 1370  | 1820 | 2,06                  | 6,27  | 8,16 | 3,65 | A     |
|                  | 13          | 13     | -      | 26    | 2,50               | 2,50   | -      | 1,8                   | 5,0   | 6,2  | 360             | 1370  | 1820 | 2,06                  | 6,27  | 8,16 | 3,65 | A     |
| 3 unit operation | 10          | 10     | 10     | 30    | 1,70               | 1,70   | 1,70   | 2,2                   | 5,1   | 6,3  | 420             | 1340  | 1970 | 2,40                  | 6,07  | 8,74 | 3,81 | A     |
|                  | 13          | 10     | 10     | 33    | 2,14               | 1,53   | 1,53   | 2,2                   | 5,2   | 6,5  | 420             | 1340  | 2100 | 2,40                  | 6,07  | 9,32 | 3,88 | A     |
|                  | 16          | 10     | 10     | 36    | 2,46               | 1,37   | 1,37   | 2,2                   | 5,2   | 6,5  | 420             | 1340  | 2100 | 2,40                  | 6,07  | 9,32 | 3,88 | A     |
|                  | 13          | 13     | 10     | 36    | 1,92               | 1,92   | 1,37   | 2,2                   | 5,2   | 6,5  | 420             | 1340  | 2100 | 2,40                  | 6,07  | 9,32 | 3,88 | A     |

Outdoor unit:

RAS-3M18SACV-E

Indoor unit:

**SKCVP****SKCV****GDCV****SMUCV**

CO

RAS-M10SKCVP-E

RAS-M10SKCV-E

RAS-M10GDCV-E

RAS-M10SMUCV-E

CO

RAS-M13SKCVP-E

RAS-M13SKCV-E

RAS-M13GDCV-E

RAS-M13SMUCV-E

CO

RAS-M16SKCVP-E

RAS-M16SKCV-E

RAS-M16GDCV-E

RAS-M16SMUCV-E

Combination ratings (size 23) **cooling only**

Cooling, 230V

| Operating status | Combination |        |        | Total | Unit capacity (kW) |        |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |       | EER  | label |
|------------------|-------------|--------|--------|-------|--------------------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|-------|------|-------|
|                  | Unit A      | Unit B | Unit C |       | Unit A             | Unit B | Unit C | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max.  |      |       |
| 1 unit operation | 10          | -      | -      | 10    | 2,70               | -      | -      | 1,4                   | 2,7   | 3,2  | 320             | 770   | 950  | 1,99                  | 3,94  | 4,86  | 3,51 | A     |
|                  | 13          | -      | -      | 13    | 3,70               | -      | -      | 1,4                   | 3,7   | 4,4  | 320             | 1200  | 1470 | 1,99                  | 6,07  | 7,18  | 3,08 | B     |
|                  | 16          | -      | -      | 16    | 4,50               | -      | -      | 1,4                   | 4,5   | 4,9  | 320             | 1600  | 1750 | 1,99                  | 7,32  | 8,01  | 2,81 | C     |
| 2 unit operation | 10          | 10     | -      | 20    | 2,70               | 2,70   | -      | 1,8                   | 5,4   | 6,0  | 360             | 1500  | 1880 | 2,24                  | 6,86  | 8,60  | 3,60 | A     |
|                  | 10          | 13     | -      | 23    | 2,45               | 3,35   | -      | 1,8                   | 5,8   | 6,3  | 360             | 1800  | 1970 | 2,24                  | 8,24  | 9,02  | 3,22 | A     |
|                  | 10          | 16     | -      | 26    | 2,21               | 3,69   | -      | 1,8                   | 5,9   | 6,4  | 360             | 1830  | 2000 | 2,24                  | 8,38  | 9,15  | 3,22 | A     |
|                  | 13          | 13     | -      | 26    | 2,95               | 2,95   | -      | 1,8                   | 5,9   | 6,4  | 360             | 1830  | 2000 | 2,24                  | 8,38  | 9,15  | 3,22 | A     |
|                  | 13          | 16     | -      | 29    | 2,71               | 3,29   | -      | 1,8                   | 6,0   | 6,4  | 360             | 1850  | 2000 | 2,24                  | 8,50  | 9,15  | 3,24 | A     |
|                  | 16          | 16     | -      | 32    | 3,05               | 3,05   | -      | 1,8                   | 6,1   | 6,5  | 360             | 1870  | 2050 | 2,24                  | 8,56  | 9,38  | 3,26 | A     |
| 3 unit operation | 10          | 10     | 10     | 30    | 2,13               | 2,13   | 2,13   | 2,2                   | 6,4   | 7,0  | 420             | 1880  | 2300 | 2,61                  | 8,60  | 10,53 | 3,40 | A     |
|                  | 10          | 10     | 13     | 33    | 1,99               | 1,99   | 2,72   | 2,2                   | 6,7   | 7,0  | 420             | 2150  | 2300 | 2,61                  | 9,84  | 10,53 | 3,12 | B     |
|                  | 10          | 13     | 13     | 36    | 1,80               | 2,45   | 2,45   | 2,2                   | 6,7   | 7,0  | 420             | 2150  | 2300 | 2,61                  | 9,84  | 10,53 | 3,12 | B     |
|                  | 10          | 10     | 16     | 36    | 1,83               | 1,83   | 3,04   | 2,2                   | 6,7   | 7,0  | 420             | 2150  | 2300 | 2,61                  | 9,84  | 10,53 | 3,12 | B     |

Outdoor unit:

RAS-3M23GACV-E

Indoor unit:

**SKCVP****SKCV****GDCV**

CO

RAS-M10SKCVP-E

RAS-M10SKCV-E

RAS-M10GDCV-E

CO

RAS-M13SKCVP-E

RAS-M13SKCV-E

RAS-M13GDCV-E

CO

RAS-M16SKCVP-E

RAS-M16SKCV-E

RAS-M16GDCV-E

Combination ratings (size 23) **cooling only**

Cooling, 230V

| Operating status | Combination |        |        |        | Total | Unit capacity (kW) |        |        |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |       | EER  | label |
|------------------|-------------|--------|--------|--------|-------|--------------------|--------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|-------|------|-------|
|                  | Unit A      | Unit B | Unit C | Unit D |       | Unit A             | Unit B | Unit C | Unit D | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max.  |      |       |
| 1 unit operation | 10          | -      | -      | -      | 10    | 2,50               | -      | -      | -      | 1,4                   | 2,5   | 3,2  | 320             | 710   | 950  | 1,86                  | 3,63  | 4,69  | 3,52 | D     |
|                  | 13          | -      | -      | -      | 13    | 3,50               | -      | -      | -      | 1,4                   | 3,5   | 4,2  | 320             | 1340  | 1750 | 1,86                  | 6,33  | 7,93  | 2,61 | D     |
|                  | 16          | -      | -      | -      | 16    | 4,50               | -      | -      | -      | 1,4                   | 4,5   | 5,0  | 320             | 2180  | 2250 | 1,86                  | 9,77  | 10,09 | 2,06 | G     |
| 2 unit operation | 10          | 10     | -      | -      | 20    | 2,60               | 2,60   | -      | -      | 1,8                   | 5,2   | 6,0  | 360             | 1500  | 1800 | 2,09                  | 7,09  | 8,24  | 3,47 | A     |
|                  | 13          | 10     | -      | -      | 23    | 3,03               | 2,17   | -      | -      | 1,8                   | 5,2   | 6,0  | 360             | 1500  | 1800 | 2,09                  | 7,09  | 8,24  | 3,47 | A     |
|                  | 16          | 10     | -      | -      | 26    | 3,60               | 2,00   | -      | -      | 1,8                   | 5,6   | 6,2  | 360             | 1790  | 1950 | 2,09                  | 8,19  | 8,74  | 3,13 | B     |
|                  | 13          | 13     | -      | -      | 26    | 2,80               | 2,80   | -      | -      | 1,8                   | 5,6   | 6,2  | 360             | 1790  | 1950 | 2,09                  | 8,19  | 8,74  | 3,13 | B     |
|                  | 16          | 13     | -      | -      | 29    | 3,15               | 2,45   | -      | -      | 1,8                   | 5,6   | 6,2  | 360             | 1790  | 1950 | 2,09                  | 8,19  | 8,74  | 3,13 | B     |
| 3 unit operation | 10          | 10     | 10     | -      | 30    | 2,03               | 2,03   | 2,03   | -      | 2,2                   | 6,1   | 6,8  | 420             | 1840  | 1950 | 2,43                  | 8,33  | 8,74  | 3,32 | A     |
|                  | 13          | 10     | 10     | -      | 33    | 2,51               | 1,79   | 1,79   | -      | 2,2                   | 6,1   | 6,8  | 420             | 1840  | 1950 | 2,43                  | 8,33  | 8,74  | 3,32 | A     |
|                  | 16          | 10     | 10     | -      | 36    | 2,98               | 1,66   | 1,66   | -      | 2,2                   | 6,3   | 7,1  | 420             | 1950  | 2200 | 2,43                  | 8,74  | 9,76  | 3,23 | A     |
|                  | 13          | 13     | 10     | -      | 36    | 2,32               | 2,32   | 1,66   | -      | 2,2                   | 6,3   | 7,1  | 420             | 1950  | 2200 | 2,43                  | 8,74  | 9,76  | 3,23 | A     |
|                  | 16          | 13     | 10     | -      | 39    | 2,70               | 2,10   | 1,50   | -      | 2,2                   | 6,3   | 7,1  | 420             | 1950  | 2200 | 2,43                  | 8,74  | 9,76  | 3,23 | A     |
|                  | 13          | 13     | 13     | -      | 39    | 2,10               | 2,10   | 2,10   | -      | 2,2                   | 6,3   | 7,1  | 420             | 1950  | 2200 | 2,43                  | 8,74  | 9,76  | 3,23 | A     |
|                  | 16          | 13     | 13     | -      | 42    | 2,47               | 2,47   | 1,37   | -      | 2,2                   | 6,3   | 7,1  | 420             | 1950  | 2200 | 2,43                  | 8,74  | 9,76  | 3,23 | A     |
| 4 unit operation | 10          | 10     | 10     | 10     | 40    | 1,70               | 1,70   | 1,70   | 1,70   | 3,0                   | 6,8   | 7,5  | 500             | 1950  | 2300 | 2,90                  | 8,74  | 10,20 | 3,49 | A     |
|                  | 13          | 10     | 10     | 10     | 43    | 2,16               | 1,55   | 1,55   | 1,55   | 3,0                   | 6,8   | 7,5  | 500             | 1950  | 2300 | 2,90                  | 8,74  | 10,20 | 3,49 | A     |
|                  | 16          | 10     | 10     | 10     | 46    | 1,75               | 1,68   | 1,68   | 1,68   | 3,0                   | 6,8   | 7,5  | 500             | 1950  | 2300 | 2,90                  | 8,74  | 10,20 | 3,49 | A     |
|                  | 13          | 13     | 10     | 10     | 46    | 1,98               | 1,98   | 1,42   | 1,42   | 3,0                   | 6,8   | 7,5  | 500             | 1950  | 2300 | 2,90                  | 8,74  | 10,20 | 3,49 | A     |

Outdoor unit:

RAS-4M23SACV-E

Indoor unit:

**SKCVP****SKCV****GDCV****SMUCV**

CO

RAS-M10SKCVP-E

RAS-M10SKCV-E

RAS-M10GDCV-E

RAS-M10SMUCV-E

CO

RAS-M13SKCVP-E

RAS-M13SKCV-E

RAS-M13GDCV-E

RAS-M13SMUCV-E

CO

RAS-M16SKCVP-E

RAS-M16SKCV-E

RAS-M16GDCV-E

RAS-M16SMUCV-E

## Combination ratings (size 27) cooling only

Cooling, 230V

| Operating status | Combination |        |        |        | Total | Unit capacity (kW) |        |        |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |       | EER  | label |
|------------------|-------------|--------|--------|--------|-------|--------------------|--------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|-------|------|-------|
|                  | Unit A      | Unit B | Unit C | Unit D |       | Unit A             | Unit B | Unit C | Unit D | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max.  |      |       |
| 1 unit operation | 10          | -      | -      | -      | 10    | 2,70               | -      | -      | -      | 1,4                   | 2,7   | 3,2  | 640             | 750   | 950  | 3,52                  | 3,71  | 4,44  | 3,60 | A     |
|                  | 13          | -      | -      | -      | 13    | 3,70               | -      | -      | -      | 1,4                   | 3,7   | 4,4  | 640             | 1200  | 1520 | 3,52                  | 5,49  | 6,88  | 3,08 | B     |
|                  | 16          | -      | -      | -      | 16    | 4,50               | -      | -      | -      | 1,4                   | 4,5   | 5,0  | 640             | 1650  | 2000 | 3,52                  | 7,47  | 8,87  | 2,73 | D     |
| 2 unit operation | 10          | 10     | -      | -      | 20    | 2,70               | 2,70   | -      | -      | 2,5                   | 5,4   | 6,3  | 640             | 1530  | 2040 | 3,48                  | 6,79  | 9,05  | 3,53 | A     |
|                  | 13          | 10     | -      | -      | 23    | 3,41               | 2,49   | -      | -      | 2,7                   | 5,9   | 6,6  | 660             | 1810  | 2220 | 3,59                  | 8,03  | 9,85  | 3,26 | A     |
|                  | 16          | 10     | -      | -      | 26    | 3,94               | 2,36   | -      | -      | 2,9                   | 6,3   | 6,9  | 670             | 2040  | 2400 | 3,64                  | 9,05  | 10,65 | 3,09 | B     |
|                  | 13          | 13     | -      | -      | 26    | 3,15               | 3,15   | -      | -      | 2,9                   | 6,3   | 6,9  | 670             | 2040  | 2400 | 3,64                  | 9,05  | 10,65 | 3,09 | B     |
|                  | 16          | 13     | -      | -      | 29    | 3,73               | 3,07   | -      | -      | 3,0                   | 6,8   | 7,2  | 690             | 2320  | 2570 | 3,75                  | 10,29 | 11,40 | 2,93 | C     |
|                  | 16          | 16     | -      | -      | 32    | 3,60               | 3,60   | -      | -      | 3,2                   | 7,2   | 7,5  | 700             | 2550  | 2750 | 3,80                  | 11,31 | 12,20 | 2,82 | C     |
| 3 unit operation | 10          | 10     | 10     | -      | 30    | 2,53               | 2,53   | 2,53   | -      | 3,8                   | 7,6   | 8,2  | 950             | 2400  | 2720 | 4,59                  | 10,65 | 12,07 | 3,17 | B     |
|                  | 13          | 10     | 10     | -      | 33    | 3,13               | 2,28   | 2,28   | -      | 3,9                   | 7,7   | 8,3  | 960             | 2410  | 2740 | 4,64                  | 10,69 | 12,16 | 3,20 | B     |
|                  | 16          | 10     | 10     | -      | 36    | 3,50               | 2,10   | 2,10   | -      | 4,0                   | 7,7   | 8,5  | 960             | 2410  | 2790 | 4,64                  | 10,69 | 12,38 | 3,20 | B     |
|                  | 13          | 13     | 10     | -      | 36    | 2,82               | 2,82   | 2,06   | -      | 4,0                   | 7,7   | 8,5  | 960             | 2410  | 2790 | 4,64                  | 10,69 | 12,38 | 3,20 | B     |
|                  | 16          | 13     | 10     | -      | 39    | 3,22               | 2,65   | 1,93   | -      | 4,1                   | 7,8   | 8,6  | 970             | 2430  | 2810 | 4,69                  | 10,78 | 12,47 | 3,21 | A     |
|                  | 13          | 13     | 13     | -      | 39    | 2,60               | 2,60   | 2,60   | -      | 4,1                   | 7,8   | 8,6  | 970             | 2430  | 2810 | 4,69                  | 10,78 | 12,47 | 3,21 | A     |
|                  | 16          | 16     | 10     | -      | 42    | 3,04               | 3,04   | 1,82   | -      | 4,1                   | 7,9   | 8,7  | 970             | 2440  | 2830 | 4,69                  | 10,83 | 12,56 | 3,24 | A     |
|                  | 16          | 13     | 13     | -      | 42    | 2,99               | 2,46   | 2,46   | -      | 4,1                   | 7,9   | 8,7  | 970             | 2440  | 2830 | 4,69                  | 10,83 | 12,56 | 3,24 | A     |
|                  | 16          | 16     | 13     | -      | 45    | 2,80               | 2,80   | 2,30   | -      | 4,2                   | 7,9   | 8,9  | 970             | 2440  | 2880 | 4,69                  | 10,83 | 12,78 | 3,24 | A     |
|                  | 16          | 16     | 16     | -      | 48    | 2,67               | 2,67   | 2,67   | -      | 4,3                   | 8,0   | 9,0  | 980             | 2450  | 2900 | 4,73                  | 10,87 | 12,87 | 3,27 | A     |
| 4 unit operation | 10          | 10     | 10     | 10     | 40    | 1,98               | 1,98   | 1,98   | 1,98   | 4,0                   | 7,9   | 8,7  | 930             | 2450  | 2800 | 4,49                  | 10,87 | 12,42 | 3,22 | A     |
|                  | 13          | 10     | 10     | 10     | 43    | 2,48               | 1,81   | 1,81   | 1,81   | 4,1                   | 7,9   | 8,8  | 940             | 2450  | 2820 | 4,54                  | 10,87 | 12,51 | 3,22 | A     |
|                  | 16          | 10     | 10     | 10     | 46    | 2,86               | 1,71   | 1,71   | 1,71   | 4,1                   | 8,0   | 9,0  | 940             | 2500  | 2860 | 4,54                  | 11,09 | 12,69 | 3,20 | B     |
|                  | 13          | 13     | 10     | 10     | 46    | 2,31               | 2,31   | 1,69   | 1,69   | 4,1                   | 8,0   | 9,0  | 940             | 2500  | 2860 | 4,54                  | 11,09 | 12,69 | 3,20 | B     |
|                  | 16          | 13     | 10     | 10     | 49    | 2,65               | 2,18   | 1,59   | 1,59   | 4,2                   | 8,0   | 9,1  | 950             | 2500  | 2880 | 4,59                  | 11,09 | 12,78 | 3,20 | B     |
|                  | 13          | 13     | 13     | 10     | 49    | 2,14               | 2,14   | 2,14   | 1,57   | 4,2                   | 8,0   | 9,1  | 950             | 2500  | 2880 | 4,59                  | 11,09 | 12,78 | 3,20 | B     |
|                  | 16          | 13     | 13     | 10     | 52    | 2,47               | 2,03   | 2,03   | 1,48   | 4,2                   | 8,0   | 9,2  | 950             | 2500  | 2900 | 4,59                  | 11,09 | 12,87 | 3,20 | B     |
|                  | 13          | 13     | 13     | 13     | 52    | 2,00               | 2,00   | 2,00   | 2,00   | 4,2                   | 8,0   | 9,2  | 950             | 2500  | 2900 | 4,59                  | 11,09 | 12,87 | 3,20 | B     |
|                  | 16          | 16     | 10     | 10     | 52    | 2,50               | 2,50   | 1,50   | 1,50   | 4,2                   | 8,0   | 9,2  | 950             | 2500  | 2900 | 4,59                  | 11,09 | 12,87 | 3,20 | B     |

Outdoor unit:

RAS-4M27GACV-E

Indoor unit:

SKCVP

SKCV

GDCV

SMUCV

CO

RAS-M10SKCVP-E

RAS-M10SKCV-E

RAS-M10GDCV-E

RAS-M10SMUCV-E

CO

RAS-M13SKCVP-E

RAS-M13SKCV-E

RAS-M13GDCV-E

RAS-M13SMUCV-E

CO

RAS-M16SKCVP-E

RAS-M16SKCV-E

RAS-M16GDCV-E

RAS-M16SMUCV-E

## Combination ratings (size 14) heat pump

Cooling, 230V

| Operating status | Combination |        | Total | Unit capacity (kW) |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |      | EER  | label |
|------------------|-------------|--------|-------|--------------------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|------|------|-------|
|                  | Unit A      | Unit B |       | Unit A             | Unit B | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max. |      |       |
| 1 unit operation | 10          | -      | 10    | 2,50               | -      | 1,1                   | 2,5   | 3,2  | 170             | 570   | 880  | 1,06                  | 2,92  | 4,11 | 4,39 | A     |
|                  | 13          | -      | 13    | 3,50               | -      | 1,1                   | 3,5   | 3,8  | 170             | 1100  | 1280 | 1,06                  | 5,03  | 5,80 | 3,18 | B     |
| 2 unit operation | 10          | 10     | 20    | 1,95               | 1,95   | 1,4                   | 3,9   | 4,4  | 230             | 1070  | 1290 | 1,43                  | 4,90  | 5,84 | 3,64 | A     |
|                  | 13          | 10     | 23    | 2,33               | 1,67   | 1,4                   | 4,0   | 4,5  | 230             | 1080  | 1300 | 1,43                  | 4,94  | 5,89 | 3,70 | A     |

Heating, 230V

| Operating status | Combination |        | Total | Unit capacity (kW) |        | Heating capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |      | EER  | label |
|------------------|-------------|--------|-------|--------------------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|------|------|-------|
|                  | Unit A      | Unit B |       | Unit A             | Unit B | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max. |      |       |
| 1 unit operation | 10          | -      | 10    | 3,20               | -      | 0,7                   | 3,2   | 4,0  | 130             | 850   | 1280 | 0,81                  | 3,89  | 6,41 | 3,76 | A     |
|                  | 13          | -      | 13    | 4,20               | -      | 0,7                   | 4,2   | 4,4  | 130             | 1250  | 1350 | 0,81                  | 5,60  | 6,86 | 3,36 | C     |
| 2 unit operation | 10          | 10     | 20    | 2,15               | 2,15   | 0,9                   | 4,3   | 4,6  | 170             | 980   | 1230 | 1,06                  | 4,39  | 5,51 | 4,39 | A     |
|                  | 13          | 10     | 23    | 2,50               | 1,90   | 0,9                   | 4,4   | 4,7  | 170             | 1010  | 1250 | 1,06                  | 4,53  | 5,60 | 4,36 | A     |

Outdoor unit:

RAS-M14GAV-E

Indoor unit:

SKVP

SKV

GDV

HP

RAS-B10SKVP-E

RAS-M10SKV-E

RAS-M10GDV-E

HP

RAS-B13SKVP-E

RAS-M13SKV-E

RAS-M13GDV-E

## Combination ratings (size 18) heat pump

Cooling, 230V

| Operating status | Combination |        | Total | Unit capacity (kW) |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |      | EER  | label |
|------------------|-------------|--------|-------|--------------------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|------|------|-------|
|                  | Unit A      | Unit B |       | Unit A             | Unit B | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max. |      |       |
| 1 unit operation | 10          | -      | 10    | 2,70               | -      | 1,1                   | 2,7   | 3,2  | 220             | 600   | 800  | 1,37                  | 2,90  | 3,66 | 4,50 | A     |
|                  | 13          | -      | 13    | 3,70               | -      | 1,1                   | 3,7   | 4,2  | 220             | 1100  | 1400 | 1,37                  | 4,88  | 6,21 | 3,36 | A     |
|                  | 16          | -      | 16    | 4,50               | -      | 1,1                   | 4,5   | 4,9  | 220             | 1500  | 1700 | 1,37                  | 6,65  | 7,54 | 3,00 | C     |
| 2 unit operation | 10          | 10     | 20    | 2,55               | 2,55   | 1,4                   | 5,1   | 6,1  | 260             | 1550  | 2150 | 1,61                  | 6,88  | 9,54 | 3,29 | A     |
|                  | 13          | 10     | 23    | 3,01               | 2,19   | 1,4                   | 5,2   | 6,2  | 260             | 1600  | 2170 | 1,61                  | 7,10  | 9,63 | 3,25 | A     |
|                  | 13          | 13     | 26    | 2,60               | 2,60   | 1,4                   | 5,2   | 6,2  | 260             | 1600  | 2170 | 1,61                  | 7,10  | 9,63 | 3,25 | A     |
|                  | 16          | 10     | 26    | 3,25               | 1,95   | 1,4                   | 5,2   | 6,2  | 260             | 1600  | 2170 | 1,61                  | 7,10  | 9,63 | 3,25 | A     |

Heating, 230V

| Operating status | Combination |        | Total | Unit capacity (kW) |        | Heating capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |       | EER  | label |
|------------------|-------------|--------|-------|--------------------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|-------|------|-------|
|                  | Unit A      | Unit B |       | Unit A             | Unit B | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max.  |      |       |
| 1 unit operation | 10          | -      | 10    | 4,00               | -      | 0,7                   | 4,0   | 5,2  | 170             | 1200  | 1700 | 1,06                  | 5,32  | 7,54  | 3,33 | C     |
|                  | 13          | -      | 13    | 5,00               | -      | 0,7                   | 5,0   | 6,5  | 170             | 1800  | 2530 | 1,06                  | 7,99  | 11,22 | 2,78 | E     |
|                  | 16          | -      | 16    | 5,50               | -      | 0,7                   | 5,5   | 6,9  | 170             | 1900  | 2530 | 1,06                  | 8,43  | 11,22 | 2,89 | D     |
| 2 unit operation | 10          | 10     | 20    | 3,20               | 3,20   | 0,9                   | 6,4   | 8,3  | 200             | 1800  | 2390 | 1,24                  | 7,99  | 10,60 | 3,56 | B     |
|                  | 13          | 10     | 23    | 3,72               | 2,98   | 0,9                   | 6,7   | 8,7  | 200             | 1850  | 2450 | 1,24                  | 8,21  | 10,87 | 3,62 | A     |
|                  | 13          | 13     | 26    | 3,35               | 3,35   | 0,9                   | 6,7   | 8,7  | 200             | 1850  | 2450 | 1,24                  | 8,21  | 10,87 | 3,62 | A     |
|                  | 16          | 10     | 26    | 3,88               | 2,82   | 0,9                   | 6,7   | 8,7  | 200             | 1850  | 2450 | 1,24                  | 8,21  | 10,87 | 3,62 | A     |

Outdoor unit:

RAS-M18GAV-E

Indoor unit:

SKVP

SKV

GDV

SMUV

HP

RAS-B10SKVP-E

RAS-M10SKV-E

RAS-M10GDV-E

RAS-M10SMUV-E

HP

RAS-B13SKVP-E

RAS-M13SKV-E

RAS-M13GDV-E

RAS-M13SMUV-E

HP

RAS-B16SKVP-E

RAS-M16SKV-E

RAS-M16GDV-E

## Combination ratings (size 18) heat pump

## Cooling, 230 V

| Operating status | Combination |        |        | Total | Unit capacity (kW) |        |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |      | EER  | label |
|------------------|-------------|--------|--------|-------|--------------------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|------|------|-------|
|                  | Unit A      | Unit B | Unit C |       | Unit A             | Unit B | Unit C | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max. |      |       |
| 1 unit operation | 10          | -      | -      | 10    | 2,50               | -      | -      | 1,4                   | 2,5   | 3,2  | 320             | 600   | 850  | 1,86                  | 3,39  | 4,02 | 4,17 | A     |
|                  | 13          | -      | -      | 13    | 3,50               | -      | -      | 1,4                   | 3,5   | 4,2  | 320             | 1070  | 1520 | 1,86                  | 4,90  | 6,81 | 3,27 | A     |
|                  | 16          | -      | -      | 16    | 4,50               | -      | -      | 1,4                   | 4,5   | 5,0  | 320             | 1670  | 1780 | 1,86                  | 7,49  | 7,98 | 2,69 | D     |
| 2 unit operation | 10          | 10     | -      | 20    | 2,40               | 2,40   | -      | 1,8                   | 4,8   | 5,9  | 360             | 1320  | 1800 | 2,06                  | 6,04  | 8,07 | 3,64 | A     |
|                  | 13          | 10     | -      | 23    | 2,80               | 2,00   | -      | 1,8                   | 4,8   | 5,9  | 360             | 1320  | 1800 | 2,06                  | 6,04  | 8,07 | 3,64 | A     |
|                  | 16          | 10     | -      | 26    | 3,21               | 1,79   | -      | 1,8                   | 5,0   | 6,2  | 360             | 1370  | 1820 | 2,06                  | 6,27  | 8,16 | 3,65 | A     |
| 3 unit operation | 13          | 13     | -      | 26    | 2,50               | 2,50   | -      | 1,8                   | 5,0   | 6,2  | 360             | 1370  | 1820 | 2,06                  | 6,27  | 8,16 | 3,65 | A     |
|                  | 10          | 10     | 10     | 30    | 1,70               | 1,70   | 1,70   | 2,2                   | 5,1   | 6,3  | 420             | 1340  | 1970 | 2,40                  | 6,07  | 8,74 | 3,81 | A     |
|                  | 13          | 10     | 10     | 33    | 2,14               | 1,53   | 1,53   | 2,2                   | 5,2   | 6,5  | 420             | 1340  | 2100 | 2,40                  | 6,07  | 9,32 | 3,88 | A     |
|                  | 16          | 10     | 10     | 36    | 2,46               | 1,37   | 1,37   | 2,2                   | 5,2   | 6,5  | 420             | 1340  | 2100 | 2,40                  | 6,07  | 9,32 | 3,88 | A     |
|                  | 13          | 13     | 10     | 36    | 1,92               | 1,92   | 1,37   | 2,2                   | 5,2   | 6,5  | 420             | 1340  | 2100 | 2,40                  | 6,07  | 9,32 | 3,88 | A     |

## Heating, 230 V

| Operating status | Combination |        |        | Total | Unit capacity (kW) |        |        | Heating capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |      | EER  | label |
|------------------|-------------|--------|--------|-------|--------------------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|------|------|-------|
|                  | Unit A      | Unit B | Unit C |       | Unit A             | Unit B | Unit C | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max. |      |       |
| 1 unit operation | 10          | -      | -      | 10    | 3,40               | -      | -      | 0,8                   | 3,4   | 4,5  | 300             | 1050  | 1500 | 1,74                  | 4,86  | 6,79 | 3,24 | C     |
|                  | 13          | -      | -      | 13    | 4,30               | -      | -      | 0,8                   | 4,3   | 4,8  | 300             | 1470  | 1600 | 1,74                  | 6,66  | 7,25 | 2,93 | D     |
|                  | 16          | -      | -      | 16    | 5,20               | -      | -      | 0,8                   | 5,2   | 6,0  | 300             | 1850  | 2050 | 1,74                  | 8,29  | 9,09 | 2,81 | D     |
| 2 unit operation | 10          | 10     | -      | 20    | 3,10               | 3,10   | -      | 1,8                   | 6,2   | 7,3  | 360             | 1720  | 2140 | 2,06                  | 7,79  | 9,49 | 3,60 | A     |
|                  | 13          | 10     | -      | 23    | 3,46               | 2,74   | -      | 1,8                   | 6,2   | 7,3  | 360             | 1720  | 2140 | 2,06                  | 7,79  | 9,49 | 3,60 | A     |
|                  | 16          | 10     | -      | 26    | 3,87               | 2,53   | -      | 1,8                   | 6,4   | 7,5  | 360             | 1840  | 2200 | 2,06                  | 8,33  | 9,76 | 3,48 | B     |
| 3 unit operation | 13          | 13     | -      | 26    | 3,20               | 3,20   | -      | 1,8                   | 6,4   | 7,5  | 360             | 1840  | 2200 | 2,06                  | 8,33  | 9,76 | 3,48 | B     |
|                  | 10          | 10     | 10     | 30    | 2,23               | 2,23   | 2,23   | 2,2                   | 6,7   | 7,5  | 420             | 1580  | 1950 | 2,40                  | 7,16  | 8,74 | 4,24 | A     |
|                  | 13          | 10     | 10     | 33    | 2,63               | 2,08   | 2,08   | 2,2                   | 6,8   | 7,7  | 420             | 1600  | 2030 | 2,40                  | 7,25  | 9,01 | 4,25 | A     |
|                  | 16          | 10     | 10     | 36    | 2,95               | 1,93   | 1,93   | 2,2                   | 6,8   | 7,7  | 420             | 1600  | 2030 | 2,40                  | 7,25  | 9,01 | 4,25 | A     |
|                  | 13          | 13     | 10     | 36    | 2,44               | 2,44   | 1,93   | 2,2                   | 6,8   | 7,7  | 420             | 1600  | 2030 | 2,40                  | 7,25  | 9,01 | 4,25 | A     |

Outdoor unit:

RAS-3M18SAV-E

Indoor unit:

HP  
HP  
HP

SKVP

RAS-B10SKVP-E  
RAS-B13SKVP-E  
RAS-B16SKVP-E

SKV

RAS-M10SKV-E  
RAS-M13SKV-E  
RAS-M16SKV-E

GDV

RAS-M10GDV-E  
RAS-M13GDV-E  
RAS-M16GDV-E

SMUV

RAS-M10SMUV-E  
RAS-M13SMUV-E  
RAS-M16SMUV-E

## Combination ratings (size 26) heat pump

## Cooling, 230 V

| Operating status | Combination |        |        | Total | Unit capacity (kW) |        |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |       | EER  | label |
|------------------|-------------|--------|--------|-------|--------------------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|-------|------|-------|
|                  | Unit A      | Unit B | Unit C |       | Unit A             | Unit B | Unit C | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max.  |      |       |
| 1 unit operation | 10          | -      | -      | 10    | 2,70               | -      | -      | 1,4                   | 2,7   | 3,2  | 640             | 750   | 950  | 3,52                  | 3,71  | 4,44  | 3,60 | A     |
|                  | 13          | -      | -      | 13    | 3,70               | -      | -      | 1,4                   | 3,7   | 4,4  | 640             | 1200  | 1520 | 3,52                  | 5,49  | 6,88  | 3,08 | B     |
|                  | 16          | -      | -      | 16    | 4,50               | -      | -      | 1,4                   | 4,5   | 5,0  | 640             | 1650  | 2000 | 3,52                  | 7,47  | 8,87  | 2,73 | D     |
| 2 unit operation | 10          | 10     | -      | 10    | 2,70               | 2,70   | -      | 2,5                   | 5,4   | 6,3  | 640             | 1530  | 2040 | 3,48                  | 6,79  | 9,05  | 3,53 | A     |
|                  | 13          | 10     | -      | 23    | 3,41               | 2,49   | -      | 2,7                   | 5,9   | 6,6  | 660             | 1810  | 2220 | 3,59                  | 8,03  | 9,85  | 3,26 | A     |
|                  | 16          | 10     | -      | 26    | 3,94               | 2,36   | -      | 2,9                   | 6,3   | 6,9  | 670             | 2040  | 2400 | 3,64                  | 9,05  | 10,65 | 3,09 | B     |
| 3 unit operation | 13          | 13     | -      | 26    | 3,15               | 3,15   | -      | 2,9                   | 6,3   | 6,9  | 670             | 2040  | 2400 | 3,64                  | 9,05  | 10,65 | 3,09 | B     |
|                  | 16          | 13     | -      | 29    | 3,73               | 3,07   | -      | 3,0                   | 6,8   | 7,2  | 690             | 2320  | 2570 | 3,75                  | 10,29 | 11,40 | 2,93 | C     |
|                  | 16          | 16     | -      | 32    | 3,60               | 3,60   | -      | 3,2                   | 7,2   | 7,5  | 700             | 2550  | 2750 | 3,80                  | 11,31 | 12,20 | 2,82 | C     |
| 3 unit operation | 10          | 10     | 10     | 30    | 2,47               | 2,47   | 2,47   | 3,6                   | 7,4   | 8,2  | 950             | 2230  | 2720 | 4,59                  | 9,89  | 12,07 | 3,32 | A     |
|                  | 13          | 10     | 10     | 33    | 3,01               | 2,20   | 2,20   | 3,9                   | 7,4   | 8,3  | 950             | 2230  | 2750 | 4,59                  | 9,89  | 12,20 | 3,32 | A     |
|                  | 16          | 10     | 10     | 36    | 3,36               | 2,02   | 2,02   | 4,0                   | 7,4   | 8,5  | 950             | 2230  | 2820 | 4,59                  | 9,89  | 12,51 | 3,32 | A     |
| 3 unit operation | 13          | 13     | 10     | 36    | 2,71               | 2,71   | 1,98   | 4,0                   | 7,4   | 8,5  | 950             | 2230  | 2820 | 4,59                  | 9,89  | 12,51 | 3,32 | A     |
|                  | 16          | 13     | 10     | 39    | 3,10               | 2,55   | 1,86   | 4,0                   | 7,5   | 8,6  | 980             | 2250  | 2850 | 4,73                  | 9,98  | 12,64 | 3,33 | A     |
|                  | 13          | 13     | 13     | 39    | 2,50               | 2,50   | 2,50   | 4,0                   | 7,5   | 8,6  | 980             | 2250  | 2850 | 4,73                  | 9,98  | 12,64 | 3,33 | A     |
| 3 unit operation | 16          | 16     | 10     | 42    | 2,88               | 2,88   | 1,73   | 4,1                   | 7,5   | 8,8  | 980             | 2250  | 2920 | 4,73                  | 9,98  | 12,95 | 3,33 | A     |
|                  | 16          | 13     | 13     | 42    | 2,84               | 2,33   | 2,33   | 4,1                   | 7,5   | 8,8  | 980             | 2250  | 2920 | 4,73                  | 9,98  | 12,95 | 3,33 | A     |
|                  | 16          | 16     | 13     | 45    | 2,66               | 2,66   | 2,19   | 4,2                   | 7,5   | 8,9  | 980             | 2250  | 2950 | 4,73                  | 9,98  | 13,09 | 3,33 | A     |

## Heating, 230 V

| Operating status | Combination |        |        | Total | Unit capacity (kW) |        |        | Heating capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |       | EER  | label |
|------------------|-------------|--------|--------|-------|--------------------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|-------|------|-------|
|                  | Unit A      | Unit B | Unit C |       | Unit A             | Unit B | Unit C | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max.  |      |       |
| 1 unit operation | 10          | -      | -      | 10    | 4,00               | -      | -      | 0,8                   | 4,0   | 5,2  | 300             | 1500  | 1980 | 1,79                  | 6,65  | 8,78  | 2,67 | E     |
|                  | 13          | -      | -      | 13    | 5,00               | -      | -      | 0,8                   | 5,0   | 6,5  | 310             | 2050  | 2750 | 1,85                  | 9,09  | 12,20 | 2,44 | F     |
|                  | 16          | -      | -      | 16    | 5,50               | -      | -      | 0,8                   | 5,5   | 6,9  | 310             | 2400  | 3000 | 1,85                  | 10,65 | 13,31 | 2,29 | G     |
| 2 unit operation | 10          | 10     | -      | 10    | 3,60               | 3,60   | -      | 1,5                   | 7,2   | 10,0 | 320             | 2050  | 3200 | 1,86                  | 9,09  | 14,20 | 3,51 | B     |
|                  | 13          | 10     | -      | 23    | 4,22               | 3,38   | -      | 1,5                   | 7,6   | 10,1 | 320             | 2240  | 3210 | 1,86                  | 9,94  | 14,24 | 3,39 | C     |
|                  | 16          | 10     | -      | 26    | 4,57               | 3,33   | -      | 1,5                   | 7,9   | 10,1 | 320             | 2380  | 3230 | 1,86                  | 10,56 | 14,33 | 3,32 | C     |
| 3 unit operation | 13          | 13     | -      | 26    | 3,95               | 3,95   | -      | 1,5                   | 7,9   | 10,1 | 320             | 2380  | 3230 | 1,86                  | 10,56 | 14,33 | 3,32 | C     |
|                  | 16          | 13     | -      | 29    | 4,35               | 3,95   | -      | 1,5                   | 8,3   | 10,2 | 320             | 2560  | 3240 | 1,86                  | 11,36 | 14,37 | 3,24 | C     |
|                  | 16          | 16     | -      | 32    | 4,30               | 4,30   | -      | 1,5                   | 8,6   | 10,2 | 320             | 2700  | 3250 | 1,86                  | 11,98 | 14,42 | 3,19 | D     |
| 3 unit operation | 10          | 10     | 10     | 30    | 2,87               | 2,87   | 2,87   | 2,0                   | 8,6   | 10,4 | 380             | 2300  | 2750 | 2,07                  | 10,20 | 12,20 | 3,74 | A     |
|                  | 13          | 10     | 10     | 33    | 3,35               | 2,68   | 2,68   | 2,0                   | 8,7   | 10,5 | 380             | 2360  | 2760 | 2,07                  | 10,47 | 12,24 | 3,69 | A     |
|                  | 16          | 10     | 10     | 36    | 3,59               | 2,61   | 2,61   | 2,0                   | 8,8   | 10,6 | 380             | 2430  | 2780 | 2,07                  | 10,78 | 12,33 | 3,62 | A     |
| 3 unit operation | 13          | 13     | 10     | 36    | 3,14               | 3,14   | 2,51   | 2,0                   | 8,8   | 10,6 | 380             | 2430  | 2780 | 2,07                  | 10,78 | 12,33 | 3,62 | A     |
|                  | 16          | 13     | 10     | 39    | 3,34               | 3,03   | 2,43   | 2,0                   | 8,8   | 10,6 | 380             | 2430  | 2780 | 2,07                  | 10,78 | 12,33 | 3,62 | A     |
|                  | 13          | 13     | 13     | 39    | 2,93               | 2,93   | 2,93   | 2,0                   | 8,8   | 10,6 | 380             | 2430  | 2780 | 2,07                  | 10,78 | 12,33 | 3,62 | A     |
| 3 unit operation | 16          | 16     | 10     | 42    | 3,26               | 3,26   | 2,37   | 2,0                   | 8,9   | 10,7 | 380             | 2490  | 2790 | 2,07                  | 11,05 | 12,38 | 3,57 | B     |
|                  | 16          | 13     | 13     | 42    | 3,16               | 2,87   | 2,87   | 2,0                   | 8,9   | 10,7 | 380             | 2490  | 2790 | 2,07                  | 11,05 | 12,38 | 3,57 | B     |
|                  | 16          | 16     | 13     | 45    | 3,09               | 3,09   | 2,81   | 2,0                   | 9,0   | 10,8 | 380             | 2550  | 2800 | 2,07                  | 11,31 | 12,42 | 3,53 | B     |

Outdoor unit:

RAS-3M26GAV-E

Indoor unit:

HP  
HP  
HP

SKVP

RAS-B10SKVP-E  
RAS-B13SKVP-E  
RAS-B16SKVP-E

SKV

RAS-M10SKV-E  
RAS-M13SKV-E  
RAS-M16SKV-E

GDV

RAS-M10GDV-E  
RAS-M13GDV-E  
RAS-M16GDV-E

SMUV

RAS-M10SMUV-E  
RAS-M13SMUV-E  
RAS-M16SMUV-E



## Combination ratings (size 23) heat pump

## Cooling, 230V

| Operating status | Combination |        |        |        | Total | Unit capacity (kW) |        |        |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |       | EER  | label |
|------------------|-------------|--------|--------|--------|-------|--------------------|--------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|-------|------|-------|
|                  | Unit A      | Unit B | Unit C | Unit D |       | Unit A             | Unit B | Unit C | Unit D | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max.  |      |       |
| 1 unit operation | 10          | -      | -      | -      | 10    | 2,50               | -      | -      | -      | 1,4                   | 2,5   | 3,2  | 320             | 710   | 950  | 1,86                  | 3,63  | 4,69  | 3,52 | A     |
|                  | 13          | -      | -      | -      | 13    | 3,50               | -      | -      | -      | 1,4                   | 3,5   | 4,2  | 320             | 1340  | 1750 | 1,86                  | 6,33  | 7,93  | 2,61 | D     |
|                  | 16          | -      | -      | -      | 16    | 4,50               | -      | -      | -      | 1,4                   | 4,5   | 5,0  | 320             | 2180  | 2250 | 1,86                  | 9,77  | 10,09 | 2,06 | G     |
| 2 unit operation | 10          | 10     | -      | -      | 20    | 2,60               | 2,60   | -      | -      | 1,8                   | 5,2   | 6,0  | 360             | 1500  | 1800 | 2,09                  | 7,09  | 8,24  | 3,47 | A     |
|                  | 13          | 10     | -      | -      | 23    | 3,03               | 2,17   | -      | -      | 1,8                   | 5,2   | 6,0  | 360             | 1500  | 1800 | 2,09                  | 7,09  | 8,24  | 3,47 | A     |
|                  | 16          | 10     | -      | -      | 26    | 3,60               | 2,00   | -      | -      | 1,8                   | 5,6   | 6,2  | 360             | 1790  | 1950 | 2,09                  | 8,19  | 8,74  | 3,13 | B     |
|                  | 13          | 13     | -      | -      | 26    | 2,80               | 2,80   | -      | -      | 1,8                   | 5,6   | 6,2  | 360             | 1790  | 1950 | 2,09                  | 8,19  | 8,74  | 3,13 | B     |
|                  | 16          | 13     | -      | -      | 29    | 3,15               | 2,45   | -      | -      | 1,8                   | 5,6   | 6,2  | 360             | 1790  | 1950 | 2,09                  | 8,19  | 8,74  | 3,13 | B     |
| 3 unit operation | 10          | 10     | 10     | -      | 30    | 2,03               | 2,03   | 2,03   | -      | 2,2                   | 6,1   | 6,8  | 420             | 1840  | 1950 | 2,43                  | 8,33  | 8,74  | 3,32 | A     |
|                  | 13          | 10     | 10     | -      | 33    | 2,51               | 1,79   | 1,79   | -      | 2,2                   | 6,1   | 6,8  | 420             | 1840  | 1950 | 2,43                  | 8,33  | 8,74  | 3,32 | A     |
|                  | 16          | 10     | 10     | -      | 36    | 2,98               | 1,66   | 1,66   | -      | 2,2                   | 6,3   | 7,1  | 420             | 1950  | 2200 | 2,43                  | 8,74  | 9,76  | 3,23 | A     |
|                  | 13          | 13     | 10     | -      | 36    | 2,32               | 2,32   | 1,66   | -      | 2,2                   | 6,3   | 7,1  | 420             | 1950  | 2200 | 2,43                  | 8,74  | 9,76  | 3,23 | A     |
|                  | 16          | 13     | 10     | -      | 39    | 2,70               | 2,10   | 1,50   | -      | 2,2                   | 6,3   | 7,1  | 420             | 1950  | 2200 | 2,43                  | 8,74  | 9,76  | 3,23 | A     |
|                  | 13          | 13     | 13     | -      | 39    | 2,10               | 2,10   | 2,10   | -      | 2,2                   | 6,3   | 7,1  | 420             | 1950  | 2200 | 2,43                  | 8,74  | 9,76  | 3,23 | A     |
|                  | 16          | 13     | 13     | -      | 42    | 2,47               | 2,47   | 1,37   | -      | 2,2                   | 6,3   | 7,1  | 420             | 1950  | 2200 | 2,43                  | 8,74  | 9,76  | 3,23 | A     |
| 4 unit operation | 10          | 10     | 10     | 10     | 40    | 1,70               | 1,70   | 1,70   | 1,70   | 3,0                   | 6,8   | 7,5  | 500             | 1950  | 2300 | 2,90                  | 8,74  | 10,20 | 3,49 | A     |
|                  | 13          | 10     | 10     | 10     | 43    | 2,16               | 1,55   | 1,55   | 1,55   | 3,0                   | 6,8   | 7,5  | 500             | 1950  | 2300 | 2,90                  | 8,74  | 10,20 | 3,49 | A     |
|                  | 16          | 10     | 10     | 10     | 46    | 1,75               | 1,68   | 1,68   | 1,68   | 3,0                   | 6,8   | 7,5  | 500             | 1950  | 2300 | 2,90                  | 8,74  | 10,20 | 3,49 | A     |
|                  | 13          | 13     | 10     | 10     | 46    | 1,98               | 1,98   | 1,42   | 1,42   | 3,0                   | 6,8   | 7,5  | 500             | 1950  | 2300 | 2,90                  | 8,74  | 10,20 | 3,49 | A     |

## Heating, 230V

| Operating status | Combination |        |        |        | Total | Unit capacity (kW) |        |        |        | Heating capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |       | EER  | label |
|------------------|-------------|--------|--------|--------|-------|--------------------|--------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|-------|------|-------|
|                  | Unit A      | Unit B | Unit C | Unit D |       | Unit A             | Unit B | Unit C | Unit D | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max.  |      |       |
| 1 unit operation | 10          | -      | -      | -      | 10    | 3,20               | -      | -      | -      | 1,4                   | 3,2   | 4,5  | 320             | 1120  | 1850 | 1,86                  | 5,29  | 8,29  | 2,86 | D     |
|                  | 13          | -      | -      | -      | 13    | 4,30               | -      | -      | -      | 1,4                   | 4,3   | 4,8  | 320             | 1800  | 2050 | 1,86                  | 8,07  | 9,19  | 2,39 | G     |
|                  | 16          | -      | -      | -      | 16    | 5,20               | -      | -      | -      | 1,4                   | 5,2   | 6,0  | 320             | 2260  | 2660 | 1,86                  | 10,03 | 11,80 | 2,30 | G     |
| 2 unit operation | 10          | 10     | -      | -      | 20    | 3,05               | 3,05   | -      | -      | 1,8                   | 6,1   | 7,1  | 360             | 1680  | 2380 | 2,09                  | 7,61  | 10,56 | 3,63 | A     |
|                  | 13          | 10     | -      | -      | 23    | 3,50               | 2,60   | -      | -      | 1,8                   | 6,1   | 7,1  | 360             | 1680  | 2380 | 2,09                  | 7,61  | 10,56 | 3,63 | A     |
|                  | 16          | 10     | -      | -      | 26    | 3,90               | 2,40   | -      | -      | 1,8                   | 6,3   | 7,3  | 360             | 1790  | 2430 | 2,09                  | 8,02  | 10,78 | 3,52 | B     |
|                  | 13          | 13     | -      | -      | 26    | 3,15               | 3,15   | -      | -      | 1,8                   | 6,3   | 7,3  | 360             | 1790  | 2430 | 2,09                  | 8,02  | 10,78 | 3,52 | B     |
|                  | 16          | 13     | -      | -      | 29    | 3,45               | 2,85   | -      | -      | 1,8                   | 6,3   | 7,3  | 360             | 1790  | 2430 | 2,09                  | 8,02  | 10,78 | 3,52 | B     |
| 3 unit operation | 10          | 10     | 10     | -      | 30    | 2,17               | 2,17   | 2,17   | -      | 2,2                   | 6,5   | 7,8  | 420             | 1650  | 2150 | 2,43                  | 7,47  | 9,54  | 3,94 | A     |
|                  | 13          | 10     | 10     | -      | 33    | 2,61               | 1,94   | 1,94   | -      | 2,2                   | 6,5   | 7,8  | 420             | 1650  | 2150 | 2,43                  | 7,47  | 9,54  | 3,94 | A     |
|                  | 16          | 10     | 10     | -      | 36    | 3,00               | 1,85   | 1,85   | -      | 2,2                   | 6,7   | 8,0  | 420             | 1750  | 2200 | 2,43                  | 7,84  | 9,76  | 3,83 | A     |
|                  | 13          | 13     | 10     | -      | 36    | 2,44               | 2,44   | 1,82   | -      | 2,2                   | 6,7   | 8,0  | 420             | 1750  | 2200 | 2,43                  | 7,84  | 9,76  | 3,83 | A     |
|                  | 16          | 13     | 10     | -      | 39    | 2,74               | 2,27   | 1,69   | -      | 2,2                   | 6,7   | 8,0  | 420             | 1750  | 2200 | 2,43                  | 7,84  | 9,76  | 3,83 | A     |
|                  | 13          | 13     | 13     | -      | 39    | 2,23               | 2,23   | 2,23   | -      | 2,2                   | 6,7   | 8,0  | 420             | 1750  | 2200 | 2,43                  | 7,84  | 9,76  | 3,83 | A     |
|                  | 16          | 13     | 13     | -      | 42    | 2,56               | 2,56   | 1,58   | -      | 2,2                   | 6,7   | 8,0  | 420             | 1750  | 2200 | 2,43                  | 7,84  | 9,76  | 3,83 | A     |
| 4 unit operation | 10          | 10     | 10     | 10     | 40    | 1,80               | 1,80   | 1,80   | 1,80   | 2,0                   | 7,2   | 8,4  | 500             | 1630  | 2110 | 2,9                   | 7,31  | 9,36  | 4,42 | A     |
|                  | 13          | 10     | 10     | 10     | 43    | 2,23               | 1,66   | 1,66   | 1,66   | 2,0                   | 7,2   | 8,4  | 500             | 1630  | 2110 | 2,9                   | 7,31  | 9,36  | 4,42 | A     |
|                  | 16          | 10     | 10     | 10     | 46    | 1,74               | 1,82   | 1,82   | 1,82   | 2,0                   | 7,2   | 8,4  | 500             | 1630  | 2110 | 2,9                   | 7,31  | 9,36  | 4,42 | A     |
|                  | 13          | 13     | 10     | 10     | 46    | 2,06               | 2,06   | 1,54   | 1,54   | 2,0                   | 7,2   | 8,4  | 500             | 1630  | 2110 | 2,9                   | 7,31  | 9,36  | 4,42 | A     |

Outdoor unit:

RAS-4M23SAV-E

Indoor unit:

HP  
HP  
HP

SKVP

RAS-B10SKVP-E  
RAS-B13SKVP-E  
RAS-B16SKVP-E

SKV

RAS-M10SKV-E  
RAS-M13SKV-E  
RAS-M16SKV-E

GDV

RAS-M10GDV-E  
RAS-M13GDV-E  
RAS-M16GDV-E

SMUV

RAS-M10SMUV-E  
RAS-M13SMUV-E  
RAS-M16SMUV-E

## Combination ratings (size 27) heat pump

## Cooling, 230V

| Operating status | Combination |        |        |        | Total | Unit capacity (kW) |        |        |        | Cooling capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |       | EER  | label |
|------------------|-------------|--------|--------|--------|-------|--------------------|--------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|-------|------|-------|
|                  | Unit A      | Unit B | Unit C | Unit D |       | Unit A             | Unit B | Unit C | Unit D | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max.  |      |       |
| 1 unit operation | 10          | -      | -      | -      | 10    | 2,70               | -      | -      | -      | 1,4                   | 2,7   | 3,2  | 640             | 750   | 950  | 3,52                  | 3,71  | 4,44  | 3,60 | A     |
|                  | 13          | -      | -      | -      | 13    | 3,70               | -      | -      | -      | 1,4                   | 3,7   | 4,4  | 640             | 1200  | 1520 | 3,52                  | 5,49  | 6,88  | 3,08 | B     |
|                  | 16          | -      | -      | -      | 16    | 4,50               | -      | -      | -      | 1,4                   | 4,5   | 5,0  | 640             | 1650  | 200  | 3,52                  | 7,47  | 8,87  | 2,73 | D     |
| 2 unit operation | 10          | 10     | -      | -      | 20    | 2,70               | 2,70   | -      | -      | 2,5                   | 5,4   | 6,3  | 640             | 1530  | 2040 | 3,48                  | 6,79  | 9,05  | 3,53 | A     |
|                  | 13          | 10     | -      | -      | 23    | 3,41               | 2,49   | -      | -      | 2,7                   | 5,9   | 6,6  | 660             | 1810  | 2220 | 3,59                  | 8,03  | 9,85  | 3,26 | A     |
|                  | 16          | 10     | -      | -      | 26    | 3,94               | 2,36   | -      | -      | 2,9                   | 6,3   | 6,9  | 670             | 2040  | 2400 | 3,64                  | 9,05  | 10,65 | 3,09 | B     |
|                  | 13          | 13     | -      | -      | 26    | 3,15               | 3,15   | -      | -      | 2,9                   | 6,3   | 6,9  | 670             | 2040  | 2400 | 3,64                  | 9,05  | 10,65 | 3,09 | B     |
|                  | 16          | 13     | -      | -      | 29    | 3,73               | 3,07   | -      | -      | 3,0                   | 6,8   | 7,2  | 690             | 2320  | 2570 | 3,75                  | 10,29 | 11,40 | 2,93 | C     |
|                  | 16          | 16     | -      | -      | 32    | 3,60               | 3,60   | -      | -      | 3,2                   | 7,2   | 7,5  | 700             | 2550  | 2750 | 3,80                  | 11,31 | 12,20 | 2,82 | C     |
| 3 unit operation | 10          | 10     | 10     | -      | 30    | 2,53               | 2,53   | 2,53   | -      | 3,6                   | 7,6   | 8,2  | 950             | 2400  | 2720 | 4,59                  | 10,65 | 12,07 | 3,17 | B     |
|                  | 13          | 10     | 10     | -      | 33    | 3,13               | 2,28   | 2,28   | -      | 3,9                   | 7,7   | 8,3  | 960             | 2410  | 2740 | 4,64                  | 10,69 | 12,16 | 3,20 | B     |
|                  | 16          | 10     | 10     | -      | 36    | 3,50               | 2,10   | 2,10   | -      | 4,0                   | 7,7   | 8,5  | 960             | 2410  | 2790 | 4,64                  | 10,69 | 12,38 | 3,20 | B     |
|                  | 13          | 13     | 10     | -      | 36    | 2,82               | 2,82   | 2,06   | -      | 4,0                   | 7,7   | 8,5  | 960             | 2410  | 2790 | 4,64                  | 10,69 | 12,38 | 3,20 | B     |
|                  | 16          | 13     | 10     | -      | 39    | 3,22               | 2,65   | 1,93   | -      | 4,1                   | 7,8   | 8,6  | 970             | 2430  | 2810 | 4,69                  | 10,78 | 12,47 | 3,21 | A     |
|                  | 13          | 13     | 13     | -      | 39    | 2,60               | 2,60   | 2,60   | -      | 4,1                   | 7,8   | 8,6  | 970             | 2430  | 2810 | 4,69                  | 10,78 | 12,47 | 3,21 | A     |
|                  | 16          | 16     | 10     | -      | 42    | 3,04               | 3,04   | 1,82   | -      | 4,1                   | 7,9   | 8,7  | 970             | 2440  | 2830 | 4,69                  | 10,83 | 12,56 | 3,24 | A     |
|                  | 16          | 13     | 13     | -      | 42    | 2,99               | 2,46   | 2,46   | -      | 4,1                   | 7,9   | 8,7  | 970             | 2440  | 2830 | 4,69                  | 10,83 | 12,56 | 3,24 | A     |
|                  | 16          | 16     | 13     | -      | 45    | 2,80               | 2,80   | 2,30   | -      | 4,2                   | 7,9   | 8,9  | 970             | 2440  | 2880 | 4,69                  | 10,83 | 12,78 | 3,24 | A     |
| 4 unit operation | 16          | 16     | 16     | -      | 48    | 2,67               | 2,67   | 2,67   | -      | 4,3                   | 8,0   | 9,0  | 980             | 2450  | 2900 | 4,73                  | 10,87 | 12,87 | 3,27 | A     |
|                  | 10          | 10     | 10     | 10     | 40    | 1,98               | 1,98   | 1,98   | 1,98   | 4,0                   | 7,9   | 8,7  | 930             | 2450  | 2800 | 4,49                  | 10,87 | 12,42 | 3,22 | A     |
|                  | 13          | 10     | 10     | 10     | 43    | 2,48               | 1,81   | 1,81   | 1,81   | 4,1                   | 7,9   | 8,8  | 940             | 2450  | 2820 | 4,54                  | 10,87 | 12,51 | 3,22 | A     |
|                  | 16          | 10     | 10     | 10     | 46    | 2,86               | 1,71   | 1,71   | 1,71   | 4,1                   | 8,0   | 9,0  | 940             | 2500  | 2860 | 4,54                  | 11,09 | 12,69 | 3,20 | B     |
|                  | 13          | 13     | 10     | 10     | 46    | 2,31               | 2,31   | 1,69   | 1,69   | 4,1                   | 8,0   | 9,0  | 940             | 2500  | 2860 | 4,54                  | 11,09 | 12,69 | 3,20 | B     |
|                  | 16          | 13     | 10     | 10     | 49    | 2,65               | 2,18   | 1,59   | 1,59   | 4,2                   | 8,0   | 9,1  | 950             | 2500  | 2880 | 4,59                  | 11,09 | 12,78 | 3,20 | B     |
|                  | 13          | 13     | 13     | 10     | 49    | 2,14               | 2,14   | 1,57   | 1,57   | 4,2                   | 8,0   | 9,1  | 950             | 2500  | 2880 | 4,59                  | 11,09 | 12,78 | 3,20 | B     |
|                  | 16          | 13     | 13     | 10     | 52    | 2,47               | 2,03   | 2,03   | 1,48   | 4,2                   | 8,0   | 9,2  | 950             | 2500  | 2900 | 4,59                  | 11,09 | 12,87 | 3,20 | B     |
|                  | 13          | 13     | 13     | 13     | 52    | 2,00               | 2,00   | 2,00   | 2,00   | 4,2                   | 8,0   | 9,2  | 950             | 2500  | 2900 | 4,59                  | 11,09 | 12,87 | 3,20 | B     |
| Heating, 230V    | 16          | 16     | 10     | 10     | 52    | 2,50               | 2,50   | 1,50   | 1,50   | 4,2                   | 8,0   | 9,2  | 950             | 2500  | 2900 | 4,59                  | 11,09 | 12,87 | 3,20 | B     |

## Heating, 230V

| Operating status | Combination |        |        |        | Total | Unit capacity (kW) |        |        |        | Heating capacity (kW) |       |      | Power input (W) |       |      | Operating current (A) |       |       | EER  | label |
|------------------|-------------|--------|--------|--------|-------|--------------------|--------|--------|--------|-----------------------|-------|------|-----------------|-------|------|-----------------------|-------|-------|------|-------|
|                  | Unit A      | Unit B | Unit C | Unit D |       | Unit A             | Unit B | Unit C | Unit D | Min.                  | Rated | Max. | Min.            | Rated | Max. | Min.                  | Rated | Max.  |      |       |
| 1 unit operation | 10          | -      | -      | -      | 10    | 4,00               | -      | -      | -      | 0,8                   | 4,0   | 5,2  | 300             | 1450  | 1980 | 1,79                  | 6,43  | 8,78  | 2,76 | E     |
|                  | 13          | -      | -      | -      | 13    | 5,00               | -      | -      | -      | 0,8                   | 5,0   | 6,5  | 310             | 2050  | 2750 | 1,85                  | 9,09  | 12,20 | 2,44 | F     |
|                  | 16          | -      | -      | -      | 16    | 5,50               | -      | -      | -      | 0,8                   | 5,5   | 6,9  | 310             | 2400  | 3000 | 1,85                  | 10,65 | 13,31 | 2,29 | G     |
| 2 unit operation | 10          | 10     | -      | -      | 20    | 3,60               | 3,60   | -      | -      | 1,5                   | 7,2   | 10,0 | 320             | 2100  | 3200 | 1,86                  | 9,32  | 14,20 | 3,43 | B     |
|                  | 13          | 10     | -      | -      | 23    | 4,22               | 3,38   | -      | -      | 1,5                   | 7,6   | 10,1 | 320             | 2320  | 3210 | 1,86                  | 10,29 | 14,24 | 3,28 | C     |
|                  | 16          | 10     | -      | -      | 26    | 4,57               | 3,33   | -      | -      | 1,5                   | 7,9   | 10,1 | 320             | 2480  | 3230 | 1,86                  | 11,00 | 14,33 | 3,19 | D     |
|                  | 13          | 13     | -      | -      | 26    | 3,95               | 3,95   | -      | -      | 1,5                   | 7,9   | 10,1 | 320             | 2480  | 3230 | 1,86                  | 11,00 | 14,33 | 3,19 | D     |
|                  | 16          | 13     | -      | -      | 29    | 4,35               | 3,95   | -      | -      | 1,5                   | 8,3   | 10,2 | 320             | 2700  | 3240 | 1,86                  | 11,98 | 14,37 | 3,07 | D     |
|                  | 16          | 16     | -      | -      | 32    | 4,30               | 4,30   | -      | -      | 1,5                   | 8,6   | 10,2 | 320             | 2860  | 3250 | 1,86                  | 12,69 | 14,42 | 3,01 | D     |
| 3 unit operation | 10          | 10     | 10     | -      | 30    | 2,87               | 2,87   | 2,87   | -      | 2,0                   | 8,6   | 10,4 | 380             | 2300  | 2750 | 2,07                  | 10,20 | 12,20 | 3,74 | A     |
|                  | 13          | 10     | 10     | -      | 33    | 3,35               | 2,68   | 2,68   | -      | 2,0                   | 8,7   | 10,5 | 380             | 2350  | 2760 | 2,07                  | 10,43 | 12,24 | 3,70 | A     |
|                  | 16          | 10     | 10     | -      | 36    | 3,54               | 2,58   | 2,58   | -      | 2,0                   | 8,7   | 10,5 | 380             | 2350  | 2760 | 2,07                  | 10,43 | 12,24 | 3,70 | A     |
|                  | 13          | 13     | 10     | -      | 36    | 3,11               | 3,11   | 2,49   | -      | 2,0                   | 8,7   | 10,5 | 380             | 2350  | 2780 | 2,07                  | 10,43 | 12,24 | 3,70 | A     |
|                  | 16          | 13     | 10     | -      | 39    | 3,34               | 3,03   | 2,43   | -      | 2,0                   | 8,8   | 10,6 | 380             | 2400  | 2780 | 2,07                  | 10,65 | 12,33 | 3,67 | A     |
|                  | 13          | 13     | 13     | -      | 39    | 2,93               | 2,93   | 2,93   | -      | 2,0                   | 8,8   | 10,6 | 380             | 2400  | 2790 | 2,07                  | 10,65 | 12,33 | 3,67 | A     |
|                  | 16          | 16     | 10     | -      | 42    | 3,26               | 2,26   | 2,37   | -      | 2,0                   | 8,9   | 10,7 | 380             | 2450  | 2790 | 2,07                  | 10,87 | 12,38 | 3,63 | A     |
|                  | 16          | 13     | 13     | -      | 42    | 3,16               | 2,87   | 2,87   | -      | 2,0                   | 8,9   | 10,7 | 380             | 2450  | 2790 | 2,07                  | 10,87 | 12,38 | 3,63 | A     |
|                  | 16          | 16     | 13     | -      | 45    | 3,06               | 3,06   | 2,78   | -      | 2,0                   | 8,9   | 10,7 | 380             | 2450  | 2790 | 2,07                  | 10,87 | 12,38 | 3,63 | A     |
| 4 unit operation | 16          | 16     | 16     | -      | 48    | 3,00               | 3,00   | 3,00   | -      | 2,0                   | 9,0   | 10,8 | 380             | 2500  | 2800 | 2,07                  | 11,09 | 12,42 | 3,60 | B     |
|                  | 10          | 10     | 10     | 10     | 40    | 2,23               | 2,23   | 2,23   | 2,23   | 2,2                   | 8,9   | 10,8 | 450             | 2100  | 2810 | 2,45                  | 9,32  | 12,47 | 4,24 | A     |
|                  | 13          | 10     | 10     | 10     | 43    | 2,62               | 2,09   | 2,09   | 2,09   | 2,2                   | 8,9   | 10,9 | 460             | 2100  | 2830 | 2,50                  | 9,32  | 12,56 | 4,24 | A     |
|                  | 16          | 10     | 10     | 10     | 46    | 2,83               | 2,06   | 2,06   | 2,06   | 2,2                   | 9,0   | 10,9 | 460             | 2250  | 2830 | 2,50                  | 9,98  | 12,56 | 4,00 | A     |
|                  | 13          | 13     | 10     | 10     | 46    | 2,50               | 2,50   | 2,00   | 2,00   | 2,2                   | 9,0   | 10,9 | 470             | 2250  | 2830 | 2,55                  | 9,98  | 12,56 | 4,00 | A     |
|                  | 16          | 13     | 10     | 10     | 49    | 2,68               | 2,43   | 1,95   | 1,95   | 2,2                   | 9,0   | 11,0 | 480             | 2250  | 2850 | 2,61                  | 9,98  | 12,64 | 4,00 | A     |
|                  | 13          | 13     | 13     | 10     | 49    | 2,37               | 2,37   | 2,37   | 1,89   | 2,2                   | 9,0   | 11,0 | 480             | 2250  | 2850 | 2,61                  | 9,98  | 12,64 | 4,00 | A     |
|                  | 16          | 13     | 13     | 10     | 52    | 2,54               | 2,31   | 2,31   | 1,85   | 2,2                   | 9,0   | 11,0 | 490             | 2250  | 2850 | 2,66                  | 9,98  | 12,64 | 4,00 | A     |
|                  | 13          | 13     | 13     | 13     | 52    | 2,25               | 2,25   | 2,25   | 2,25   | 2,2                   | 9,0   | 11,0 | 490             | 2250  | 2850 | 2,66                  | 9,98  | 12,64 | 4,00 | A     |
| Outdoor unit:    | 16          | 16     | 10     | 10     | 52    | 2,61               | 2,61   | 1,89   | 1,89   | 2,2                   | 9,0   | 11,0 | 500             | 2250  | 2850 | 2,72                  | 9,98  | 12,64 | 4,00 | A     |

RAS-4M27GAV-E

Indoor unit:

HP  
HP  
HP

SKVP

RAS-B10SKVP-E  
RAS-B13SKVP-E  
RAS-B16SKVP-E

SKV

RAS-M10SKV-E  
RAS-M13SKV-E  
RAS-M16SKV-E

GDV

RAS-M10GDV-E  
RAS-M13GDV-E  
RAS-M16GDV-E

SMUV

RAS-M10SMUV-E  
RAS-M13SMUV-E  
RAS-M16SMUV-E

**R-410A****FIXED-SPEED HIGH WALL****COOLING ONLY / HEAT PUMPS**

## SKHP Fixed-speed high-wall

### Features

These high wall units are compact, slim line and elegant and ideal for any residential or small commercial application.

They offer improved efficiency levels and air quality features.

A/A for all sizes below 24 k.

7 in 1 filter system.

### Key features

Elegant and attractive design with clean, simple lines of the flat panel.

A energy class in cooling and heating (capacities below 24 k): high energy saving.

Low noise level: it operates quietly for a comfortable night's sleep.

7 in 1 filtering system: Zeolite Plus and Sasa filter, the new anti-oxidant Vitamin C filter and the Ginkgo Filter combined with the Bio Enzyme filter.

One touch pre-set memory to recall your favourite settings.

**A/A for all sizes below 6 kW**

**7 in 1 filter system**

**Modern Front Panel**

**AC Rotary**



### Technical specifications **cooling only**

| Outdoor unit              |     | RAS-10SA-ES  | RAS-13SA-ES  | RAS-18SA-ES  | RAS-24SA-ES  |
|---------------------------|-----|--------------|--------------|--------------|--------------|
| Indoor unit               |     | RAS-10SKP-ES | RAS-13SKP-ES | RAS-18SKP-ES | RAS-24SKP-ES |
| Cooling capacity          | kW  | 2,73         | 3,75         | 5,33         | 6,83         |
| Power input               | kW  | 0,84         | 1,15         | 1,66         | 2,43         |
| EER                       | W/W | 3,25         | 3,26         | 3,21         | 2,81         |
| Energy efficiency class   |     | A            | A            | A            | C            |
| Annual energy consumption | kWh | 420          | 575          | 830          | 1215         |
| Running current           | A   | 3,76         | 5,40         | 7,40         | 11,10        |

### Technical specifications **heat pump**

| Outdoor unit              |     |   | RAS-10S2AH-ES | RAS-13S2AH-ES | RAS-18S2AH-ES | RAS-24S2AH-ES |
|---------------------------|-----|---|---------------|---------------|---------------|---------------|
| Indoor unit               |     |   | RAS-10SKHP-ES | RAS-13SKHP-ES | RAS-18SKHP-ES | RAS-24SKHP-ES |
| Cooling capacity          | kW  |   | 2,73          | 3,78          | 5,11          | 6,33          |
| Power input               | kW  | C | 0,84          | 1,17          | 1,55          | 2,21          |
| EER                       | W/W |   | 3,25          | 3,23          | 3,30          | 2,86          |
| Energy efficiency class   |     | C | A             | A             | A             | C             |
| Annual energy consumption | kWh |   | 420           | 585           | 775           | 1105          |
| Running current           | A   | C | 3,76          | 5,25          | 7,00          | 10,15         |
| Heating capacity          | kW  |   | 2,94          | 4,23          | 5,49          | 6,85          |
| Power input               | kW  | H | 0,81          | 1,16          | 1,50          | 2,07          |
| COP                       | W/W |   | 3,63          | 3,65          | 3,66          | 3,31          |
| Energy efficiency class   |     | H | A             | A             | A             | C             |

| Indoor unit                |                         | CO | RAS-10SKP-ES    | RAS-13SKP-ES    | RAS-18SKP-ES     | RAS-24SKP-ES     |
|----------------------------|-------------------------|----|-----------------|-----------------|------------------|------------------|
|                            |                         | HP | RAS-10SKHP-ES   | RAS-13SKHP-ES   | RAS-18SKHP-ES    | RAS-24SKHP-ES    |
| Air flow                   | m <sup>3</sup> /h – l/s | C  | 510 – 142       | 620 – 172       | 1000 – 278       | 1100 – 306       |
| Air flow                   | m <sup>3</sup> /h – l/s | H  | 560 – 156       | 650 – 181       | 1000 – 278       | 1100 – 306       |
| Sound pressure level (h/l) | dB(A)                   |    | 39/31           | 41/31           | 44/35            | 45/37            |
| Sound power level          | dB(A)                   |    | 52              | 54              | 57               | 58               |
| Dimensions (h × d × w)     | mm                      |    | 250 × 740 × 195 | 275 × 790 × 205 | 320 × 1050 × 228 | 320 × 1050 × 228 |
| Weight                     | kg                      |    | 8               | 9               | 13               | 13               |

### Technical specifications **cooling only and heat pump**

| Outdoor unit                     |                         | CO    | RAS-10SA-ES     | RAS-13SA-ES     | RAS-18SA-ES     | RAS-24SA-ES     |
|----------------------------------|-------------------------|-------|-----------------|-----------------|-----------------|-----------------|
|                                  |                         | HP    | RAS-10S2AH-ES   | RAS-13S2AH-ES   | RAS-18S2AH-ES   | RAS-24S2AH-ES   |
| Air flow                         | m <sup>3</sup> /h – l/s |       | 1795 – 499      | 2160 – 600      | 2475 – 688      | 2525 – 701      |
| Sound pressure level             | dB(A)                   | C (H) | 47              | 51              | 57              | 57              |
| Sound power level                | dB(A)                   | C (H) | 60              | 64              | 70              | 70              |
| Operating range                  | °C                      | C     | 15 ÷ 43         | 15 ÷ 43         | 15 ÷ 43         | 15 ÷ 43         |
| Operating range                  | °C                      | H     | -10 ÷ 24        | -10 ÷ 24        | -10 ÷ 24        | -10 ÷ 24        |
| Dimensions (h × w × d)           | mm                      |       | 550 × 780 × 290 | 550 × 780 × 290 | 715 × 780 × 290 | 715 × 780 × 290 |
| Weight                           | kg                      | C (H) | 30              | 38              | 45              | 50              |
| Flare connections (gas – liquid) |                         |       | 3/8" – 1/4"     | 1/2" – 1/4"     | 1/2" – 1/4"     | 1/2" – 1/4"     |
| Maximum pipe length              | m                       |       | 10              | 15              | 20              | 25              |
| Maximum height difference        | m                       |       | 5               | 6               | 8               | 10              |
| Chargeless pipe length           | m                       |       | 10              | 15              | 15              | 15              |
| Power supply                     | V-ph-Hz                 |       | 220/240-1-50    | 220/240-1-50    | 220/240-1-50    | 220/240-1-50    |

C = cooling mode  
H = heating mode



**R-410A****FIXED-SPEED FLEXI UNIT****SINGLE SPLIT****6 in 1 filtering solution****Hi power**

## GFP Fixed-speed flexi unit

### Features

The stylish GF-series console and ceiling-suspended units blend unobtrusively with any room decoration, thanks to their elegant design.

They incorporate the latest Toshiba technology and are ideal for residential applications, but also for offices and shops.

For additional flexibility, this unit can be used for low-wall or ceiling suspended mounting.

### Key features

Flexible for both low-wall and ceiling-suspended installations.

Efficient filtration system: Zeolite Plus and Sasa filter, the new anti-oxidant Vitamin C filter and the Gingko filter combined with the Bio-Enzyme filter.

Hi Power: to achieve the desired temperatures faster.

Light, compact and attractive design.

Fresh air intake facility at the back of the unit.



Technical specifications **cooling only**

| Outdoor unit              |     | RAS-18GA-ES  | RAS-24GA-ES  |
|---------------------------|-----|--------------|--------------|
| Indoor unit               |     | RAS-18GFP-ES | RAS-24GFP-ES |
| Cooling capacity          | kW  | 5,33         | 6,83         |
| Power input               | kW  | 2,00         | 2,72         |
| EER                       | W/W | 2,67         | 2,51         |
| Energy efficiency class   |     | D            | E            |
| Annual energy consumption | kWh | 1000         | 1360         |
| Running current           | A   | 5,50         | 12,70        |

Technical specifications **heat pump**

| Outdoor unit              |      | RAS-18GAH-ES  | RAS-24GAH-ES  |
|---------------------------|------|---------------|---------------|
| Indoor unit               |      | RAS-18GFHP-ES | RAS-24GFHP-ES |
| Cooling capacity          | kW   | 5,05          | 6,25          |
| Power input               | kW C | 1,99          | 2,55          |
| EER                       | W/W  | 2,54          | 2,45          |
| Energy efficiency class   | C    | E             | E             |
| Annual energy consumption | kWh  | 995           | 1275          |
| Running current           | A C  | 8,85          | 11,73         |
| Heating capacity          | kW   | 5,73          | 7,05          |
| Power input               | kW H | 1,86          | 2,48          |
| COP                       | W/W  | 3,08          | 2,84          |
| Energy efficiency class   | H    | B             | C             |

| Indoor unit                |                         | CO | RAS-18GFP-ES      | RAS-24GFP-ES      |
|----------------------------|-------------------------|----|-------------------|-------------------|
|                            |                         | HP | RAS-18GFHP-ES     | RAS-24GFHP-ES     |
| Air flow (h/l)             | m <sup>3</sup> /h – l/s | C  | 800/580 – 220/160 | 930/650 – 250/180 |
| Air flow (h/l)             | m <sup>3</sup> /h – l/s | H  | 830/650 – 230/180 | 930/660 – 260/183 |
| Sound pressure level (h/l) | dB(A)                   |    | 43/36             | 46/37             |
| Sound power level          | dB(A)                   |    | 56                | 59                |
| Dimensions (h × w × d)     | mm                      |    | 633 × 1093 × 208  | 633 × 1093 × 208  |
| Weight                     | kg                      |    | 23                | 23                |

Technical specifications **cooling only and heat pump**

| Outdoor unit                     |                         | CO    | RAS-18GA-ES        | RAS-24GA-ES        |
|----------------------------------|-------------------------|-------|--------------------|--------------------|
|                                  |                         | HP    | RAS-18GAH-ES       | RAS-24GAH-ES       |
| Air flow                         | m <sup>3</sup> /h – l/s |       | 1920 – 535         | 3470 – 965         |
| Sound pressure level             | dB(A)                   | C (H) | 51 (53)            | 56 (58)            |
| Sound power level                | dB(A)                   | C (H) | 65 (66)            | 70 (71)            |
| Operating range                  | °C                      | C (H) | 15 ÷ 43 (–10 ÷ 24) | 15 ÷ 43 (–10 ÷ 24) |
| Dimensions (h × w × d)           | mm                      |       | 538 × 780 × 300    | 690 × 880 × 310    |
| Weight                           | kg                      | C (H) | 39 (50)            | 63 (65)            |
| Flare connections (gas – liquid) |                         |       | 1/2" – 1/4"        | 1/2" – 1/4"        |
| Maximum pipe length              | m                       |       | 20                 | 25                 |
| Maximum height difference        | m                       |       | 8                  | 10                 |
| Chargeless pipe length           | m                       |       | 15                 | 15                 |
| Power supply                     | V-ph-Hz                 |       | 220/240-1-50       | 220/240-1-50       |

C = cooling mode  
H = heating mode

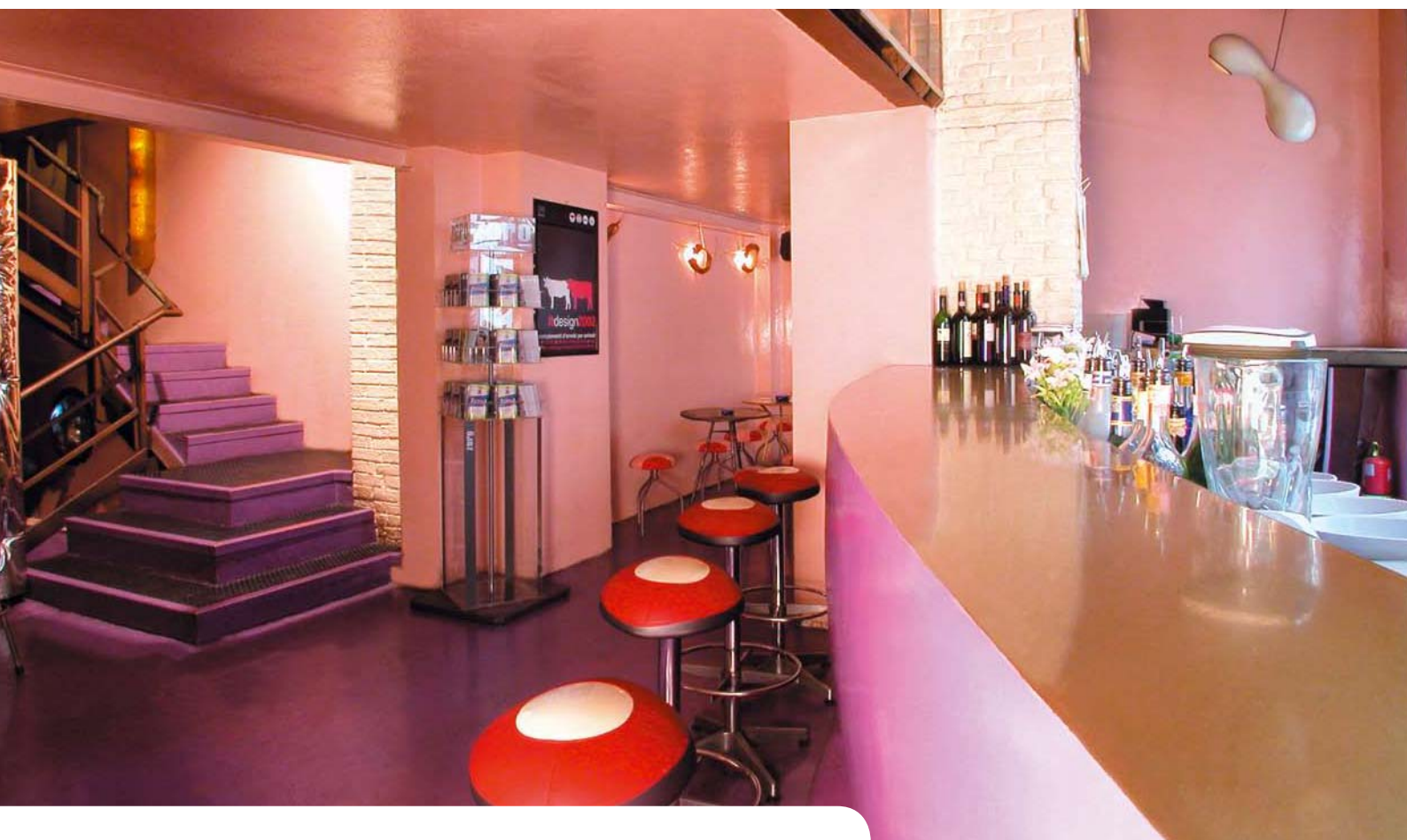
## The light commercial range. The applications

Digital and Super Digital Inverter air conditioners perfectly satisfy the requirement of the commercial sector for the best return on your investment.

Toshiba offers the best options on operating costs, flexibility and maintenance.

Moreover, thanks to its flexibility, Toshiba can always find the ideal product for any requirement:

high performance, technology, compactness, optimum comfort.



## Simplicity and beyond

The Digital Inverter range for business applications provides compact, light weight units with exceptional performance.

Thanks to the TCC-Link communication system, the systems suit any installation with little business disruption.

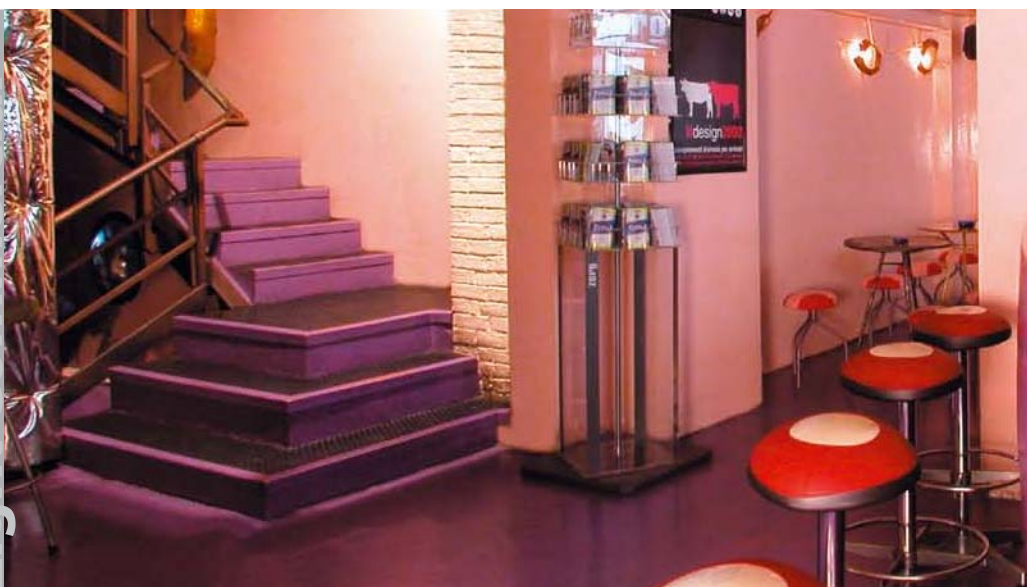
In addition, the Super Digital Inverter boasts energy efficiency class A and enables an even larger range of applications.

The wide range of indoor units is able to satisfy any kind of requirement and the enhanced DC twin rotary compressor delivers stable

performance with less friction, making this system really silent.







## Solutions for professionals, from professionals

Toshiba Digital and Super Digital Inverter systems deliver exceptional operating savings and ecological features in extremely compact units. With state-of-the-art technologies, flexible controls and improved installation they bring comfort

and convenience to any business installation. A complete range of indoor units satisfies all commercial applications: ceiling, cassette, ducted, suspended, high-wall and flexi units.

The Digital and Super Digital Inverter systems represent the most economical and ecological approach to air conditioning.

## When the inverter becomes digital

The technology of the Digital Inverter control module ensures optimised reproduction of the supply sine wave at the desired frequency, in order to reduce inefficient harmonics that inverters normally emit.

With this innovative control method, the Toshiba Digital Inverter brings state-of-the-art inverter technology to the commercial sector, offering considerable advantages in terms

of capacity, energy savings and optimised comfort. Compressor operation is practically imperceptible.

## Who says that you must choose between improved performance and minimised consumption?

The Toshiba Digital and Super Digital Inverters are powerful and extremely efficient. They provide air conditioning with great energy savings. The new series 4 of Super Digital Inverter provides the best efficiency performance in the industry: up to 4,52 EER in cooling mode and up to

4,79 COP in heating mode. In most applications, these systems can reduce energy consumption by up to 50% compared with traditional fixed speed units. The variable capacity level of the compressor allows the Digital and Super Digital Inverter to maintain

room temperature control and to ensure minimum energy wastage. The new series Super Digital Inverter 4 and Digital Inverter 3 can fit R22 or R407C old pipes in case of replacement of high-consuming fixed speed systems.



Absolute ease-of-installation

Superior compactness and capacity

Maximised operating economy

Total unit control



## All the flexibility you have ever dreamt of

If you want high performance, compact units and optimum comfort, Toshiba has the ideal product for your requirements.

With the continuous improvement of the inverter control system, Toshiba offers vector control for its DC hybrid inverter, which enhances system efficiency and reduces noise levels in the power lines.

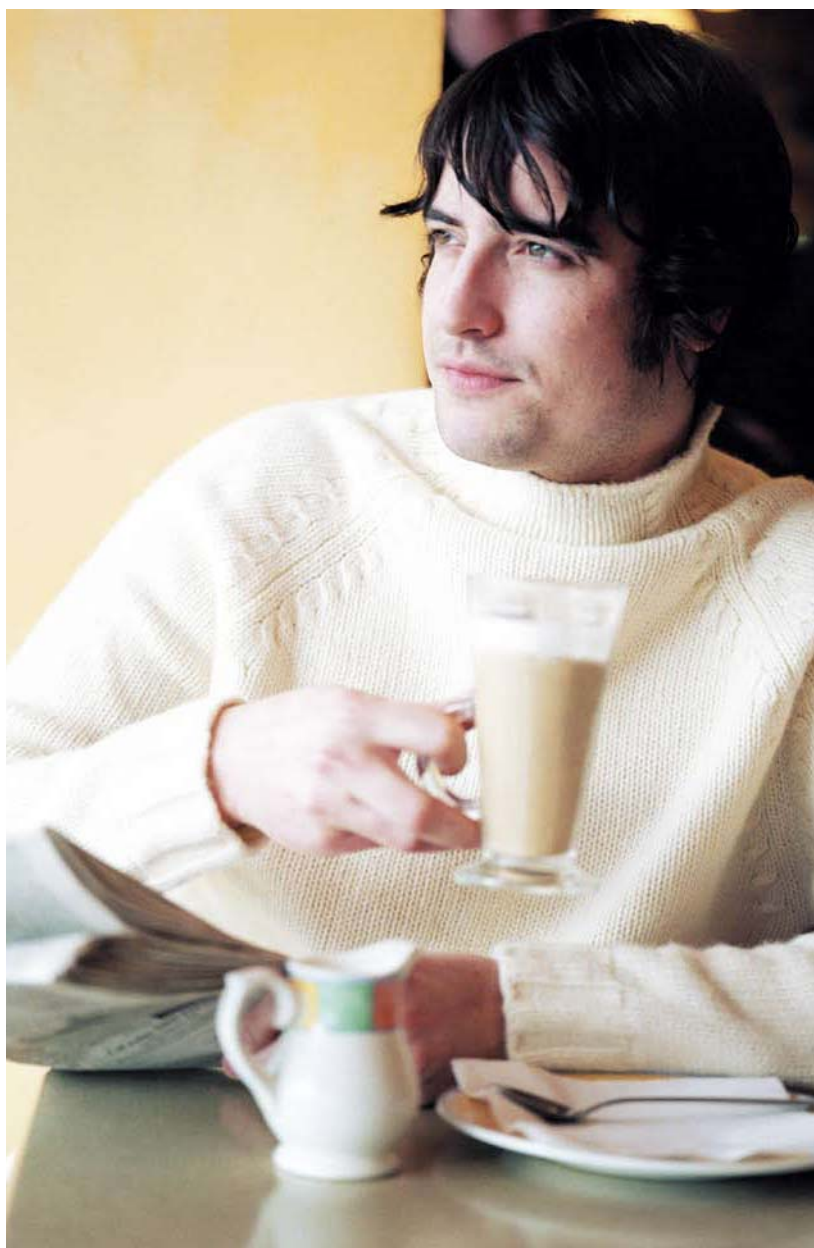
The new developments in electronics have been complemented by compressor innovation.

High-tech elements include improved coils, high precision components and higher refrigerant compression thanks to redesigned compression channels.

The new technology applied to Super Digital Inverter series 4 is able to satisfy applications that require cooling at low operating conditions down to  $-15^{\circ}\text{C}$ .

Powerful heating capacities are possible at  $-20^{\circ}\text{C}$  outdoor temperature.

The enhanced Eco-driving DC twin-rotary compressor delivers stable performance with extremely low rotor friction, making it ideal for noise-sensitive applications as well as for efficient operations in partial load conditions.





R-410A HEAT PUMPS

ECO-DRIVING COMPRESSORS

A-CLASS SYSTEM



## Super Digital Inverter 4 Outdoor unit

### Features

The new Super Digital Inverter, series 4, set a new limit for the industry energy performance.

The seasonal and the rated efficiencies are the highest in the industry for capacities from 10 to 12,5 kW.

The air management system has been improved: high efficiency fan motors, larger fans and new fan grille design also contribute to the exceptional energy performance.

Piping and operating limits improved. The new system can work at extremely low temperatures, in cooling and heating. Admitted pipe length is up to 75 m.

SDI4 are designed for R410A refrigerant, but can also fit R22 or R407C piping refrigerants in case of refurbishment.

### Key features

The best EER/COP values in the industry.

All systems are A-class rated in cooling and heating operation. The new systems provide excellent seasonal energy performance, thus saving up to 70% annual energy cost compared to fixed speed systems.

The structure and magnetic action of the new Eco-driving twin-rotary compressors provide excellent energy performance at full load as well as in partial load conditions (operation down to 10 rps).

Longer pipe runs, up to 75 m length and 30 m elevation for increased installation flexibility.

Wide operating range: down to  $-15^{\circ}\text{C}$  in cooling mode and down to  $-20^{\circ}\text{C}$  in heating mode.

The Toshiba "anti-ice" circuit prevents the condenser to be caught in ice in winter.

### The best EER and COP

**Wide operating range;**  
down to  $-15^{\circ}\text{C}$  in cooling  
and  $-20^{\circ}\text{C}$  in heating



Technical specifications **heat pump**

| Outdoor unit                     |                         |   | RAV-SP404AT-E   | RAV-SP454AT-E   | RAV-SP564AT-E   | RAV-SP804AT-E   | RAV-SP1104AT-E   | RAV-SP1404AT-E   |
|----------------------------------|-------------------------|---|-----------------|-----------------|-----------------|-----------------|------------------|------------------|
|                                  |                         |   | 1,5 HP          | 1,7 HP          | 2 HP            | 3 HP            | 4 HP             | 5 HP             |
| Air flow                         | m <sup>3</sup> /h – l/s |   | 2400 – 667      | 2400 – 667      | 2400 – 667      | 3000 – 833      | 6060 – 1683      | 6180 – 1716      |
| Sound pressure level             | dB(A)                   | C | 45              | 45              | 47              | 48              | 49               | 51               |
| Sound power level                | dB(A)                   | C | 62              | 62              | 64              | 65              | 66               | 68               |
| Operating range                  | °C                      | C | –15 ÷ 43        | –15 ÷ 43        | –15 ÷ 43        | –15 ÷ 43        | –15 ÷ 43         | –15 ÷ 43         |
| Sound pressure level             | dB(A)                   | H | 47              | 47              | 48              | 49              | 50               | 52               |
| Sound power level                | dB(A)                   | H | 64              | 64              | 65              | 66              | 67               | 69               |
| Operating range                  | °C                      | H | –15 ÷ 15        | –15 ÷ 15        | –20 ÷ 15        | –20 ÷ 15        | –20 ÷ 15         | –20 ÷ 15         |
| Dimensions (h × w × d)           | mm                      |   | 550 × 780 × 290 | 550 × 780 × 290 | 550 × 780 × 290 | 890 × 900 × 320 | 1340 × 900 × 320 | 1340 × 900 × 320 |
| Weight                           | kg                      |   | 40              | 40              | 44              | 63              | 93               | 93               |
| Compressor type                  |                         |   | DC Twin Rotary  | DC Twin Rotary  | DC Twin Rotary  | DC Twin Rotary  | DC Twin Rotary   | DC Twin Rotary   |
| Flare connections (gas – liquid) |                         |   | 1/2" – 1/4"     | 1/2" – 1/4"     | 1/2" – 1/4"     | 5/8" – 3/8"     | 5/8" – 3/8"      | 5/8" – 3/8"      |
| Minimum pipe length              | m                       |   | 5               | 5               | 5               | 5               | 3                | 3                |
| Maximum pipe length              | m                       |   | 30              | 30              | 50              | 50              | 75               | 75               |
| Maximum height difference        | m                       |   | 30              | 30              | 30              | 30              | 30               | 30               |
| Chargeless pipe length           | m                       |   | 20              | 20              | 20              | 30              | 30               | 30               |
| Power supply                     | V-ph-Hz                 |   | 220/240-1-50    | 220/240-1-50    | 220/240-1-50    | 220/240-1-50    | 220/240-1-50     | 220/240-1-50     |

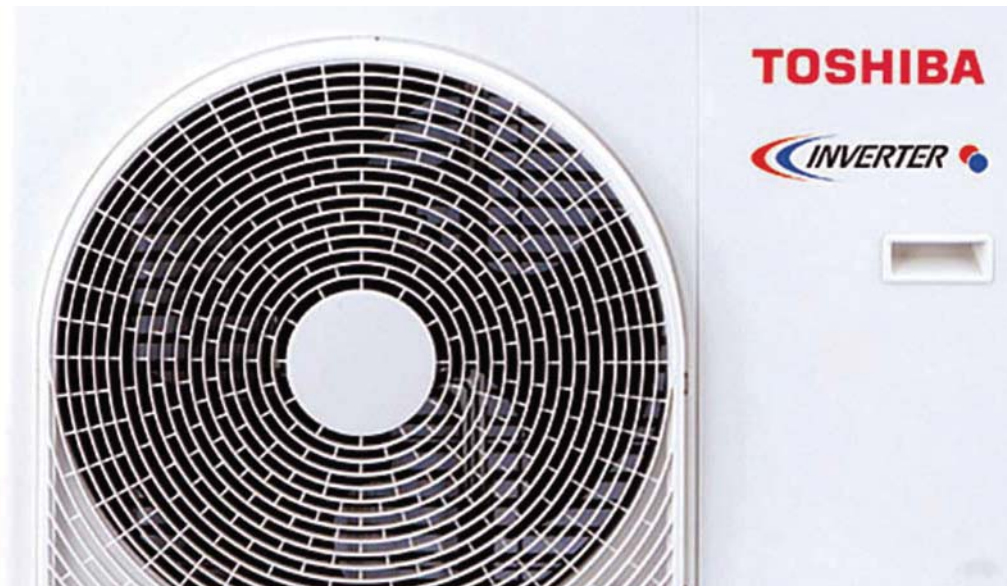
C = cooling mode

H = heating mode

R-410A HEAT PUMPS

VECTOR IPDU INVERTER

HIGH PERFORMANCE



## Digital Inverter 3 Outdoor unit

### Features

Toshiba Digital Inverter 3 is an advanced solution that can meet the growing market need for ease of installation and increased consumer comfort.

This new system, which uses the environmentally friendly refrigerant R410A, can use piping designed for old R22 or R407C refrigerants.

This new solution enables old high-energy consumption air conditioning systems to be replaced by the most advanced inverter units, with significant benefits in terms of performance, acoustic comfort and energy-efficiency.

### Key features

Extremely light and compact condensing units: easy to install in small spaces.

Excellent EER with significant savings in annual power consumption.

Compatible with a wide choice of indoor units: ceiling, 4-way cassette, compact 4-way cassette, ducted, high-wall and flexi units.

The Vector Intelligent Drive Unit (IPDU) technology ensures high performance.

Simplified maintenance using the new TCC-Link wired remote control panel.

Performing EER

Light and compact: easy to install

Compatible with a wide range of indoor units





Technical specifications **heat pump**

| Outdoor unit                     |                         |   | RAV-SM563AT-E   | RAV-SM803AT-E   | RAV-SM1103AT-E  | RAV-SM1403AT-E  | RAV-SM1603AT-E   |
|----------------------------------|-------------------------|---|-----------------|-----------------|-----------------|-----------------|------------------|
|                                  |                         |   | 2 HP            | 3 HP            | 4 HP            | 5 HP            | 6 HP             |
| Air flow                         | m <sup>3</sup> /h – l/s |   | 2400 – 667      | 2700 – 750      | 4500 – 1250     | 4500 – 1250     | 6180 – 1717      |
| Sound pressure level             | dB(A)                   | C | 46              | 47              | 53              | 54              | 51               |
| Sound power level                | dB(A)                   | C | 63              | 64              | 70              | 71              | 68               |
| Operating range                  | °C                      | C | –15 ÷ 43        | –15 ÷ 43        | –15 ÷ 43        | –15 ÷ 43        | –15 ÷ 43         |
| Sound pressure level             | dB(A)                   | H | 48              | 50              | 54              | 54              | 53               |
| Sound power level                | dB(A)                   | H | 65              | 67              | 71              | 71              | 70               |
| Operating range                  | °C                      | H | –15 ÷ 15        | –15 ÷ 15        | –15 ÷ 15        | –15 ÷ 15        | –15 ÷ 15         |
| Dimensions (h × w × d)           | mm                      |   | 550 × 780 × 290 | 550 × 780 × 290 | 795 × 900 × 320 | 795 × 900 × 320 | 1340 × 900 × 320 |
| Weight                           | kg                      |   | 38              | 42              | 77              | 77              | 99               |
| Compressor type                  |                         |   | DC Twin Rotary  | DC Twin Rotary  | DC Twin Rotary  | DC Twin Rotary  | DC Twin Rotary   |
| Flare connections (gas – liquid) |                         |   | 1/2" – 1/4"     | 5/8" – 3/8"     | 5/8" – 3/8"     | 5/8" – 3/8"     | 5/8" – 3/8"      |
| Minimum pipe length              | m                       |   | 5               | 5               | 5               | 5               | 5                |
| Maximum pipe length              | m                       |   | 30              | 30              | 50              | 50              | 50               |
| Maximum height difference        | m                       |   | 30              | 30              | 30              | 30              | 30               |
| Chargeless pipe length           | m                       |   | 20              | 20              | 30              | 30              | 30               |
| Power supply                     | V-ph-Hz                 |   | 220/240-1-50    | 220/240-1-50    | 220/240-1-50    | 220/240-1-50    | 220/240-1-50     |

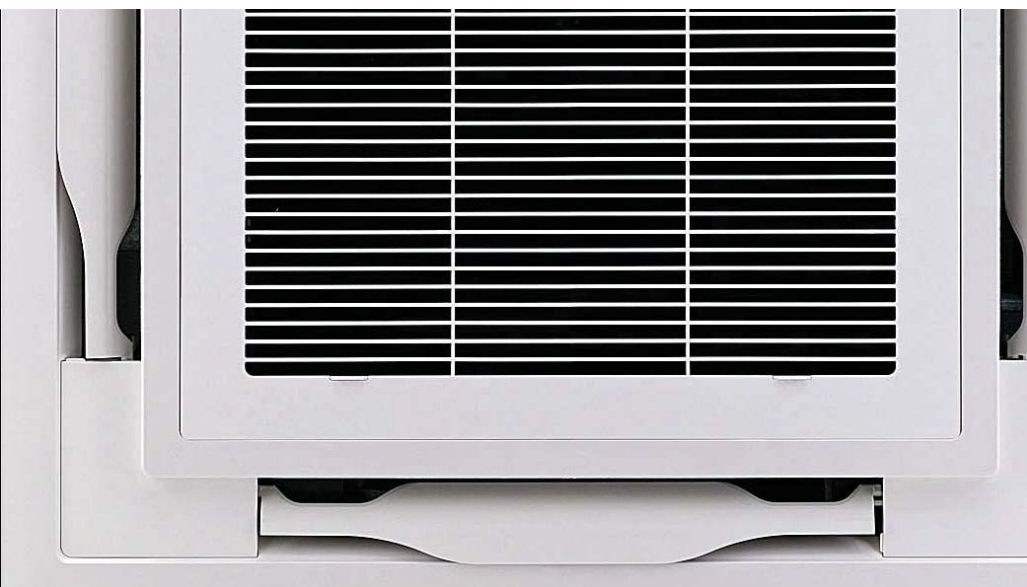
C = cooling mode

H = heating mode

**R-410A HEAT PUMPS**

**DIGITAL/SUPER DIGITAL  
INVERTER**

**NEW DESIGN CEILING PANEL**



## SM\_UT New 4-way cassette

### Features

This new 4-way cassette is unobtrusive and flexible and can easily blend in with any room interior.

Thanks to the new ceiling panel, it grants uniform air distribution, providing total comfort. This system is ideal for small commercial applications.

### Key features

Two louver shape options: straight flow louver and wide flow louver; optimum air distribution.

Light-weight unit, for easy and quick installation.

Built-in drain pump.

Simple maintenance, thanks to the Self-cleaning function and the Ag-ion tip for anti-mould in drain cap.

Individual setting of louver position: 3 different Swing modes: standard, diagonally opposite, turn-around.

Optional wired remote control RBC-AX31U(W)-E.

**Uniform air distribution**

**Light and easy to install**

**Three different setting  
modes**



Technical specifications **heat pump**

| Indoor unit                  |                         | RAV-SM564UT-E      | RAV-SM804UT-E      | RAV-SM1104UT-E      | RAV-SM1404UT-E      | RAV-SM1604UT-E      |
|------------------------------|-------------------------|--------------------|--------------------|---------------------|---------------------|---------------------|
| Air flow (h/l)               | m <sup>3</sup> /h – l/s | 1080/780 – 300/217 | 1260/840 – 350/233 | 2040/1200 – 567/333 | 2100/1260 – 583/350 | 2130/1500 – 592/417 |
| Sound pressure level (h/m/l) | dB(A)                   | 32/29/28           | 35/31/28           | 43/38/33            | 44/38/34            | 45/40/36            |
| Sound power level (h/m/l)    | dB(A)                   | 47/44/43           | 50/46/43           | 58/53/48            | 59/53/49            | 60/55/51            |
| Dimensions (h × w × d)       | mm                      | 256 × 840 × 840    | 256 × 840 × 840    | 319 × 840 × 840     | 319 × 840 × 840     | 319 × 840 × 840     |
| Weight                       | kg                      | 20                 | 20                 | 24                  | 24                  | 24                  |
| Panel dimensions (h × w × d) | mm                      | 30 × 950 × 950     | 30 × 950 × 950     | 30 × 950 × 950      | 30 × 950 × 950      | 30 × 950 × 950      |
| Panel weight                 | kg                      | 4,2                | 4,2                | 4,2                 | 4,2                 | 4,2                 |

Super Digital Inverter (series 4) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SP564AT-E      | RAV-SP804AT-E      | RAV-SP1104AT-E     | RAV-SP1404AT-E     |
|-----------------------------------|------|--------------------|--------------------|--------------------|--------------------|
| Indoor unit (cassette)            |      | RAV-SM564UT-E      | RAV-SM804UT-E      | RAV-SM1104UT-E     | RAV-SM1404UT-E     |
| Cooling capacity                  | kW   | 5,3                | 7,1                | 10,0               | 12,5               |
| Cooling range (min. – max.)       | kW   | 1,2 – 5,6          | 1,9 – 8,0          | 2,6 – 12,0         | 2,6 – 14,0         |
| Power input (min. – rated – max.) | kW C | 0,20 – 1,47 – 1,95 | 0,30 – 1,86 – 2,52 | 0,65 – 2,21 – 3,60 | 0,64 – 3,16 – 4,40 |
| EER                               | W/W  | 3,61               | 3,82               | 4,52               | 3,96               |
| Energy efficiency class           | C    | A                  | A                  | A                  | -                  |
| Annual energy consumption         | kWh  | 735                | 930                | 1105               | 1580               |
| Heating capacity                  | kW   | 5,6                | 8,0                | 11,2               | 14,0               |
| Heating range (min. – max.)       | kW   | 0,9 – 8,1          | 1,3 – 11,3         | 2,4 – 13,0         | 2,4 – 16,5         |
| Power input (min. – rated – max.) | kW H | 0,15 – 1,21 – 2,40 | 0,25 – 1,91 – 3,52 | 0,52 – 2,34 – 4,20 | 0,52 – 3,21 – 4,50 |
| COP                               | W/W  | 4,63               | 4,19               | 4,79               | 4,36               |
| Energy efficiency class           | H    | A                  | A                  | A                  | -                  |

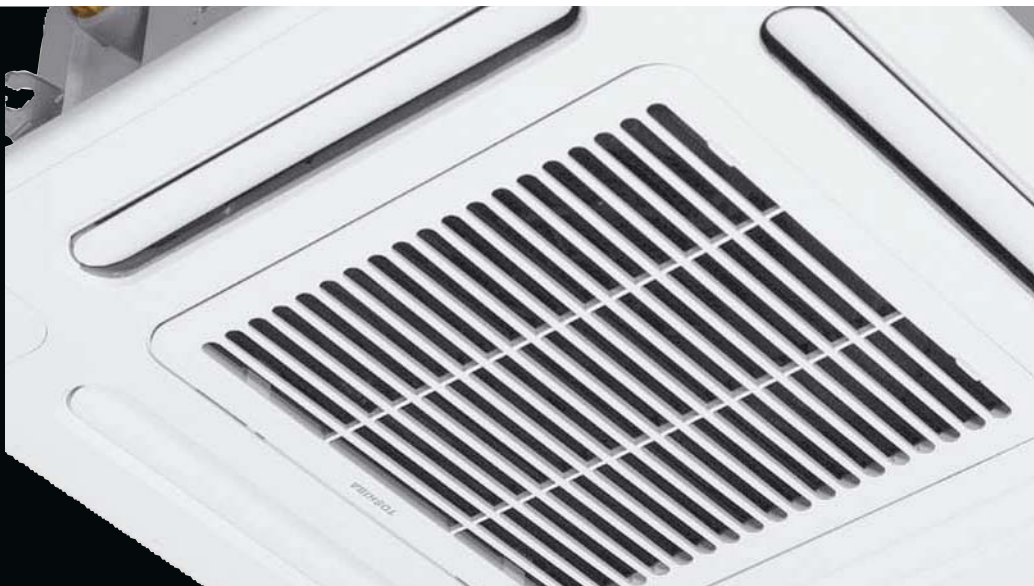
Digital Inverter (series 3) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SM563AT-E      | RAV-SM803AT-E      | RAV-SM1103AT-E     | RAV-SM1403AT-E     | RAV-SM1603AT-E     |
|-----------------------------------|------|--------------------|--------------------|--------------------|--------------------|--------------------|
| Indoor unit (4-way cassette)      |      | RAV-SM564UT-E      | RAV-SM804UT-E      | RAV-SM1104UT-E     | RAV-SM1404UT-E     | RAV-SM1604UT-E     |
| Cooling capacity                  | kW   | 5,3                | 6,7                | 10                 | 12                 | 14                 |
| Cooling range (min. – max.)       | kW   | 1,5 – 5,6          | 1,5 – 8,0          | 3,0 – 11,2         | 3,0 – 13,2         | 3,0 – 16,0         |
| Power input (min. – rated – max.) | kW C | 0,35 – 1,65 – 1,86 | 0,45 – 2,09 – 2,60 | 0,60 – 3,11 – 4,10 | 0,65 – 3,74 – 4,50 | 0,65 – 4,49 – 5,70 |
| EER                               | W/W  | 3,21               | 3,21               | 3,22               | 3,21               | 3,12               |
| Energy efficiency class           | C    | A                  | A                  | A                  | A                  | -                  |
| Annual energy consumption         | kWh  | 825                | 1045               | 1555               | 1870               | 2245               |
| Heating capacity                  | kW   | 5,6                | 8                  | 11,2               | 14                 | 16                 |
| Heating range (min. – max.)       | kW   | 1,5 – 6,3          | 1,5 – 9            | 3,0 – 13           | 3,0 – 16           | 3,0 – 18           |
| Power input (min. – rated – max.) | kW H | 0,35 – 1,44 – 2,08 | 0,45 – 2,21 – 3,03 | 0,60 – 2,93 – 4,30 | 0,65 – 3,8 – 4,50  | 0,65 – 4,43 – 6,51 |
| COP                               | W/W  | 3,89               | 3,62               | 3,82               | 3,68               | 3,61               |
| Energy efficiency class           | H    | A                  | A                  | A                  | A                  | -                  |

C = cooling mode

H = heating mode

h/m/l = high / medium / low speed

**R-410A HEAT PUMPS****DIGITAL/SUPER DIGITAL  
INVERTER****COMPACT DIMENSIONS****Suitable for standard false ceiling****Easy installation and maintenance****Slim and unobtrusive design**

## SM\_MUT Compact 4-way cassette

### Features

This 4-way cassette has been designed to suit all standard 600 × 600 mm grid ceilings, for easy installation and maintenance.

Its compact design blends with any room interior, where appearance is as important as functionality.

Draught prevention and clean ceiling functions make this unit ideal for the most demanding applications.

When combined with the Super Digital Inverter outdoor unit, the highest energy label (A) is achieved.

### Key features

Slim-line dimensions: this cassette is suitable for the most demanding installations where ceiling height is reduced.

Same dimensions for all capacities: the installation is always smart and elegant.

Easy maintenance: ease of access to the corner pockets facilitates installation and adjustment for perfect ceiling fitting.

Easy installation with a built-in drain pump.

TCC Link control panel makes control of the system flexible and simplifies maintenance.





### Super Digital Inverter (series 4) – Technical specifications **heat pump**

| Indoor unit                  |                         | RAV-SM402MUT-E    | RAV-SM452MUT-E    | RAV-SM562MUT-E    |
|------------------------------|-------------------------|-------------------|-------------------|-------------------|
| Air flow (h/l)               | m <sup>3</sup> /h – l/s | 660/468 – 183/130 | 660/468 – 183/130 | 798/546 – 222/152 |
| Sound pressure level (h/m/l) | dB(A)                   | 40/36/31          | 40/36/31          | 43/39/34          |
| Sound power level (h/m/l)    | dB(A)                   | 55/51/46          | 55/51/46          | 58/54/49          |
| Dimensions (h × w × d)       | mm                      | 268 × 575 × 575   | 268 × 575 × 575   | 268 × 575 × 575   |
| Weight                       | kg                      | 17                | 17                | 17                |
| Panel dimensions (h × w × d) | mm                      | 27 × 700 × 700    | 27 × 700 × 700    | 27 × 700 × 700    |
| Panel weight                 | kg                      | 3                 | 3                 | 3                 |

### Super Digital Inverter (series 4) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SP404AT-E      | RAV-SP454AT-E      | RAV-SP564AT-E      |
|-----------------------------------|------|--------------------|--------------------|--------------------|
| Indoor unit (600 × 600 cassette)  |      | RAV-SM402MUT-E     | RAV-SM452MUT-E     | RAV-SM562MUT-E     |
| Cooling capacity                  | kW   | 3,6                | 4,0                | 5,0                |
| Cooling range (min. – max.)       | kW   | 1,5 – 4,0          | 1,5 – 4,5          | 1,2 – 5,6          |
| Power input (min. – rated – max.) | kW C | 0,36 – 1,00 – 1,25 | 0,36 – 1,19 – 1,49 | 0,36 – 1,19 – 1,50 |
| EER                               | W/W  | 3,60               | 3,36               | 3,21               |
| Energy efficiency class           | C    | A                  | A                  | A                  |
| Annual energy consumption         | kWh  | 500                | 595                | 780                |
| Heating capacity                  | kW   | 4                  | 4,5                | 5,6                |
| Heating range (min. – max.)       | kW   | 1,5 – 5,0          | 1,5 – 5,6          | 0,9 – 7,4          |
| Power input (min. – rated – max.) | kW H | 0,36 – 0,97 – 2,20 | 0,36 – 1,16 – 2,30 | 0,17 – 1,54 – 2,37 |
| COP                               | W/W  | 4,12               | 3,88               | 3,64               |
| Energy efficiency class           | H    | A                  | A                  | A                  |

### Digital Inverter (series 3) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SM563AT-E      |
|-----------------------------------|------|--------------------|
| Indoor unit (600 × 600 cassette)  |      | RAV-SM562MUT-E     |
| Cooling capacity                  | kW   | 5,0                |
| Cooling range (min. – max.)       | kW   | 1,5 – 5,6          |
| Power input (min. – rated – max.) | kW C | 0,40 – 1,61 – 1,86 |
| EER                               | W/W  | 3,11               |
| Energy efficiency class           | C    | B                  |
| Annual energy consumption         | kWh  | 805                |
| Heating capacity                  | kW   | 5,6                |
| Heating range (min. – max.)       | kW   | 1,5 – 6,3          |
| Power input (min. – rated – max.) | kW H | 0,40 – 1,61 – 2,40 |
| COP                               | W/W  | 3,48               |
| Energy efficiency class           | H    | B                  |

C = cooling mode

H = heating mode

h/m/l = high / medium / low speed

**R-410A HEAT PUMPS**

**DIGITAL/SUPER DIGITAL  
INVERTER**

**DISCREET DUCTED UNITS**



## SM\_BT Ducted

### Features

Whatever the shape of the room could be, ducted units ensure uniform temperatures in it.

The unit can be discreetly positioned in the walls or ceiling.

It's ideal for hotels, banks and similar applications, where very low noise levels are needed.

### Key features

Wide range of applications: the use of ducts ensures flexible installations.

Improved room aesthetic: it's unobtrusive and discreet.

High static pressure: 98 Pa can be achieved and all the areas of the room can have the same temperature simultaneously.

High-lift drain pump kit: raises drain up to 290 mm for flexible condensate piping layout.

**Infra-red control option**

**Discharge spigots as  
standard**

**High static pressure up to  
98 Pa**



Technical specifications **heat pump**

| Indoor unit                                  |                         | RAV-SM562BT-E     | RAV-SM802BT-E      | RAV-SM1102BT-E      | RAV-SM1402BT-E      |
|----------------------------------------------|-------------------------|-------------------|--------------------|---------------------|---------------------|
| Air flow (h/l)                               | m <sup>3</sup> /h – l/s | 780/588 – 217/163 | 1140/798 – 317/222 | 1620/1134 – 450/315 | 1980/1386 – 550/385 |
| Sound pressure level (h/m/l)                 | dB(A)                   | 40/37/33          | 40/37/34           | 42/39/36            | 44/41/38            |
| Sound power level (h/m/l)                    | dB(A)                   | 55/52/48          | 55/52/49           | 57/54/51            | 59/56/53            |
| Dimensions (h × w × d)                       | mm                      | 320 × 700 × 800   | 320 × 1000 × 800   | 320 × 1350 × 800    | 320 × 1350 × 800    |
| Weight                                       | kg                      | 30                | 39                 | 54                  | 54                  |
| External static pressure (stand/upper limit) | Pa                      | 40/100            | 40/100             | 40/100              | 40/90               |

Super Digital Inverter (series 4) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SP564AT-E      | RAV-SP804AT-E      | RAV-SP1104AT-E     | RAV-SP1404AT-E     |
|-----------------------------------|------|--------------------|--------------------|--------------------|--------------------|
| Indoor unit (standard duct)       |      | RAV-SM562BT-E      | RAV-SM802BT-E      | RAV-SM1102BT-E     | RAV-SM1402BT-E     |
| Cooling capacity                  | kW   | 5,0                | 7,1                | 10,0               | 12,5               |
| Cooling range (min. – max.)       | kW   | 1,2 – 5,6          | 1,9 – 8,0          | 2,6 – 12,0         | 2,6 – 14,0         |
| Power input (min. – rated – max.) | kW C | 0,21 – 1,56 – 2,75 | 0,30 – 2,21 – 3,34 | 0,64 – 2,94 – 3,80 | 0,64 – 3,83 – 4,47 |
| EER                               | W/W  | 3,21               | 3,21               | 3,40               | 3,26               |
| Energy efficiency class           | C    | A                  | A                  | A                  | -                  |
| Annual energy consumption         | kWh  | 780                | 1105               | 1470               | 1915               |
| Heating capacity                  | kW   | 5,6                | 8,0                | 11,2               | 14,0               |
| Heating range (min. – max.)       | kW   | 0,9 – 7,4          | 1,3 – 10,6         | 2,4 – 13,0         | 2,4 – 16,5         |
| Power input (min. – rated – max.) | kW H | 0,17 – 1,55 – 2,51 | 0,27 – 2,21 – 3,50 | 0,52 – 2,77 – 4,00 | 0,52 – 3,41 – 4,50 |
| COP                               | W/W  | 3,61               | 3,62               | 4,04               | 4,11               |
| Energy efficiency class           | H    | A                  | A                  | A                  | -                  |

Digital Inverter (series 3) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SM563AT-E      | RAV-SM803AT-E      | RAV-SM1103AT-E     | RAV-SM1403AT-E     |
|-----------------------------------|------|--------------------|--------------------|--------------------|--------------------|
| Indoor unit (standard duct)       |      | RAV-SM562BT-E      | RAV-SM802BT-E      | RAV-SM1102BT-E     | RAV-SM1402BT-E     |
| Cooling capacity                  | kW   | 5,0                | 7,1                | 10,0               | 12,5               |
| Cooling range (min. – max.)       | kW   | 1,5 – 5,6          | 1,5 – 7,4          | 3,0 – 12,5         | 3,0 – 13,2         |
| Power input (min. – rated – max.) | kW C | 0,45 – 1,78 – 1,95 | 0,50 – 2,53 – 2,76 | 0,60 – 3,56 – 4,50 | 0,65 – 4,42 – 4,85 |
| EER                               | W/W  | 2,81               | 2,81               | 2,81               | 2,83               |
| Energy efficiency class           | C    | C                  | C                  | C                  | -                  |
| Annual energy consumption         | kWh  | 890                | 1265               | 1780               | 2210               |
| Heating capacity                  | kW   | 5,6                | 8,0                | 11,2               | 14,0               |
| Heating range (min. – max.)       | kW   | 1,5 – 6,3          | 1,5 – 9,0          | 3,0 – 12,5         | 3,0 – 16,0         |
| Power input (min. – rated – max.) | kW H | 0,45 – 1,71 – 2,47 | 0,50 – 2,41 – 3,18 | 0,60 – 3,14 – 4,00 | 0,65 – 4,03 – 4,55 |
| COP                               | W/W  | 3,27               | 3,32               | 3,57               | 3,47               |
| Energy efficiency class           | H    | C                  | C                  | B                  | -                  |

C = cooling mode

H = heating mode

h/m/l = high / medium / low speed

**R-410A****DIGITAL/SUPER DIGITAL  
INVERTER****1,5 – 2 HP RANGE**

## SM\_SDT Slim Duct

### Features

Very compact design unit with low height dimension for flexible ceiling installations.

Combines with the new small sizes of the high performances Super Digital Inverter outdoor units, for commercial applications which require limited power (1,5 HP).

### Key features

High energy efficiency performances in cooling and heating.

Two choice of selection for the air inlet section: bottom or back side.

Natural drain discharge port and drain pump (up to 850 mm).

Easy to install and service.

Cleanable prefilter included.

Fresh air inlet possibility via a pre-punched knock hole.

**Slim profile**

**Self cleaning**





Super Digital Inverter (series 4) – Technical specifications **heat pump**

| Indoor unit                                  |                         | RAV-SM404SDT-E    | RAV-SM454SDT-E    | RAV-SM564SDT-E    |
|----------------------------------------------|-------------------------|-------------------|-------------------|-------------------|
| Air flow (h/l)                               | m <sup>3</sup> /h – l/s | 690/522 – 192/145 | 690/522 – 192/145 | 780/582 – 217/162 |
| Sound pressure level (h/m/l)*                | dB(A)                   | 39/36/33          | 39/36/33          | 44/40/36          |
| Sound power level (h/m/l)                    | dB(A)                   | 54/51/48          | 54/51/48          | 59/55/51          |
| Dimensions (h × w × d)                       | mm                      | 210 × 845 × 645   | 210 × 845 × 645   | 210 × 845 × 645   |
| Weight                                       | kg                      | 22                | 22                | 22                |
| External static pressure (stand/upper limit) | Pa                      | 5/24              | 5/24              | 4/24              |

\*bottom air inlet

Super Digital Inverter (series 4) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SP404AT-E      | RAV-SP454AT-E      | RAV-SP564AT-E      |
|-----------------------------------|------|--------------------|--------------------|--------------------|
| Indoor unit (slim duct)           |      | RAV-SM404SDT-E     | RAV-SM454SDT-E     | RAV-SM564SDT-E     |
| Cooling capacity                  | kW   | 3,6                | 4,0                | 5,0                |
| Cooling range (min. – max.)       | kW   | 1,5 – 4,0          | 1,5 – 4,5          | 1,2 – 5,6          |
| Power input (min. – rated – max.) | kW C | 0,37 – 1,03 – 1,25 | 0,37 – 1,20 – 1,49 | 0,21 – 1,56 – 2,78 |
| EER                               | W/W  | 3,50               | 3,33               | 3,21               |
| Energy efficiency class           | C    | A                  | A                  | A                  |
| Annual energy consumption         | kWh  | 515                | 600                | 780                |
| Heating capacity                  | kW   | 4,0                | 4,5                | 5,6                |
| Heating range (min. – max.)       | kW   | 1,5 – 5,0          | 1,5 – 5,6          | 0,9 – 7,4          |
| Power input (min. – rated – max.) | kW H | 0,37 – 1,00 – 2,20 | 0,37 – 1,15 – 2,30 | 0,17 – 1,44 – 2,37 |
| COP                               | W/W  | 4,00               | 3,91               | 3,89               |
| Energy efficiency class           | H    | A                  | A                  | A                  |

Digital Inverter (series 3) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SM563AT-E      |
|-----------------------------------|------|--------------------|
| Indoor unit (slim duct)           |      | RAV-SM564SDT-E     |
| Cooling capacity                  | kW   | 5,0                |
| Cooling range (min. – max.)       | kW   | 1,5 – 5,6          |
| Power input (min. – rated – max.) | kW C | 0,38 – 1,66 – 1,97 |
| EER                               | W/W  | 3,01               |
| Energy efficiency class           | C    | B                  |
| Annual energy consumption         | kWh  | 830                |
| Heating capacity                  | kW   | 5,6                |
| Heating range (min. – max.)       | kW   | 1,5 – 6,3          |
| Power input (min. – rated – max.) | kW H | 0,38 – 1,59 – 2,40 |
| COP                               | W/W  | 3,52               |
| Energy efficiency class           | H    | B                  |

C = cooling mode

H = heating mode

h/m/l = high / medium / low speed

**R-410A HEAT PUMPS****DIGITAL/SUPER DIGITAL  
INVERTER****CEILING-SUSPENDED UNITS**

## SM\_CT Ceiling

**Low noise levels****Slim-line design****Automatic louvre control  
plus auto-swing**

### Features

These ceiling suspended units are the ideal solution for offices, classrooms and restaurants.

The automatic louvre control and low noise levels are the key characteristics of this state-of-the-art unit.

In addition, the drain pan inside the unit ensures the maximum hygiene and is easily recyclable thanks to its stain resistant polypropylene resin body.

### Key features

It operates at only 30 dB(A) (in 2HP) – twice as quietly as conventional units, thanks to its new design.

Optimum louvre control: the air flow angle is automatically set to the most suitable setting, and an automatic swing mode ensures air flow reaches all areas of the room.

Installation efficiency: for ceiling mounting, the unit can be suspended simply by adjusting 2 screws on the intake grille (compared to a dozen screws for conventional models).



Technical specifications **heat pump**

| Indoor unit                      |                         | RAV-SM562CT-E     | RAV-SM802CT-E      | RAV-SM1102CT-E      | RAV-SM1402CT-E      |
|----------------------------------|-------------------------|-------------------|--------------------|---------------------|---------------------|
| Air flow (h/l)                   | m <sup>3</sup> /h – l/s | 780/600 – 217/167 | 1110/876 – 308/243 | 1650/1272 – 458/353 | 1800/1386 – 500/385 |
| Sound pressure level (h / m / l) | dB(A)                   | 38/33/30          | 38/33/30           | 41/38/35            | 43/40/37            |
| Sound power level (h / m / l)    | dB(A)                   | 51/48/45          | 51/48/45           | 56/53/50            | 58/55/52            |
| Dimensions (h × w × d)           | mm                      | 210 × 910 × 680   | 210 × 1180 × 680   | 210 × 1595 × 680    | 210 × 1595 × 680    |
| Weight                           | kg                      | 21                | 25                 | 33                  | 33                  |

Super Digital Inverter (series 4) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SP564AT-E      | RAV-SP804AT-E      | RAV-SP1104AT-E     | RAV-SP1404AT-E     |
|-----------------------------------|------|--------------------|--------------------|--------------------|--------------------|
| Indoor unit (ceiling)             |      | RAV-SM562CT-E      | RAV-SM802CT-E      | RAV-SM1102CT-E     | RAV-SM1402CT-E     |
| Cooling capacity                  | kW   | 5,0                | 7,0                | 10                 | 12,5               |
| Cooling range (min. – max.)       | kW   | 1,2 – 5,6          | 1,9 – 8,0          | 2,6 – 12,0         | 2,6 – 14,0         |
| Power input (min. – rated – max.) | kW C | 0,21 – 1,56 – 2,75 | 0,30 – 2,21 – 3,34 | 0,64 – 2,67 – 3,70 | 0,64 – 3,73 – 4,47 |
| EER                               | W/W  | 3,21               | 3,21               | 3,75               | 3,35               |
| Energy efficiency class           | C    | A                  | A                  | A                  | -                  |
| Annual energy consumption         | kWh  | 780                | 1105               | 1335               | 1865               |
| Heating capacity                  | kW   | 5,6                | 8,0                | 11,2               | 14,0               |
| Heating range (min. – max.)       | kW   | 0,9 – 7,4          | 1,3 – 10,6         | 2,4 – 13,0         | 2,4 – 16,5         |
| Power input (min. – rated – max.) | kW H | 0,17 – 1,47 – 2,34 | 0,27 – 2,16 – 3,50 | 0,52 – 2,62 – 4,00 | 0,52 – 3,65 – 4,60 |
| COP                               | W/W  | 3,81               | 3,70               | 4,27               | 3,84               |
| Energy efficiency class           | H    | A                  | A                  | A                  | -                  |

Digital Inverter (series 3) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SM563AT-E      | RAV-SM803AT-E      | RAV-SM1103AT-E     | RAV-SM1403AT-E     |
|-----------------------------------|------|--------------------|--------------------|--------------------|--------------------|
| Indoor unit (ceiling)             |      | RAV-SM562CT-E      | RAV-SM802CT-E      | RAV-SM1102CT-E     | RAV-SM1402CT-E     |
| Cooling capacity                  | kW   | 5,0                | 7,0                | 10,0               | 12,3               |
| Cooling range (min. – max.)       | kW   | 1,5 – 5,6          | 1,5 – 7,4          | 3,0 – 11,2         | 3,0 – 13,2         |
| Power input (min. – rated – max.) | kW C | 0,45 – 1,82 – 1,95 | 0,50 – 2,53 – 2,76 | 0,60 – 3,51 – 4,10 | 0,65 – 4,52 – 4,85 |
| EER                               | W/W  | 2,75               | 2,77               | 2,85               | 2,72               |
| Energy efficiency class           | C    | D                  | D                  | C                  | -                  |
| Annual energy consumption         | kWh  | 910                | 1265               | 1755               | 2260               |
| Heating capacity                  | kW   | 5,6                | 8,0                | 11,2               | 14,0               |
| Heating range (min. – max.)       | kW   | 1,5 – 6,3          | 1,5 – 9,0          | 3,0 – 12,5         | 3,0 – 16,0         |
| Power input (min. – rated – max.) | kW H | 0,45 – 1,64 – 2,40 | 0,50 – 2,47 – 3,20 | 0,60 – 3,20 – 4,10 | 0,65 – 4,14 – 4,60 |
| COP                               | W/W  | 3,41               | 3,24               | 3,50               | 3,38               |
| Energy efficiency class           | H    | B                  | C                  | B                  | -                  |

C = cooling mode

H = heating mode

h/m/l = high / medium / low speed

**R-410A HEAT PUMPS**

**DIGITAL/SUPER DIGITAL  
INVERTER**

**STYLISH DESIGN**



## SM\_2KRT High-wall

### Features

With its attractive and slim-line design, this high-wall is suitable for offices, restaurants and other applications where elegance is required.

The filtration system further improves the indoor air quality benefits of this high-wall unit.

### Key features

With its slim design, this compact and stylish unit blends with any room setting.

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Enhanced filtration system: Zeolite Plus + Sasa filter to deodorise, Bio-Enzyme filter + Gingko filter to purify and new antioxidant Vitamin C filter.

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Auto louvre mode allows optimum air distribution throughout.

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Wireless controller included.

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TCC Link remote control (optional).

**Slim design**

**4 levels of filtration**



Technical specifications **heat pump**

| Indoor unit                  |                         | RAV-SM562KRT-E    | RAV-SM802KRT-E     |
|------------------------------|-------------------------|-------------------|--------------------|
| Air flow (h/l)               | m <sup>3</sup> /h – l/s | 840/642 – 233/178 | 1110/732 – 308/203 |
| Sound pressure level (h/m/l) | dB(A)                   | 39/36/33          | 45/41/36           |
| Sound power level (h/m/l)    | dB(A)                   | 54/51/48          | 60/56/51           |
| Dimensions (h × w × d)       | mm                      | 298 × 998 × 221   | 298 × 998 × 221    |
| Weight                       | kg                      | 12                | 12                 |

Super Digital Inverter (series 4) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SP562AT-E      | RAV-SP804AT-E      |
|-----------------------------------|------|--------------------|--------------------|
| Indoor unit (high-wall)           |      | RAV-SM562KRT-E     | RAV-SM-802KRT-E    |
| Cooling capacity                  | kW   | 5,0                | 6,9                |
| Cooling range (min. – max.)       | kW   | 1,2 – 5,6          | 1,9 – 8,0          |
| Power input (min. – rated – max.) | kW C | 0,21 – 1,56 – 2,75 | 0,3 – 2,40 – 3,34  |
| EER                               | W/W  | 3,21               | 2,88               |
| Energy efficiency class           | C    | A                  | C                  |
| Annual energy consumption         | kWh  | 780                | 1200               |
| Heating capacity                  | kW   | 5,6                | 8,0                |
| Heating range (min. – max.)       | kW   | 0,9 – 7,3          | 1,3 – 10,6         |
| Power input (min. – rated – max.) | kW H | 0,17 – 1,55 – 2,57 | 0,27 – 2,40 – 3,87 |
| COP                               | W/W  | 3,61               | 3,33               |
| Energy efficiency class           | H    | A                  | C                  |

Digital Inverter (series 3) – Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SM563AT-E      | RAV-SM803AT-E      |
|-----------------------------------|------|--------------------|--------------------|
| Indoor unit (high-wall)           |      | RAV-SM562KRT-E     | RAV-SM802KRT-E     |
| Cooling capacity                  | kW   | 5,0                | 7,0                |
| Cooling range (min. – max.)       | kW   | 1,5 – 5,6          | 1,5 – 8,0          |
| Power input (min. – rated – max.) | kW C | 0,40 – 1,74 – 1,86 | 0,50 – 2,72 – 2,85 |
| EER                               | W/W  | 2,93               | 2,46               |
| Energy efficiency class           | C    | C                  | E                  |
| Annual energy consumption         | kWh  | 870                | 1360               |
| Heating capacity                  | kW   | 5,6                | 8,0                |
| Heating range (min. – max.)       | kW   | 1,5 – 6,3          | 1,5 – 9,0          |
| Power input (min. – rated – max.) | kW H | 0,40 – 1,70 – 2,40 | 0,50 – 2,67 – 3,46 |
| COP                               | W/W  | 3,29               | 3,00               |
| Energy efficiency class           | H    | C                  | D                  |

C = cooling mode

H = heating mode

h/m/l = high / medium / low speed



**R-410A HEAT PUMPS****DIGITAL INVERTER****CEILING-SUSPENDED/  
CONSOLE UNIT****Triple action filtration system****Two installation layouts**

## SM\_XT Flexi

### Features

The stylish Flexi unit can add a touch of elegance to a commercial space. Installation flexibility and indoor air quality are two of the many advantages of the system.

### Key features

Ultimate flexibility: low wall and ceiling-suspended installations are both possible, with no modification or additional accessories.

Triple-action filtration system: the first step removes large particles, then a passive electrostatic filter removes solid particles down to 0,01 micron in size and finally the Zeolite Plus filter absorbs even smaller airborne pollutants.

Natural air distribution: in ceiling-suspended applications, air can be directed either horizontally, parallel to the ceiling or away from the room occupants for non intrusive air distribution.

High-lift drain pump kit: raises drain piping up to 290 mm for flexible condensate piping layout (option suitable for ceiling suspended unit only).



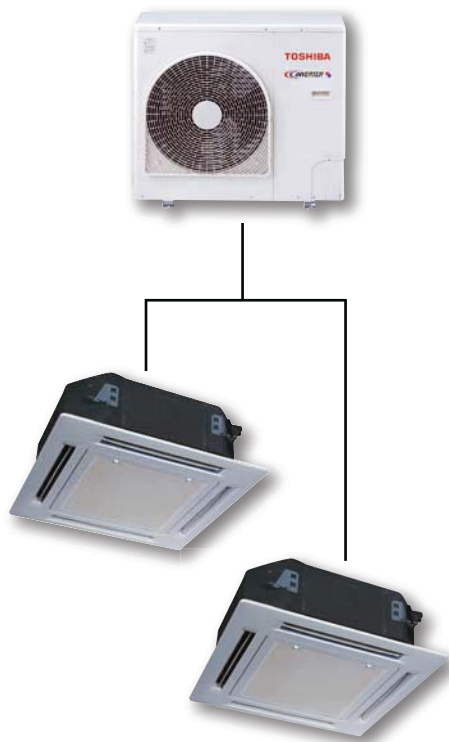
### Technical specifications **heat pump**

| Indoor unit                |                         | RAV-SM562XT-E     | RAV-SM802XT-E      |
|----------------------------|-------------------------|-------------------|--------------------|
| Air flow (h/l)             | m <sup>3</sup> /h – l/s | 840/600 – 233/178 | 1110/642 – 308/178 |
| Sound pressure level (h/l) | dB(A)                   | 43/36             | 46/37              |
| Sound power level (h/l)    | dB(A)                   | 58/51             | 61/52              |
| Dimensions (h × w × d)     | mm                      | 208 × 1093 × 633  | 208 × 1093 × 633   |
| Weight                     | kg                      | 23                | 23                 |

### Digital Inverter (series 3 ) - Technical specifications **heat pump**

| Outdoor unit                      |      | RAV-SM563AT-E      | RAV-SM803AT-E      |
|-----------------------------------|------|--------------------|--------------------|
| Indoor unit (flexi)               |      | RAV-SM562XT-E      | RAV-SM802XT-E      |
| Cooling capacity                  | kW   | 5,0                | 7,0                |
| Cooling range (min. – max.)       | kW   | 1,5 – 5,6          | 1,5 – 8,0          |
| Power input (min. – rated – max.) | kW C | 0,55 – 1,87 – 2,01 | 0,55 – 2,72 – 2,85 |
| EER                               | W/W  | 2,67               | 2,46               |
| Energy efficiency class           | C    | D                  | E                  |
| Annual energy consumption         | kWh  | 935                | 1360               |
| Heating capacity                  | kW   | 5,6                | 8,0                |
| Heating range (min. – max.)       | kW   | 1,5 – 6,3          | 1,5 – 9,0          |
| Power input (min. – rated – max.) | kW H | 0,55 – 1,70 – 2,40 | 0,55 – 2,67 – 3,46 |
| COP                               | W/W  | 3,29               | 3,00               |
| Energy efficiency class           | H    | C                  | D                  |

C = cooling mode  
H = heating mode

**R-410A****DIGITAL/SUPER DIGITAL  
INVERTER**

## Twin Split SDI/DI

### Features

The Twin Split system connects two indoor units of the same type and capacity to one outdoor unit in order to ensure more air distribution in a large zone.

One of the units is designated the master unit and it is managing the room temperature reference for both indoor units.

These should be installed in the same room, operate simultaneously and share a single controller.

### Key features

Twin operation is possible with all Light Commercial indoor units.

Precise capacity control at all conditions.

Ideal for large shops, open-plan offices and other similar applications.

User-friendly controls.

Compact outdoor unit for easy installation.

Twinning requires a connection kit that includes an electromagnetic noise filter and pipe joint.

### Super Digital Inverter (series 4) – Technical specifications **heat pump - twin system**

|                                | HP                | 3                      | 3          | 4              | 4                      | 4          | 4          | 4          |
|--------------------------------|-------------------|------------------------|------------|----------------|------------------------|------------|------------|------------|
| Outdoor unit                   | RAV-              | SP804AT-E              | SP804AT-E  | SP1104AT-E     | SP1104AT-E             | SP1104AT-E | SP1104AT-E | SP1104AT-E |
| Indoor unit                    | RAV-              | SM402MUT-E             | SM404SDT-E | SM564UT-E      | SM562MUT-E             | SM562BT-E  | SM564SDT-E | SM562CT-E  |
|                                | Indoor unit model | Compact 4-way cassette | Slim duct  | 4-way cassette | Compact 4-way cassette | Ducted     | Slim duct  | Ceiling    |
| Cooling capacity               | kW                | 7,1                    | 7,1        | 10,0           | 10,0                   | 10,0       | 10,0       | 10,0       |
| Cooling capacity (min. – max.) | kW                | 1,9 – 8,0              | 1,9 – 8,0  | 2,6 – 12,0     | 2,6 – 12,0             | 2,6 – 12,0 | 2,6 – 12,0 | 2,6 – 12,0 |
| Power input                    | kW C              | 2,21                   | 2,21       | 2,21           | 2,67                   | 2,94       | 2,77       | 2,67       |
| EER                            | W/W               | 3,21                   | 3,21       | 4,52           | 3,75                   | 3,40       | 3,61       | 3,75       |
| Energy efficiency class        | C                 | A                      | A          | A              | A                      | A          | A          | A          |
| Annual energy consumption      | kWh               | 1105                   | 1105       | 1105           | 1335                   | 1470       | 1385       | 1335       |
| Heating capacity               | kW                | 8,0                    | 8,0        | 11,2           | 11,2                   | 11,2       | 11,2       | 11,2       |
| Heating range (min. – max.)    | kW                | 1,3 – 10,6             | 1,3 – 10,6 | 2,4 – 13,0     | 2,4 – 13,0             | 2,4 – 13,0 | 2,4 – 13,0 | 2,4 – 13,0 |
| Power input                    | kW H              | 2,16                   | 2,16       | 2,34           | 2,67                   | 2,77       | 2,67       | 2,62       |
| COP                            | W/W               | 3,70                   | 3,70       | 4,79           | 4,19                   | 4,04       | 4,19       | 4,27       |
| Energy efficiency class        | H                 | A                      | A          | A              | A                      | A          | A          | A          |

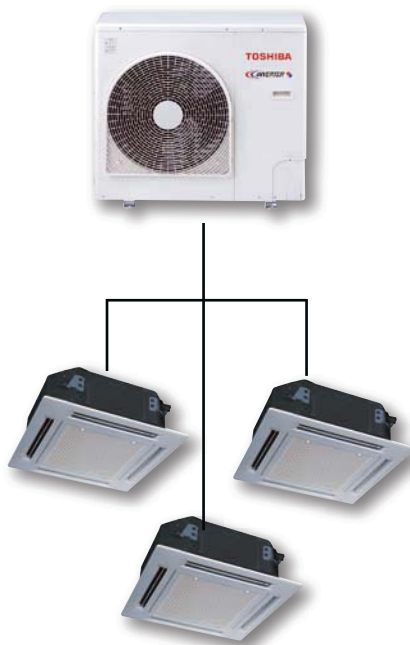
### Super Digital Inverter (series 4) – Technical specifications **heat pump - twin system**

|                                | HP                | 5          | 5              | 5          | 5          | 5          |
|--------------------------------|-------------------|------------|----------------|------------|------------|------------|
| Outdoor unit                   | RAV-              | SP1104AT-E | SP1404AT-E     | SP1404AT-E | SP1404AT-E | SP1404AT-E |
| Indoor unit                    | RAV-              | SM562KRT-E | SM804UT-E      | SM802BT-E  | SM802CT-E  | SM802KRT-E |
|                                | Indoor unit model | High-wall  | 4-way cassette | Ducted     | Ceiling    | High-wall  |
| Cooling capacity               | kW                | 10,0       | 12,5           | 12,5       | 12,5       | 12,3       |
| Cooling capacity (min. – max.) | kW                | 2,6 – 12,0 | 2,6 – 14,0     | 2,6 – 14,0 | 2,6 – 14,0 | 2,6 – 13,5 |
| Power input                    | kW C              | 2,77       | 3,16           | 3,83       | 3,73       | 3,88       |
| EER                            | W/W               | 3,61       | 3,96           | 3,26       | 3,35       | 3,17       |
| Energy efficiency class        | C                 | A          | -              | -          | -          | -          |
| Annual energy consumption      | kWh               | 1385       | 1580           | 1915       | 1865       | 1940       |
| Heating capacity               | kW                | 11,2       | 14,0           | 14,0       | 14,0       | 14,0       |
| Heating range (min. – max.)    | kW                | 2,4 – 13,0 | 2,4 – 16,5     | 2,4 – 16,5 | 2,4 – 16,5 | 2,4 – 16,5 |
| Power input                    | kW H              | 2,8        | 3,21           | 3,41       | 3,65       | 3,83       |
| COP                            | W/W               | 4,00       | 4,36           | 4,11       | 3,84       | 3,66       |
| Energy efficiency class        | H                 | A          | -              | -          | -          | -          |

### Digital Inverter (series 3) – Technical specifications **heat pump - twin system**

|                                | HP                | 5              | 5          | 5          | 5          | 6              | 6          | 6          | 6          |
|--------------------------------|-------------------|----------------|------------|------------|------------|----------------|------------|------------|------------|
| Outdoor unit                   | RAV-              | SM1403AT-E     | SM1403AT-E | SM1403AT-E | SM1403AT-E | SM1603AT-E     | SM1603AT-E | SM1603AT-E | SM1603AT-E |
| Indoor unit                    | RAV-              | SM804UT-E      | SM802BT-E  | SM802CT-E  | SM802KRT-E | SM804UT-E      | SM802BT-E  | SM802CT-E  | SM802KRT-E |
|                                | Indoor unit model | 4-way cassette | Ducted     | Ceiling    | High-wall  | 4-way cassette | Ducted     | Ceiling    | High-wall  |
| Cooling capacity               | kW                | 12,0           | 12,5       | 12,3       | 12,0       | 14,0           | 14,0       | 14,0       | 14,0       |
| Cooling capacity (min. – max.) | kW                | 3,0 – 13,2     | 3,0 – 13,2 | 3,0 – 13,2 | 3,0 – 13,0 | 3,0 – 16,0     | 3,0 – 16,0 | 3,0 – 16,0 | 3,0 – 16,0 |
| Power input                    | kW C              | 3,74           | 4,42       | 4,52       | 4,52       | 4,49           | 5,12       | 4,99       | 5,10       |
| EER                            | W/W               | 3,21           | 2,83       | 2,72       | 2,65       | 3,12           | 2,73       | 2,81       | 2,75       |
| Energy efficiency class        | C                 | -              | -          | -          | -          | -              | -          | -          | -          |
| Annual energy consumption      | kWh               | 1870           | 2210       | 2260       | 2260       | 2245           | 2560       | 2495       | 2550       |
| Heating capacity               | kW                | 14,0           | 14,0       | 14,0       | 14,0       | 16,0           | 16,0       | 16,0       | 16,0       |
| Heating range (min. – max.)    | kW                | 3,0 – 16,0     | 3,0 – 16,0 | 3,0 – 16,0 | 3,0 – 16,0 | 3,0 – 18,0     | 3,0 – 18,0 | 3,0 – 18,0 | 3,0 – 18,0 |
| Power input                    | kW H              | 3,80           | 4,14       | 4,14       | 4,24       | 4,43           | 4,69       | 4,69       | 4,98       |
| COP                            | W/W               | 3,68           | 3,38       | 3,38       | 3,30       | 3,61           | 3,41       | 3,41       | 3,21       |
| Energy efficiency class        | H                 | -              | -          | -          | -          | -              | -          | -          | -          |

C = cooling mode  
H = heating mode

**R-410A****DIGITAL INVERTER****MORE COMBINATIONS  
POSSIBLE**

## Triple Combination DI

### Features

The Triple Combination system connects three indoor units of the same type and capacity to one outdoor unit in order to ensure more air distribution in a large zone.

One of the units is designated the master unit and it is managing the room temperature reference for the other indoor units.

These should be installed in the same room, operate simultaneously and share a single controller.

### Key features

Triple operation is possible with DI Light Commercial indoor units.

Precise capacity control at all conditions.

Ideal for large shops, open-plan offices and other similar applications.

User-friendly controls.

Compact outdoor unit for easy installation.

Triple combination requires a piping connection kit to optimize refrigerant flow.



**Digital Inverter (series 3): RAV-SM1603AT-E – Technical specifications heat pump - triple system**

| Outdoor unit                   | RAV-              | SM1603AT-E     | SM1603AT-E             | SM1603AT-E | SM1603AT-E | SM1603AT-E | SM1603AT-E |
|--------------------------------|-------------------|----------------|------------------------|------------|------------|------------|------------|
| Indoor unit                    | RAV-              | SM564UT-E      | SM562MUT-E             | SM562BT-E  | SM564SDT-E | SM562CT-E  | SM562KRT-E |
|                                | Indoor unit model | 4-way cassette | Compact 4-way cassette | Ducted     | Slim duct  | Ceiling    | High-wall  |
| Cooling capacity               | kW                | 14,0           | 14,0                   | 14,0       | 14,0       | 14,0       | 14,0       |
| Cooling capacity (min. – max.) | kW                | 3,0 – 16,0     | 3,0 – 16,0             | 3,0 – 16,0 | 3,0 – 16,0 | 3,0 – 16,0 | 3,0 – 16,0 |
| Power input                    | kW C              | 4,49           | 4,99                   | 5,12       | 4,99       | 4,99       | 5,10       |
| EER                            | W/W               | 3,12           | 2,81                   | 2,73       | 2,81       | 2,81       | 2,75       |
| Energy efficiency class        | C                 | -              | -                      | -          | -          | -          | -          |
| Annual energy consumption      | kWh               | 2245           | 2495                   | 2560       | 2495       | 2495       | 2550       |
| Heating capacity               | kW                | 16,0           | 16,0                   | 16,0       | 16,0       | 16,0       | 16,0       |
| Heating range (min. – max.)    | kW                | 3,0 – 18,0     | 3,0 – 18,0             | 3,0 – 18,0 | 3,0 – 18,0 | 3,0 – 18,0 | 3,0 – 18,0 |
| Power input                    | kW H              | 4,43           | 4,69                   | 4,69       | 4,69       | 4,69       | 4,98       |
| COP                            | W/W               | 3,61           | 3,41                   | 3,41       | 3,41       | 3,41       | 3,21       |
| Energy efficiency class        | H                 | -              | -                      | -          | -          | -          | -          |

C = cooling mode  
H = heating mod

## The business range. The applications

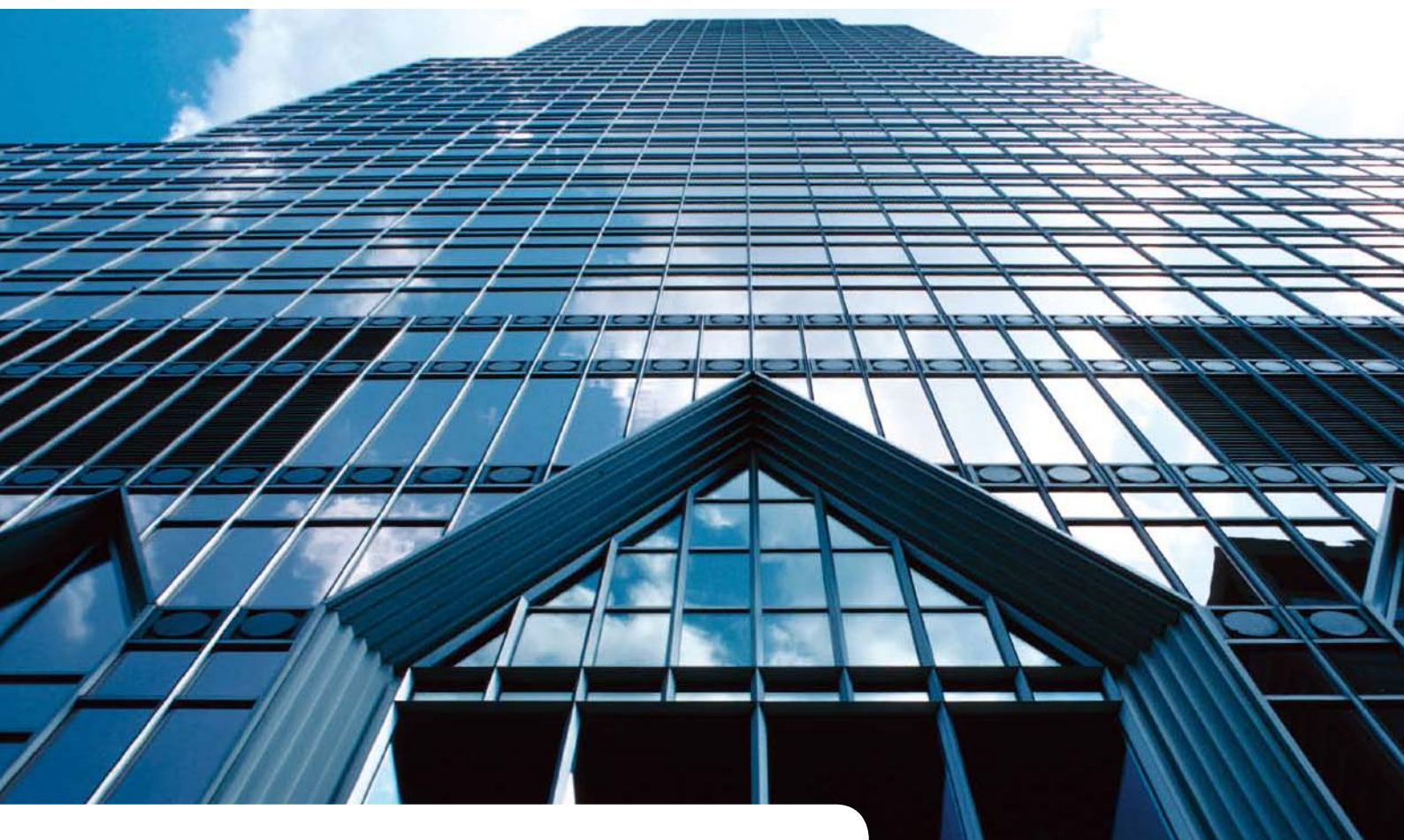
VRF technology offers the best solution for large commercial and industrial buildings: including hotels, hospitals, leisure and shopping centres.

The dual inverter compressor guarantees high efficiency levels,

operating flexibility and reduced maintenance requirements.

Moreover, the wide range of indoor units makes VRF system the most flexible choice to satisfy any kind of requirement and to be ideal for many installations. The MiNi-SMMS systems,

for instance, are the ideal solution both commercial and for private spaces.



## Compactness and beyond

In 2006, Toshiba completed the VRF range with the new compact MiNi-SMMS system. This new range has been designed to guarantee the most flexible solutions for both commercial and large spaces applications.

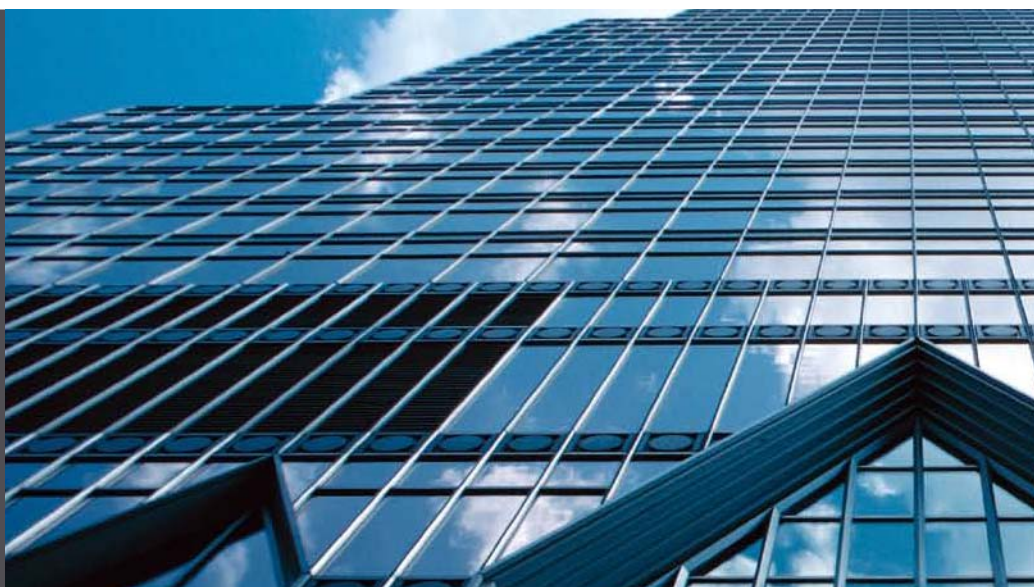
VRF offers important features: performance, low noise levels, sophisticated and precise control

systems, energy savings and respect for the environment.

In fact, Toshiba's commitment to the research and the development of new technologies grants always the maximum care of the environment: all the VRF systems are endowed with non-ozone-depleting refrigerant

(R-410A) and the sophisticated dual inverter control.





## VRF. The freedom of choice

Variable refrigerant flow benefits from the advantages of direct expansion linked to inverter control and the most sophisticated electronics.

This technology has many advantages, from the system design to the installation and operation

phase. The wide range of indoor units makes VRF the most flexible choice to satisfy any requirement.

In fact, Toshiba has three VRF system ranges: SMMS (Super Modular Multi System) that provides cooling or heating, SHRM (Super Heat Recovery

Multi) that provides simultaneous heating and cooling and MiNi-SMMS, the new compact system, ideal for small spaces.

## Typical applications and advantages

VRF systems offer safety, reliability, comfort, flexibility, ease-of-installation, durability and energy savings.

More and more commercial centres, office towers, hospitals and hotels,

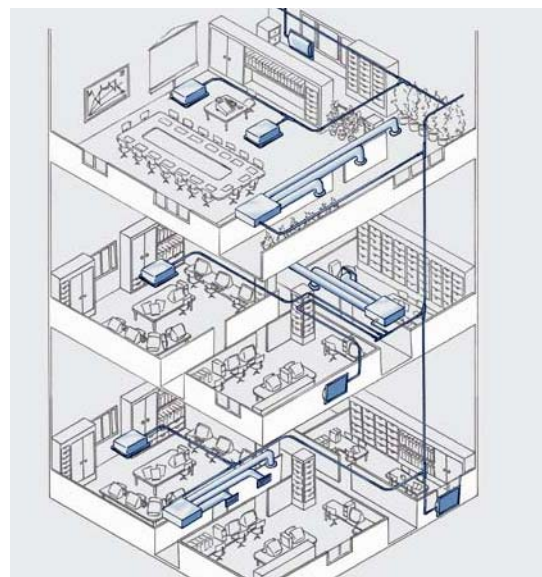
all typically requiring the benefit of energy savings, have selected this type of system.

## Energy savings according to Toshiba

The advanced electronic technology in these systems permits capacity control that results in significant energy savings. This objective is achieved thanks to the use of **sophisticated inverter control** and modulating control valves in

each indoor unit. In addition, the power input of the outdoor unit is dramatically reduced with the heat load reduction in the areas served. No particular routine maintenance is required, except periodic cleaning of

the indoor unit filters: this also means that maintenance costs are minimised.







Unlimited flexibility

Optimised product choice

The ultimate inverter system

Minimised consumption



## Precision is our top priority

Sophisticated inverter control permits matching of the actual refrigerant flow to the capacity required by each indoor unit in an application.

This results in optimised efficiency of the refrigerant cycle and increased precision in maintaining the required temperature, improving comfort for the occupants.

The required capacity and the related technical parameters for each indoor unit are electronically transmitted to the outdoor unit in order to optimise the zone load calculation and to control the actual refrigerant flow to each indoor unit, using the special Pulsed Modulating Valves (PMV).





**R-410A****WIDE RANGE OF INDOOR UNITS****Superior EER and COP****Quiet operation**

## MiNi-SMMS VRF Outdoor unit

### Features

The MiNi-SMMS system has been developed to achieve the best performance in a wide variety of commercial applications including shops, offices and large apartments, where unobtrusive appearance and quiet operation are important advantages.

The extraordinary flexibility of this Toshiba system is guaranteed by the breadth of the range of SMMS indoor units – up to 13 models with a combination of 81 units. MiNi-SMMS can be easily installed.

## PMV Kit

- The PMV kit is an option for super-silent operation, available for hotel rooms and residential applications where noise levels are critical
- Ease of installation
- Integral condensate pump
- Low cost

### Key features

Best COP (4,61 for 4HP): represents state-of-art energy saving efficiency.

Wide range: up to 9 indoor units may be connected with a single outdoor unit.

DC Twin Rotary compressor delivers high efficiency and complete reliability.

Full SMMS indoor and control units available.

The compact design of the outdoor unit (70% smaller overall than standard VRF unit) means it can be easily installed virtually anywhere; including on a balcony.

Technical specifications **heat pump**

| Outdoor unit                           |                         |     | HP | MCY-MAP0401HT    | MCY-MAP0501HT    | MCY-MAP0601HT    |
|----------------------------------------|-------------------------|-----|----|------------------|------------------|------------------|
|                                        |                         |     |    | 4 HP             | 5 HP             | 6 HP             |
| Cooling capacity                       | kW                      |     |    | 12,1             | 14,0             | 15,5             |
| Power input                            | kW                      | C   |    | 2,82             | 3,47             | 4,63             |
| EER                                    | W/W                     |     |    | 4,29             | 4,03             | 3,35             |
| Running current                        | A                       | C   |    | 13,2             | 16,1             | 21,4             |
| Heating capacity                       | kW                      |     |    | 12,5             | 16,0             | 18,0             |
| Power input                            | kW                      | H   |    | 2,71             | 4,00             | 4,85             |
| COP                                    | W/W                     |     |    | 4,61             | 4,00             | 3,71             |
| Running current                        | A                       | H   |    | 12,5             | 18,3             | 22,2             |
| Peak demand current                    | A                       |     |    | 25               | 28               | 31               |
| Air flow                               | m <sup>3</sup> /h – l/s |     |    | 5820 – 1617      | 6120 – 1700      | 6420 – 1783      |
| Sound pressure level                   | dB(A)                   | C/H |    | 49/50            | 50/52            | 51/53            |
| Operating range – db                   | °C                      | C   |    | -5 ÷ 43          | -5 ÷ 43          | -5 ÷ 43          |
| Operating range – wb                   | °C                      | H   |    | -15,0 ÷ 15,5     | -15,0 ÷ 15,5     | -15,0 ÷ 15,5     |
| Dimensions (h × w × d)                 | mm                      |     |    | 1340 × 900 × 320 | 1340 × 900 × 320 | 1340 × 900 × 320 |
| Weight                                 | kg                      |     |    | 117              | 117              | 117              |
| Compressor type                        |                         |     |    | Twin Rotary      | Twin Rotary      | Twin Rotary      |
| Refrigerant charge R410A               | kg                      |     |    | 7,2              | 7,2              | 7,2              |
| Suction line type – diameter           |                         |     |    | Flare – 5/8"     | Flare – 5/8"     | Brazing – 3/4"   |
| Liquid line type – diameter            |                         |     |    | Flare – 3/8"     | Flare – 3/8"     | Flare – 3/8"     |
| Maximum equivalent length separation*  | m                       |     |    | 125              | 125              | 125              |
| Maximum actual piping separation*      | m                       |     |    | 100              | 100              | 100              |
| Maximum pipe length*                   | m                       |     |    | 180              | 180              | 180              |
| Maximum lift (Indoor unit above/below) | m                       |     |    | 20/30            | 20/30            | 20/30            |
| Power supply                           | V-ph-Hz                 |     |    | 230-1-50         | 230-1-50         | 230-1-50         |

\* when PMV Kit is used: Maximum equivalent length separation (80 m); Maximum actual piping separation (65 m); Maximum pipe length (150 m)

C = cooling mode  
H = heating mode

Technical specifications **outdoor units**

| Model Name    |      | Cooling capacity | Heating capacity | Number of indoor units | Total capacity of connectable indoor units |        |
|---------------|------|------------------|------------------|------------------------|--------------------------------------------|--------|
|               |      |                  |                  | Max                    | Min*                                       | Max*   |
| MCY-MAP0401HT | 4 HP | 12,1 kW          | 12,5 kW          | 6                      | 3,2 HP                                     | 5,2 HP |
| MCY-MAP0501HT | 5 HP | 14,0 kW          | 16,0 kW          | 8                      | 4,0 HP                                     | 6,5 HP |
| MCY-MAP0601HT | 6 HP | 15,5 kW          | 18,0 kW          | 9                      | 4,8 HP                                     | 7,8 HP |

Technical specifications **PMV kit**

|                                                                                     | Model Name   | Indoor unit capacity code                         |
|-------------------------------------------------------------------------------------|--------------|---------------------------------------------------|
|  | RBM-PMV0361E | 0,8 – 1 – 1,25                                    |
|                                                                                     | RBM-PMV0362E | For new High wall MMK-AP***3H - 0,8 – 1 – 1,25 HP |
|                                                                                     | RBM-PMV0901E | 1,7 – 2 – 2,5                                     |
|                                                                                     | RBM-PMV0902E | For new High Wall MMK-AP***3H - 1,7 – 2 – 2,5 HP  |

**R-410A****SUPER MODULAR MULTI  
SYSTEM****Dual Inverter VRF Heat  
Pump**

## SMMS VRF Outdoor unit

### Features

The 2-pipe VRF Super Modular Multi System (SMMS) operates with R-410A refrigerant and incorporates the inverter technology in all outdoor unit models.

In addition, the SMMS incorporates twin-inverter compressors in every outdoor unit. The capacities range from 14 to 135 kW in cooling mode and 16 to 150 kW in heating mode with a capability to serve up to 48 indoor units.

## Protection devices

- Discharge and suction temperature sensors
- Internal overload relay
- Compressor over current relay
- Over current sensor
- High pressure switch
- Low pressure sensors

### Key features

High COP: (4,25 with 22,4 kW size) for reduced energy consumption and increased savings.

Advanced compressor Oil Management System: it guarantees improved reliability.

TCC Link: state-of-the-art communication bus system with automatically configured addressing.

Up to 48 indoor units can be connected – Toshiba exclusivity.

Latest Inverter Technology with the Intelligent Power Drive Unit (IPDU).

Pipe runs up to 300 m: greater application flexibility.

Technical specifications **heat pump**

| Outdoor unit                                        | CO<br>HP                | MMY-<br>MMY- | MAP0501T8-E<br>MAP0501HT8-E<br>5 HP | MAP0601T8-E<br>MAP0601HT8-E<br>6 HP | MAP0801T8-E<br>MAP0801HT8-E<br>8 HP | MAP1001T8-E<br>MAP1001HT8-E<br>10 HP | MAP1201T8-E<br>MAP1201HT8-E<br>12 HP |
|-----------------------------------------------------|-------------------------|--------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------------------|--------------------------------------|
| Cooling capacity <sup>1</sup>                       | kW                      |              | 14                                  | 16                                  | 22,4                                | 28                                   | 33,5                                 |
| Power input                                         | kW                      | C            | 3,65                                | 4,64                                | 5,67                                | 7,67                                 | 11,92                                |
| EER                                                 | W/W                     |              | 3,84                                | 3,45                                | 3,95                                | 3,65                                 | 2,81                                 |
| Running current                                     | A                       | C            | 5,85                                | 7,28                                | 8,62                                | 11,55                                | 18,30                                |
| Heating capacity <sup>2</sup>                       | kW                      |              | 16                                  | 18                                  | 25                                  | 31,5                                 | 37,5                                 |
| Power input                                         | kW                      | H            | 3,84                                | 4,56                                | 5,88                                | 7,97                                 | 10,19                                |
| COP                                                 | W/W                     |              | 4,17                                | 3,95                                | 4,25                                | 3,95                                 | 3,68                                 |
| Running current                                     | A                       | H            | 6,09                                | 7,08                                | 8,93                                | 11,98                                | 15,65                                |
| Peak demand current <sup>3</sup>                    | A                       |              | 20                                  | 20                                  | 30                                  | 30                                   | 30                                   |
| Air flow                                            | m <sup>3</sup> /h – l/s |              | 9000 – 2520                         | 9000 – 2520                         | 9900 – 2750                         | 10500 – 2917                         | 10500 – 2917                         |
| Sound pressure level                                | dB(A)                   |              | 55                                  | 56                                  | 57                                  | 58                                   | 59                                   |
| Operating range – db                                | °C                      | C            | -5 ÷ 43                             | -5 ÷ 43                             | -5 ÷ 43                             | -5 ÷ 43                              | -5 ÷ 43                              |
| Operating range – wb <sup>4</sup>                   | °C                      | H            | -20 ÷ 16                            | -20 ÷ 16                            | -20 ÷ 16                            | -20 ÷ 16                             | -20 ÷ 16                             |
| Dimensions (h x w x d)                              | mm                      |              | 1800 x 990 x 750                    | 1800 x 990 x 750                    | 1800 x 990 x 750                    | 1800 x 990 x 750                     | 1800 x 990 x 750                     |
| Weight                                              | kg                      |              | 228                                 | 228                                 | 258                                 | 258                                  | 258                                  |
| Compressor type                                     |                         |              | Twin Rotary                         | Twin Rotary                         | Twin Rotary                         | Twin Rotary                          | Twin Rotary                          |
| Refrigerant charge R410A                            | kg                      |              | 8,5                                 | 8,5                                 | 11,8                                | 11,8                                 | 11,8                                 |
| Suction line type – diameter                        |                         |              | Flare – 5/8"                        | Brazed – 3/4"                       | Brazed – 7/8"                       | Brazed – 7/8"                        | Brazed – 1-1/8"                      |
| Liquid line type – diameter                         |                         |              | Flare – 3/8"                        | Flare – 3/8"                        | Flare – 1/2"                        | Flare – 1/2"                         | Flare – 1/2"                         |
| Discharge line connection type – diameter           |                         |              | Flare – 3/8"                        | Flare – 3/8"                        | Flare – 3/8"                        | Flare – 3/8"                         | Flare – 3/8"                         |
| Maximum equivalent length separation                | m                       |              | 175                                 | 175                                 | 175                                 | 175                                  | 175                                  |
| Maximum actual piping separation                    | m                       |              | 150                                 | 150                                 | 150                                 | 150                                  | 150                                  |
| Maximum pipe length                                 | m                       |              | 300                                 | 300                                 | 300                                 | 300                                  | 300                                  |
| Maximum lift (Indoor unit above/below) <sup>5</sup> | m                       |              | 40/50                               | 40/50                               | 40/50                               | 40/50                                | 40/50                                |
| Power supply                                        | V-ph-Hz                 |              | 400-3-50                            | 400-3-50                            | 400-3-50                            | 400-3-50                             | 400-3-50                             |

<sup>1</sup> based on an indoor air temperature of 27 °C db/19 °C wb and an outdoor air temperature of 35 °C db

<sup>2</sup> based on an indoor air temperature of 20 °C db and an outdoor air temperature of 7 °C db/6 °C wb

<sup>3</sup> if outdoor units are combined, refer to the installation manual

<sup>4</sup> the unit can be operated even if outdoor temperature gets down to -20 °C, however note that the warranty covers only up to -15 °C because operation beyond that temperature is out of specification.

When outdoor air temperature falls to under -15 °C, it may cause shortening the product lifetime

<sup>5</sup> if the height difference between indoor units exceeds 3 m and if the indoor unit is above, max. lift is reduced to 30 m

C = cooling mode

H = heating mode

Technical specifications **outdoor units**

|                                                                                     | Model Name       |       | Cooling capacity | Heating capacity | Outdoor units in combination              | Number of indoor units<br>Max |
|-------------------------------------------------------------------------------------|------------------|-------|------------------|------------------|-------------------------------------------|-------------------------------|
|  | MMY-MAP0501HT8-E | 5 HP  | 14 kW            | 16 kW            | 1                                         | 8                             |
|                                                                                     | MMY-MAP0601HT8-E | 6 HP  | 16 kW            | 18 kW            | 1                                         | 10                            |
|                                                                                     | MMY-MAP0801HT8-E | 8 HP  | 22,4 kW          | 25 kW            | 1                                         | 13                            |
|                                                                                     | MMY-MAP1001HT8-E | 10 HP | 28 kW            | 31,5 kW          | 1                                         | 16                            |
|                                                                                     | MMY-MAP1201HT8-E | 12 HP | 33,5 kW          | 37,5 kW          | 1                                         | 20                            |
|  | MMY-AP1401HT8-E  | 14 HP | 38,4 kW          | 43 kW            | 2 (22,4 kW + 16 kW)                       | 23                            |
|                                                                                     | MMY-AP1601HT8-E  | 16 HP | 45 kW            | 50 kW            | 2 (22,4 kW + 22,4 kW)                     | 27                            |
|                                                                                     | MMY-AP1801HT8-E  | 18 HP | 50,4 kW          | 56,5 kW          | 2 (28 kW + 22,4 kW)                       | 30                            |
|                                                                                     | MMY-AP2001HT8-E  | 20 HP | 56 kW            | 63 kW            | 2 (28 kW + 28 kW)                         | 33                            |
|                                                                                     | MMY-AP2201HT8-E  | 22 HP | 61,5 kW          | 69 kW            | 3 (22,4 kW + 22,4 kW + 16 kW)             | 37                            |
|                                                                                     | MMY-AP2211HT8-E  | 22 HP | 61,5 kW          | 69 kW            | 2 (33,5 kW + 28 kW)                       | 37                            |
|                                                                                     | MMY-AP2401HT8-E  | 24 HP | 68 kW            | 76,5 kW          | 3 (22,4 kW + 22,4 kW + 22,4 kW)           | 40                            |
|                                                                                     | MMY-AP2411HT8-E  | 24 HP | 68 kW            | 76,5 kW          | 2 (33,5 kW + 33,5 kW)                     | 40                            |
|  | MMY-AP2601HT8-E  | 26 HP | 73 kW            | 81,5 kW          | 3 (28 kW + 22,4 kW + 22,4 kW)             | 43                            |
|                                                                                     | MMY-AP2801HT8-E  | 28 HP | 78,5 kW          | 88 kW            | 3 (28 kW + 28 kW + 22,4 kW)               | 47                            |
|                                                                                     | MMY-AP3001HT8-E  | 30 HP | 84 kW            | 95 kW            | 3 (28 kW + 28 kW + 28 kW)                 | 48                            |
|                                                                                     | MMY-AP3201HT8-E  | 32 HP | 90 kW            | 100 kW           | 4 (22,4 kW + 22,4 kW + 22,4 kW + 22,4 kW) | 48                            |
|                                                                                     | MMY-AP3211HT8-E  | 32 HP | 90 kW            | 100 kW           | 3 (33,5 kW + 28 kW + 28 kW)               | 48                            |
|                                                                                     | MMY-AP3401HT8-E  | 34 HP | 96 kW            | 108 kW           | 4 (28 kW + 22,4 kW + 22,4 kW + 22,4 kW)   | 48                            |
|                                                                                     | MMY-AP3411HT8-E  | 34 HP | 96 kW            | 108 kW           | 3 (33,5 kW + 33,5 kW + 28 kW)             | 48                            |
|                                                                                     | MMY-AP3601HT8-E  | 36 HP | 101 kW           | 113 kW           | 4 (28 kW + 28 kW + 22,4 kW + 22,4 kW)     | 48                            |
|                                                                                     | MMY-AP3611HT8-E  | 36 HP | 101 kW           | 113 kW           | 3 (33,5 kW + 33,5 kW + 33,5 kW)           | 48                            |
|  | MMY-AP3801HT8-E  | 38 HP | 106,5 kW         | 119,5 kW         | 4 (28 kW + 28 kW + 28 kW + 22,4 kW)       | 48                            |
|                                                                                     | MMY-AP4001HT8-E  | 40 HP | 112 kW           | 126,5 kW         | 4 (28 kW + 28 kW + 28 kW + 28 kW)         | 48                            |
|                                                                                     | MMY-AP4201HT8-E  | 42 HP | 118 kW           | 132 kW           | 4 (33,5 kW + 28 kW + 28 kW + 28 kW)       | 48                            |
|                                                                                     | MMY-AP4401HT8-E  | 44 HP | 123,5 kW         | 138 kW           | 4 (33,5 kW + 33,5 kW + 28 kW + 28 kW)     | 48                            |
|                                                                                     | MMY-AP4601HT8-E  | 46 HP | 130 kW           | 145 kW           | 4 (33,5 kW + 33,5 kW + 33,5 kW + 28 kW)   | 48                            |
|                                                                                     | MMY-AP4801HT8-E  | 48 HP | 135 kW           | 150 kW           | 4 (33,5 kW + 33,5 kW + 33,5 kW + 33,5 kW) | 48                            |

**R-410A****SUPER HEAT RECOVERY MULTI  
SYSTEM****Dual Inverter heat  
recovery VRF**

## SHRM VRF Outdoor unit

### Features

The three-pipe VRF Super Heat Recovery Multi System (SHRM) delivers simultaneously cooling and heating and has exceptional energy efficiency.

### Key features

Unbeatable energy consumption efficiency: average COP of 3,97 (22,4 kW).

Compact flow selector unit: it automatically adjusts the temperature either by unit or by area.

Piping branch flexibility: the three-pipe connection allows installation height variation of 35 m (equivalent to a 9-story building).

Active Oil Management system: it increases the operation reliability.

Wide control applications: Artificial Intelligence network system available and Building Management System (BMS) compatible.

## Protection devices

- Discharge and suction temperature sensors
- Internal overload relay
- Compressor over current relay
- Over current sensor
- High pressure switch
- Low pressure sensors



Technical specifications **heat pump**

| Outdoor unit                                        |                         |   | MMY-MAP0802FT8-E | MMY-MAP1002FT8-E | MMY-MAP1202FT8-E  |
|-----------------------------------------------------|-------------------------|---|------------------|------------------|-------------------|
|                                                     |                         |   | 8 HP             | 10 HP            | 12 HP             |
| Cooling capacity <sup>1</sup>                       | kW                      |   | 22,4             | 28               | 33,5              |
| Power input                                         | kW                      | C | 6,07             | 8,54             | 12,9              |
| EER                                                 | W/W                     |   | 3,69             | 3,18             | 2,6               |
| Running current                                     | A                       | C | 9,25             | 13,15            | 19,85             |
| Heating capacity <sup>2</sup>                       | kW                      |   | 25               | 31,5             | 35,5              |
| Power input                                         | kW                      | H | 6,29             | 8,73             | 9,65              |
| COP                                                 | W/W                     |   | 3,97             | 3,61             | 3,68              |
| Running current                                     | A                       | H | 9,55             | 13,4             | 14,85             |
| Peak demand current <sup>3</sup>                    | A                       |   | 30               | 30               | 30                |
| Air flow                                            | m <sup>3</sup> /h – l/s |   | 9900 – 2750      | 10500 – 2917     | 10500 – 2917      |
| Sound pressure level – at 1 m                       | dB(A)                   |   | 57               | 58               | 59                |
| Operating range – db                                | °C                      | C | -10 ÷ 43         | -10 ÷ 43         | -10 ÷ 43          |
| Operating range – wb <sup>4</sup>                   | °C                      | H | -20 ÷ 16         | -20 ÷ 16         | -20 ÷ 16          |
| Dimensions (h x w x d)                              | mm                      |   | 1800 x 990 x 750 | 1800 x 990 x 750 | 1800 x 990 x 750  |
| Weight                                              | kg                      |   | 263              | 263              | 263               |
| Compressor type                                     |                         |   | Twin Rotary      | Twin Rotary      | Twin Rotary       |
| Refrigerant charge R410A                            | kg                      |   | 11,5             | 11,5             | 11,5              |
| Suction line type – diameter                        |                         |   | Brazed – 7/8"    | Brazed – 7/8"    | Brazed – 1 - 1/8" |
| Liquid line type – diameter                         |                         |   | Flare – 1/2"     | Flare – 1/2"     | Flare – 1/2"      |
| Discharge line connection type – diameter           |                         |   | Brazed – 3/4"    | Brazed – 3/4"    | Brazed – 3/4"     |
| Maximum equivalent length separation                | m                       |   | 150              | 150              | 150               |
| Maximum actual piping separation                    | m                       |   | 125              | 125              | 125               |
| Maximum pipe length                                 | m                       |   | 300              | 300              | 300               |
| Maximum lift (Indoor unit above/below) <sup>5</sup> | m                       |   | 30/50            | 30/50            | 30/50             |
| Power supply                                        | V-ph-Hz                 |   | 400-3-50         | 400-3-50         | 400-3-50          |

<sup>1</sup> based on an indoor air temperature of 27 °C db/19 °C wb and an outdoor air temperature of 35 °C db

<sup>2</sup> based on an indoor air temperature of 20 °C db and an outdoor air temperature of 7 °C db/6 °C wb

<sup>3</sup> if outdoor units are combined, refer to the installation manual

<sup>4</sup> the unit can be operated even if outdoor temperature gets down to -20 °C, however note that the warranty covers only up to -15 °C because operation beyond that temperature is out of specification. When outdoor air temperature falls to under -15 °C, it may cause shortening the product lifetime

<sup>5</sup> if the height difference between indoor units exceeds 3 m and if the indoor unit is above, max. lift is reduced to 30 m

C = cooling mode

H = heating mode

Technical specifications **outdoor units**

|                                                                                     | Model Name       |       | Cooling capacity | Heating capacity | Outdoor units in combination    | Number of indoor units<br>Max | Total cap. of connectable indoor units |         |
|-------------------------------------------------------------------------------------|------------------|-------|------------------|------------------|---------------------------------|-------------------------------|----------------------------------------|---------|
|                                                                                     |                  |       |                  |                  |                                 |                               | Min                                    | Max     |
|  | MMY-MAP0802FT8-E | 8 HP  | 22,4 kW          | 25 kW            | 1                               | 13                            | 5,6 HP                                 | 10,8 HP |
|                                                                                     | MMY-MAP1002FT8-E | 10 HP | 28 kW            | 31,5 kW          | 1                               | 16                            | 7 HP                                   | 13,5 HP |
|                                                                                     | MMY-MAP1202FT8-E | 12 HP | 33,5 kW          | 35,5 kW          | 1                               | 16                            | 8,4 HP                                 | 14,4 HP |
|  | MMY-AP1602FT8-E  | 16 HP | 45 kW            | 50 kW            | 2 (22,4 kW + 22,4 kW)           | 27                            | 11,2 HP                                | 21,6 HP |
|                                                                                     | MMY-AP1802FT8-E  | 18 HP | 50,4 kW          | 56,5 kW          | 2 (22,4 kW + 28 kW)             | 30                            | 12,6 HP                                | 24,3 HP |
|                                                                                     | MMY-AP2002FT8-E  | 20 HP | 56 kW            | 63 kW            | 2 (28 kW+28 kW)                 | 33                            | 14 HP                                  | 27 HP   |
|   | MMY-AP2402FT8-E  | 24 HP | 68 kW            | 76,5 kW          | 3 (22,4 kW + 22,4 kW + 22,4 kW) | 40                            | 16,8 HP                                | 32,4 HP |
|                                                                                     | MMY-AP2602FT8-E  | 26 HP | 73 kW            | 81,5 kW          | 3 (22,4 kW + 22,4 kW + 28 kW)   | 43                            | 18,2 HP                                | 35,1 HP |
|                                                                                     | MMY-AP2802FT8-E  | 28 HP | 78,5 kW          | 88 kW            | 3 (22,4 kW + 28 kW + 28 kW)     | 47                            | 19,6 HP                                | 37,8 HP |
|                                                                                     | MMY-AP3002FT8-E  | 30 HP | 84 kW            | 95 kW            | 3 (28 kW + 28 kW + 28kW)        | 48                            | 21 HP                                  | 40,5 HP |



## Compact 4-way cassette

This 4-way cassette has been designed to suit all standard 600 × 600 mm grid ceilings, for easy installation and maintenance. Its compact design blends with any room interior, where appearance is as important as functionality.

- STANDARD 600 × 600 mm GRID
- EASY ACCESS TO CONTROL, VIA CORNER POCKETS
- FRESH AIR INTRODUCTION POSSIBILITY



## 4-way cassette

Unobtrusive and flexible, it blends in with any room interior, and offers the ideal solution for small commercial applications where space is limited.

- CLEAN CEILING
- CLEAN UNIT: WASHABLE LOUVRE AND GRILLE
- FLEXIBLE INSTALLATION



## 2-way cassette

Discreet and easy to install, it fits any room interior. Thanks to its silent operation, it creates a very quiet and comfortable environment.

- SLIM DESIGN
- LOW NOISE LEVEL
- ENHANCED INDOOR AIR QUALITY



## 1-way cassette

It's simple to install and is suitable for small areas, such as hotels or offices.

- COMPACT HI-TECH DESIGN
- LOW NOISE LEVEL
- EASY INSTALLATION

## Standard ducted unit



It can easily be installed in ceiling voids or false ceiling and operates very quietly. Whatever the shape of the room could be, this flexible unit ensures uniform temperatures in it.

- COMPACT DESIGN
- UNIFORM AIR DISTRIBUTION
- QUIET OPERATION

## Slim duct



It offers the ultimate technology with exceptional energy savings, high performance and easy installation. Ideal for hotel rooms.

- UNOBTRUSIVE DESIGN
- LOW NOISE LEVEL
- FLEXIBLE INSTALLATION

## High static pressure ducted unit



It's Toshiba most powerful ducted unit. Flexible and compact, it can be installed discretely in any interior.

- WIDE RANGE OF OPTIONS AVAILABLE
- EASY MAINTENANCE
- EASY INSTALLATION

## Ceiling Suspended unit



Thanks to its simple suspension, the installation is very easy. It creates a very pleasing environment, diffusing rapidly and uniformly the required temperature.

- OPTIMUM LOUVRE CONTROL
- PIPING FLEXIBILITY
- COMPACT DIMENSIONS

## Compact high-wall



It's compact and lightweight, perfect for modest spaces, such as offices, shops and hotel rooms.

- COMPACT AND STYLISH DESIGN
- LOW NOISE LEVEL
- AUTO-SWING MECHANISM

## High-wall



NEW

With its attractive and slim-line design, it compliments any room interior.

- REDUCED DIMENSIONS
- PIPING FLEXIBILITY
- OPTIMUM AIR DISTRIBUTION

## Floor mounted console



It's suitable for refurbishment projects of small spaces.

- WIDE CHOICE OF INSTALLATION SETTINGS
- PIPING FLEXIBILITY
- COMPACT DIMENSIONS

## Concealed chassis unit



It's the perfect solution for perimeter walls and can be hidden behind a decorative panel to blend with any room interior.

- VERY COMPACT DESIGN
- UNOBTRUSIVE OPERATION
- EASY MAINTENANCE

## Floor standing



It's particularly suitable for large rooms with low ceilings. The units offer high air flow rates and superior air throw values.  
Ideal for refurbishment.

- HIGH AIR FLOWS
- WIDE AIR DISTRIBUTION ANGLE
- REDUCED DIMENSIONS

## Fresh Air Intake



It offers the possibility to introduce in the building pre-conditioned external fresh air. Ideal solution for school, hospital, offices and all the buildings that require a fresh air ventilation without any further exclusive system.

- EXTERNAL STATIC PRESSURE AVAILABLE UP TO 230 Pa
- COMPACT DIMENSION
- TCC-LINK CONTROL CONNECTION

Technical specifications **heat pump**

| Model Type                                                                                                                  | Model Name                                                                                       | Mini-SMMS     | SMMS and SHRM | Capacity Code | Cooling cap. (kW) | Heating cap. (kW) | Height (mm) | Width (mm) | Depth (mm) | Weight (kg) |     |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------|---------------|---------------|-------------------|-------------------|-------------|------------|------------|-------------|-----|
| <br>4-way cassette                         | MMU-AP0092H                                                                                      | ●             | ●             | 1,00          | 2,80              | 3,20              | 256         | 840        | 840        | 18          |     |
|                                                                                                                             | MMU-AP0122H                                                                                      | ●             | ●             | 1,25          | 3,60              | 4,00              |             |            |            | 20          |     |
|                                                                                                                             | MMU-AP0152H                                                                                      | ●             | ●             | 1,70          | 4,50              | 5,00              |             |            |            | 20          |     |
|                                                                                                                             | MMU-AP0182H                                                                                      | ●             | ●             | 2,00          | 5,60              | 6,30              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0242H                                                                                      | ●             | ●             | 2,50          | 7,10              | 8,00              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0272H                                                                                      | ●             | ●             | 3,00          | 8,00              | 9,00              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0302H                                                                                      | ●             | ●             | 3,20          | 9,00              | 10,00             |             |            |            |             |     |
|                                                                                                                             | MMU-AP0362H                                                                                      | ●             | ●             | 4,00          | 11,20             | 12,50             | 319         | 840        | 840        | 25          |     |
| MMU-AP0482H                                                                                                                 | ●                                                                                                | ●             | 5,00          | 14,00         | 16,00             |                   |             |            |            |             |     |
| MMU-AP0562H                                                                                                                 | ●                                                                                                | ●             | 6,00          | 16,00         | 18,00             |                   |             |            |            |             |     |
| <br>Compact 4-way cassette                 | MMU-AP0071MH                                                                                     | ●             | ●             | 0,80          | 2,20              | 2,50              | 268         | 575        | 575        | 17          |     |
|                                                                                                                             | MMU-AP0091MH                                                                                     | ●             | ●             | 1,00          | 2,80              | 3,20              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0121MH                                                                                     | ●             | ●             | 1,25          | 3,60              | 4,00              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0151MH                                                                                     | ●             | ●             | 1,70          | 4,50              | 5,00              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0181MH                                                                                     | ●             | ●             | 2,00          | 5,60              | 6,30              |             |            |            |             |     |
| <br>2-way cassette                         | MMU-AP0071WH                                                                                     | ●             | ●             | 0,80          | 2,20              | 2,50              | 398         | 830        | 550        | 33          |     |
|                                                                                                                             | MMU-AP0091WH                                                                                     | ●             | ●             | 1,00          | 2,80              | 3,20              |             | 1350       |            | 44          |     |
|                                                                                                                             | MMU-AP0121WH                                                                                     | ●             | ●             | 1,25          | 3,60              | 4,00              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0151WH                                                                                     | ●             | ●             | 1,70          | 4,50              | 5,00              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0181WH                                                                                     | ●             | ●             | 2,00          | 5,60              | 6,30              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0241WH                                                                                     | ●             | ●             | 2,50          | 7,10              | 8,00              |             |            |            |             | 48  |
|                                                                                                                             | MMU-AP0271WH                                                                                     | ●             | ●             | 3,00          | 8,00              | 9,00              |             |            |            |             |     |
| MMU-AP0301WH                                                                                                                | ●                                                                                                | ●             | 3,20          | 9,00          | 10,00             |                   |             |            |            |             |     |
| <br>1-way cassette                         | MMU-AP0071YH                                                                                     | ●             | ●             | 0,80          | 2,20              | 2,50              | 235         | 850        | 400        | 22          |     |
|                                                                                                                             | MMU-AP0091YH                                                                                     | ●             | ●             | 1,00          | 2,80              | 3,20              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0121YH                                                                                     | ●             | ●             | 1,25          | 3,60              | 4,00              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0152SH                                                                                     | ●             | ●             | 1,70          | 4,50              | 5,00              | 200         | 1000       | 710        | 21          |     |
|                                                                                                                             | MMU-AP0182SH                                                                                     | ●             | ●             | 2,00          | 5,60              | 6,30              |             |            |            |             |     |
|                                                                                                                             | MMU-AP0242SH                                                                                     | ●             | ●             | 2,50          | 7,10              | 8,00              |             |            |            |             |     |
| <br>Concealed duct, stand type            | MMD-AP0071BH                                                                                     | ●             | ●             | 0,80          | 2,20              | 2,50              | 320         | 550        | 800        | 27          |     |
|                                                                                                                             | MMD-AP0091BH                                                                                     | ●             | ●             | 1,00          | 2,80              | 3,20              |             | 700        |            |             | 30  |
|                                                                                                                             | MMD-AP0121BH                                                                                     | ●             | ●             | 1,25          | 3,60              | 4,00              |             |            |            |             |     |
|                                                                                                                             | MMD-AP0151BH                                                                                     | ●             | ●             | 1,70          | 4,50              | 5,00              |             | 1000       |            | 39          |     |
|                                                                                                                             | MMD-AP0181BH                                                                                     | ●             | ●             | 2,00          | 5,60              | 6,30              |             |            |            |             |     |
|                                                                                                                             | MMD-AP0241BH                                                                                     | ●             | ●             | 2,50          | 7,10              | 8,00              |             |            |            |             |     |
|                                                                                                                             | MMD-AP0271BH                                                                                     | ●             | ●             | 3,00          | 8,00              | 9,00              |             |            |            |             |     |
|                                                                                                                             | MMD-AP0301BH                                                                                     | ●             | ●             | 3,20          | 9,00              | 10,00             |             |            |            |             |     |
|                                                                                                                             | MMD-AP0361BH                                                                                     | ●             | ●             | 4,00          | 11,20             | 12,50             |             | 1350       |            | 51          |     |
|                                                                                                                             | MMD-AP0481BH                                                                                     | ●             | ●             | 5,00          | 14,00             | 16,00             |             |            |            |             |     |
| MMD-AP0561BH                                                                                                                | ●                                                                                                | ●             | 6,00          | 16,00         | 18,00             |                   |             |            |            |             |     |
| <br>Concealed duct, high static pressure | MMD-AP0181H                                                                                      | ●             | ●             | 2,00          | 5,60              | 6,30              | 380         | 850        | 660        | 50          |     |
|                                                                                                                             | MMD-AP0241H                                                                                      | ●             | ●             | 2,50          | 7,10              | 8,00              |             |            |            | 52          |     |
|                                                                                                                             | MMD-AP0271H                                                                                      | ●             | ●             | 3,00          | 8,00              | 9,00              |             |            |            | 56          |     |
|                                                                                                                             | MMD-AP0361H                                                                                      | ●             | ●             | 4,00          | 11,20             | 12,50             |             |            |            | 67          |     |
|                                                                                                                             | MMD-AP0481H                                                                                      | ●             | ●             | 5,00          | 14,00             | 16,00             |             | 1200       |            | 1250        | 155 |
|                                                                                                                             | MMD-AP0721H                                                                                      | ●             | ●             | 8,00          | 22,40             | 25,00             |             |            |            |             |     |
|                                                                                                                             | MMD-AP0961H                                                                                      | ●             | ●             | 10,00         | 28,00             | 31,50             |             |            |            |             |     |
|                                                                                                                             | <br>Slim Duct | MMD-AP0071SPH | ●             | ●             | 0,80              | 2,20              |             | 2,50       |            |             |     |
| MMD-AP0091SPH                                                                                                               |                                                                                                  | ●             | ●             | 1,00          | 2,80              | 3,20              | 23          |            |            |             |     |
| MMD-AP0121SPH                                                                                                               |                                                                                                  | ●             | ●             | 1,25          | 3,60              | 4,00              |             |            |            |             |     |
| MMD-AP0151SPH                                                                                                               |                                                                                                  | ●             | ●             | 1,70          | 4,50              | 5,00              |             |            |            |             |     |
| MMD-AP0181SPH                                                                                                               |                                                                                                  | ●             | ●             | 2,00          | 5,60              | 6,30              |             |            |            |             |     |
| <br>Under-ceiling                        |                                                                                                  | MMC-AP0151H   | ●             | ●             | 1,70              | 4,50              |             | 5,00       | 210        | 910         | 680 |
|                                                                                                                             | MMC-AP0181H                                                                                      | ●             | ●             | 2,00          | 5,60              | 6,30              | 1180        | 25         |            |             |     |
|                                                                                                                             | MMC-AP0241H                                                                                      | ●             | ●             | 2,50          | 7,10              | 8,00              |             |            |            |             |     |
|                                                                                                                             | MMC-AP0271H                                                                                      | ●             | ●             | 3,00          | 8,00              | 9,00              | 1595        | 33         |            |             |     |
|                                                                                                                             | MMC-AP0361H                                                                                      | ●             | ●             | 4,00          | 11,20             | 12,50             |             |            |            |             |     |
|                                                                                                                             | MMC-AP0481H                                                                                      | ●             | ●             | 5,00          | 14,00             | 16,00             |             |            |            |             |     |
| <br>High-wall compact                    | MMK-AP0072H                                                                                      | ●             | ●             | 0,80          | 2,20              | 2,50              | 275         | 790        | 208        | 11          |     |
|                                                                                                                             | MMK-AP0092H                                                                                      | ●             | ●             | 1,00          | 2,80              | 3,20              |             |            |            |             |     |
|                                                                                                                             | MMK-AP0122H                                                                                      | ●             | ●             | 1,25          | 3,60              | 4,00              |             |            |            |             |     |
| <br>High-wall                            | MMK-AP0073H                                                                                      | ●             | ●             | 0,80          | 2,20              | 2,50              | 320         | 1050       | 228        | 15          |     |
|                                                                                                                             | MMK-AP0093H                                                                                      | ●             | ●             | 1,00          | 2,80              | 3,20              |             |            |            |             |     |
|                                                                                                                             | MMK-AP0123H                                                                                      | ●             | ●             | 1,25          | 3,60              | 4,00              |             |            |            |             |     |
|                                                                                                                             | MMK-AP0153H                                                                                      | ●             | ●             | 1,70          | 4,50              | 5,00              |             |            |            |             |     |
|                                                                                                                             | MMK-AP0183H                                                                                      | ●             | ●             | 2,00          | 5,60              | 6,30              |             |            |            |             |     |
|                                                                                                                             | MMK-AP0243H                                                                                      | ●             | ●             | 2,50          | 7,10              | 8,00              |             |            |            |             |     |
| <br>Floor standing cabinet type          | MML-AP0071H                                                                                      | ●             | ●             | 0,80          | 2,20              | 2,50              | 630         | 950        | 230        | 37          |     |
|                                                                                                                             | MML-AP0091H                                                                                      | ●             | ●             | 1,00          | 2,80              | 3,20              |             |            |            | 40          |     |
|                                                                                                                             | MML-AP0121H                                                                                      | ●             | ●             | 1,25          | 3,60              | 4,00              |             |            |            |             |     |
|                                                                                                                             | MML-AP0151H                                                                                      | ●             | ●             | 1,70          | 4,50              | 5,00              |             |            |            |             |     |
|                                                                                                                             | MML-AP0181H                                                                                      | ●             | ●             | 2,00          | 5,60              | 6,30              |             |            |            |             |     |
|                                                                                                                             | MML-AP0241H                                                                                      | ●             | ●             | 2,50          | 7,10              | 8,00              |             |            |            |             |     |
| <br>Floor standing Concealed type        | MML-AP0071BH                                                                                     | ●             | ●             | 0,80          | 2,20              | 2,50              | 600         | 745        | 220        | 21          |     |
|                                                                                                                             | MML-AP0091BH                                                                                     | ●             | ●             | 1,00          | 2,80              | 3,20              |             | 1045       |            | 29          |     |
|                                                                                                                             | MML-AP0121BH                                                                                     | ●             | ●             | 1,25          | 3,60              | 4,00              |             |            |            |             |     |
|                                                                                                                             | MML-AP0151BH                                                                                     | ●             | ●             | 1,70          | 4,50              | 5,00              |             |            |            |             |     |
|                                                                                                                             | MML-AP0181BH                                                                                     | ●             | ●             | 2,00          | 5,60              | 6,30              |             |            |            |             |     |
|                                                                                                                             | MML-AP0241BH                                                                                     | ●             | ●             | 2,50          | 7,10              | 8,00              |             |            |            |             |     |
| <br>Tall floor-standing                  | MMF-AP0151H                                                                                      | ●             | ●             | 1,70          | 4,50              | 5,00              | 1750        | 600        | 210        | 48          |     |
|                                                                                                                             | MMF-AP0181H                                                                                      | ●             | ●             | 2,00          | 5,60              | 6,30              |             |            |            | 49          |     |
|                                                                                                                             | MMF-AP0241H                                                                                      | ●             | ●             | 2,50          | 7,10              | 8,00              |             |            |            |             |     |
|                                                                                                                             | MMF-AP0271H                                                                                      | ●             | ●             | 3,00          | 8,00              | 9,00              |             |            | 390        | 65          |     |
|                                                                                                                             | MMF-AP0361H                                                                                      | ●             | ●             | 4,00          | 11,20             | 12,50             |             |            |            |             |     |
|                                                                                                                             | MMF-AP0481H                                                                                      | ●             | ●             | 5,00          | 14,00             | 16,00             |             |            |            |             |     |
|                                                                                                                             | MMF-AP0561H                                                                                      | ●             | ●             | 6,00          | 16,00             | 18,00             |             |            |            |             |     |
| <br>Fresh Air Intake                     | MMD-AP0481HFE                                                                                    | ●             | ●             | 5,00          | 14,00             | 8,90              | 492         | 892        | 1292       | 93          |     |
|                                                                                                                             | MMD-AP0721HFE                                                                                    | ●             | ●             | 8,00          | 22,40             | 13,90             |             | 1392       |            | 144         |     |
|                                                                                                                             | MMD-AP0961HFF                                                                                    | ●             | ●             | 10,00         | 28,00             | 17,40             |             |            |            |             |     |



**FRESH AIR VENTILATION AND  
HEAT RECOVERY UNIT**

**COMPATIBLE WITH LIGHT  
COMMERCIAL AND VRF  
SYSTEMS**



## Air-to-Air Heat Exchangers

### Features

The air-to-air heat exchangers can be integrated with the air conditioning system.

They use exhaust air to pre-condition the incoming air, thus reducing the cooling or heating load and the overall size of the required air conditioning system.

A range of electric heaters is available, together with controls that enable integration with both Light Commercial and VRF indoor units.

### Key features

5 models available with air flow ranges from 70 to 280 l/s (250 ÷ 1000 m<sup>3</sup>/h).

Fresh air ventilation: increasingly required in rooms with no window access.

Temperature and humidity: changed by the entering fresh air.

Improved energy efficiency, particularly during extremes of heat and cold.

Recovers up to 75% heat from exhaust air.

**High efficiency**

**Wide range**

**Electric heaters (optional)**

Technical specifications **heat pump**

| Model                                 |                         | VN-250TE        | VN-350TE        | VN-500TE          | VN-800TE          | VN-1KTAE           |
|---------------------------------------|-------------------------|-----------------|-----------------|-------------------|-------------------|--------------------|
| Air flow (h/l)                        | m <sup>3</sup> /h – l/s | 250/170 – 70/48 | 350/280 – 98/78 | 500/370 – 140/104 | 800/650 – 224/182 | 1000/810 – 218/227 |
| Temperature exchange efficiency (h/l) | %                       | 75/77           | 75/77           | 75/77             | 75/77             | 75/77              |
| Heat reclaim mode (h/l)               | dB(A)                   | 28/21           | 32/26           | 34/25             | 39/32             | 38,5/31            |
| Bypass mode (h/l)                     | dB(A)                   | 28/22,5         | 32/26           | 34/26,5           | 38,5/33           | 39/31,5            |
| Operating range                       | °C                      | –10 ÷ 40        | –10 ÷ 40        | –10 ÷ 40          | –10 ÷ 40          | –10 ÷ 40           |
| Power Input (h/l)                     |                         |                 |                 |                   |                   |                    |
| Heat reclaim mode                     | W                       | 119/79          | 154/117         | 214/151           | 347/302           | 445/332            |
| Bypass mode                           | W                       | 119/79          | 151/113         | 210/145           | 337/297           | 438/326            |
| Enthalpy exchange efficiency (h/l)    |                         |                 |                 |                   |                   |                    |
| Heating                               | %                       | 70/73           | 69/71           | 67/71             | 71/74             | 71/73              |
| Cooling                               | %                       | 66/63           | 69/66           | 67/62             | 68/65             | 68/65              |
| Max. external static pressure (h/l)   | Pa                      | 90/37           | 95/42           | 105/38            | 140/70            | 90/35              |
| Dimensions (h × w × d)                | mm                      | 270 × 599 × 882 | 270 × 804 × 882 | 270 × 904 × 962   | 388 × 884 × 1322  | 388 × 1134 × 1322  |
| Weight                                | kg                      | 29              | 37              | 43                | 71                | 83                 |
| Duct diameter                         | mm                      | 150             | 150             | 200               | 250               | 250                |
| Filtration efficiency grade (EU3)     | %                       | 82              | 82              | 82                | 82                | 82                 |
| Power supply                          | V-ph-Hz                 | 220/240-1-50    | 220/240-1-50    | 220/240-1-50      | 220/240-1-50      | 220/240-1-50       |
| Maximum relative humidity             | %                       | 85              | 85              | 85                | 85                | 85                 |

## With Toshiba everything is easier

Toshiba's commitment to the development of technological and innovative products with improved performances is complemented

by a responsibility to supply more sophisticated and functional tools for the design, installation and control of these systems.

## Selection Software: everything at the click of a button

Sophisticated system design software has been developed for the whole Mini-SMMS, SMMS and SHRM range and is a useful and irreplaceable support tool for engineers, architects, installers and, in general, for anyone who wants to apply innovative Toshiba solutions.

With this software, the user can create a complete VRF system by simply

clicking on the icons for the indoor units and the other connection components. It is also possible to define, in advance, relevant parameters such as outside and inside temperatures, fan speed, pipe system length and routing etc.

The software automatically manages all the parameters entered, and the actual system capacity for the

conditions required can be quickly calculated and simulated during the design stage. Using this software, the design of VRF systems is guaranteed for the project at the given conditions.

The software constantly monitors possible design errors and warns the user, when it reaches the system limits.

Graphical representation of the required pipe connection system and pipe sizing.

Specific details and data of the system selected: heating capacity, sensible and total cooling capacity, actual cooling capacity, additional refrigerant charge and pricing indications.

Multiple system management as a single project.

Export function to transfer the project report

using standard Microsoft® Word® and Adobe® Acrobat® (PDF). The images can also be exported to an AutoCAD® (DXF) file.

Automatic regeneration when adding or amending an existing project selection.

Indoor unit fan speed indication (high/medium/low) on the system report.



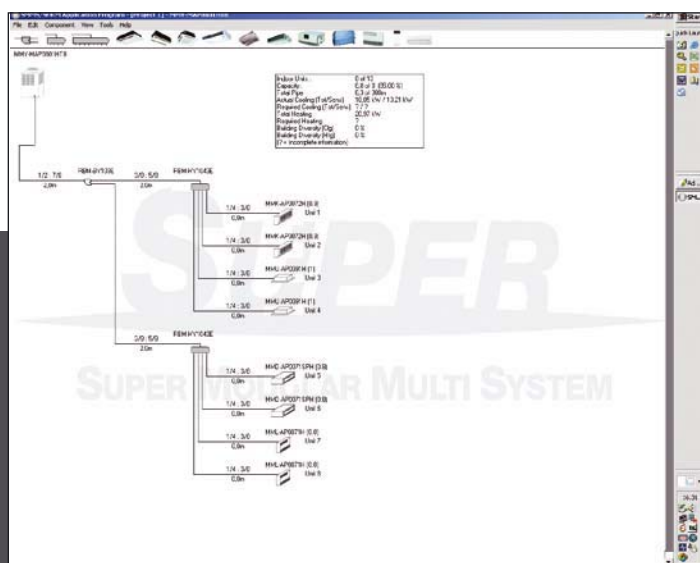
# Diagnostic software

The correct operation of sophisticated systems such as VRF is important to the long-term reliability of the system.

In order to assist with the correct commissioning of MiNi-SMMS, SMMS and SHRM systems, Toshiba has developed a diagnostic software programme – a valuable tool for the commissioning and service engineer.

The engineer can connect to the VRF system using a dedicated interface – enabling the download of all operating parameters and providing the engineer with detailed information for instant analysis or record.

Diagnostic software is distributed exclusively by the Toshiba EMEA RLC Technical Department.



Selection software screenshots









# Controls

Light commercial and VRF control options.

# Technology is nothing without control

An innovative and complete range of integrated controls for application in the Toshiba VRF MiNi-SMMS, SMMS, SHRM systems ensures maximum comfort and excellent performance

by perfectly matching the different control requirements. The range is composed of three control types: local, central and network controls.

Compact design and minimised installation space

Simplified display using icons

Automatic network addressing

TCC-Link connections with non-polarised wiring

## Local control systems

The wired local controller RBC-AMT32E (or simplified model RBC-AS21E2) can monitor a single unit or a group of a maximum of eight indoor units.

It offers the following functions: start/stop, operating mode change, temperature and fan speed adjustment, timer, auto-diagnostics and fault code display.

To define a timer scheduled for each day of the week, the schedule timer TBC-EXS21TLE can be used with a single local or central controller.

You can also simply use the complete remote controller with weekly timer, RBC-AMS41E.

To facilitate application flexibility, a range of wireless controls is also available (TBC-AX21E2, RBC-AX22CE2, RBC-AX31U(W/WS)E) to manage the main control functions.



## Central control systems

With the Toshiba central controller TCB-SC642TLE2, up to 64 indoor units can be monitored individually.

The central Toshiba controllers are compact and user-friendly and can also be combined with local controls and a weekly timer to guarantee

optimised user comfort under any conditions.



## Toshiba network solutions

Toshiba offers precise control of new VRF systems in both stand-alone applications for autonomous monitoring of the air conditioning system, and integrated into a central control scheme together with Super

Digital Inverter and Digital Inverter split systems. The innovative solutions of the Toshiba Network guarantee maximised integration with other building systems such as elevators, fire protection systems, lighting etc.

Open-network controls are specifically designed for Building Management Systems.

LonWorks Interface

BACnet System

Windows™ gateway

Touch screen control saver

# Toshiba units have everything under control



## Wired Control

RBC-AMT31E / RBC-AMT32E

The standard remote controller can control an individual indoor unit or a group of 8 indoor units. The remote control allows the operating parameters to be set for the indoor unit. It also allows faults to be displayed and unit configurations to be set up. The weekly timer can be fitted to this remote control.



## Simplified Control

RBC-AS21E2

The simplified remote controller is connected in the same way as the standard remote controller, but offers reduced functionality. The remote controller does not have the lapse timer and the ability to set up the indoor unit. Unit fault codes are still displayed.



## Central Controller

TCB-SC642TLE2

The central controller can control up to 64 indoor units individually. All their functions can be controlled. Malfunction checks are available for each indoor unit. This controller can also connect to the weekly timer and has volt-free inputs to enable the indoor units and indicate a fault. It has the ability to shut down all units in the event of a fire. Up to four controllers can be connected to the network.



## IR Remote Control

TCB-AX21E2

The wireless remote controller can be used with the appropriate indoor units to give full control of the indoor units. Two remote controllers can be used on the same indoor unit. The sensor on the remote controller can be used to control the system. Fault codes are displayed.



## Remote controller with weekly timer (7-day timer function)

RBC-AMS41E

Clock display

Schedule timer: possible to program schedule timer (7-day timer) function, or to set 8 different functions for each day of the week

Wide range of programs: Operation time, Operation start/stop, Operation mode, Temperature setting, Restriction on button operation



## Schedule Timer

TCB-EXS21TLE

Operating in weekly timer mode and schedule timer mode

6 programming per day

Enabling 8 groups to be programmed

100 hrs max back up power supply

7 types of weekly schedule and 3 programming per day (weekly timer mode): combination of operation ON and OFF with a timer that can be set by a minute unit

Holidays special program



## VRF, DI, SDI Control Options

| Model number    | Reference                             | Description                                                                                                               | Used with                                                                                  |
|-----------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| RBC-AMT31E      | Wired Remote Controller               | Main wired remote controller                                                                                              | All indoor units (Excluding DI Flexi type)                                                 |
| RBC-AMT32E      | Wired Remote Controller               | Main wired remote controller                                                                                              | All indoor units (Excluding DI Flexi type)                                                 |
| RBC-AS21E2      | Simplified Wired Remote Controller    | As above but designed for hotel and domestic applications                                                                 | All indoor units (Excluding DI Flexi type)                                                 |
| TCB-EXS21TLE    | Schedule timer                        | Operating in weekly timer mode or schedule timer mode                                                                     | All indoor units (Excluding DI Flexi type)                                                 |
| RBC-AMS41E      | Remote controller with schedule timer | Enables to control indoor unit operation with schedule timer (7-days) allowing to program 8 functions/day + clock display | VRF, SDI, DI (except flexi) indoor units                                                   |
| TCB-AX21U(W)-E2 | Infra-red Remote Kit                  | Wireless remote controller                                                                                                | 4-way cassette                                                                             |
| RBC-AX22CE2     | Infra-red Remote Kit                  | Wireless remote controller                                                                                                | All ceiling units and one-way cassettes (SH2 series)                                       |
| TCB-AX21E2      | Infra-red Remote Kit                  | Wireless remote controller                                                                                                | All other units (including compact 4-way cassette)                                         |
| RBC-AX31U(W)-E  | Wireless remote unit kit              | Wireless remote unit kit for 4-way cassette                                                                               | MMU-AP***2H & RAW-SM***4UT-E with RBC-U31PG(W)-E & RBC-U31PGS(W)-E panels                  |
| RBC-AX31U(W)-E  | Wireless remote unit kit              | Wireless remote unit kit for 4-way cassette                                                                               | MMU-AP***2H & RAW-SM***4UT-E with RBC-U31PGS(W)-E panels                                   |
| TCB-TC21LE2     | Remote temperature sensor             | Remote temperature sensor for cassette & duct                                                                             | VRF, DI, SDI                                                                               |
| WH-H2UE         | Infra-red Remote Controller           | Wireless remote unit kit for Flexi units                                                                                  | DI Flexi                                                                                   |
| TCB-SC642TLE2   | Central Remote Controller             | Enables the control of up to 64 individual units                                                                          | SMMS/SHRM, 1:1 model connection interface required for DI/SDI (Excluding high-wall type)   |
| TCB-CC163TLE2   | On / Off Controller                   | Enables On/Off control (Max. 16 units)                                                                                    | All indoor units (Excluding DI Flexi type)                                                 |
| TCB-IFCB4E2     | Remote location On/Off Control Box    | Enables remote location On/Off control                                                                                    | All indoor units (Excluding DI Flexi type)                                                 |
| TCB-PCMO2E      | External Master On/Off control board  | External Master On/Off control board                                                                                      | VRF outdoor units                                                                          |
| TCB-PCIN2E      | Error Output Control Board            | Error output control board                                                                                                | VRF outdoor units                                                                          |
| TCB-PCDM2E      | Power Peak Cut Control Board          | Power Peak Cut Control Board                                                                                              | VRF outdoor units                                                                          |
| BMS-CM1280TLE   | Compliant Manager                     | Central controller, 10 times per day operation, operation mode, setting temp, remote ctrl restriction, up to 128 IDUs     | VRF outdoor units                                                                          |
| BMS-CM1280FTLE  | Compliant Manager                     | As BMS-CM1280TLE with Web connection and energy monitoring function                                                       | VRF outdoor units                                                                          |
| BMS-WB2561PWE   | Web Based Controller                  | Gateway server, Network Intranet connection, yearly schedule, error message history, up to 256 IDUs                       |                                                                                            |
| BMS-WE01GTE     | Web Based Controller                  | Web server, Network Intranet connection, yearly schedule, error message history, up to 512 IDUs                           | Web control operates with BMS-WB2561PWE (up to 2) & BMS-IFLSV3E                            |
| BMS-TP0641ACE   | Touch Screen Controller               | Enables full control of up to 64 indoor units, ML                                                                         | VRF, network 1:1 model connection interface required for DI/SDI (Excluding high-wall type) |
| BMS-TP5121ACE   | Touch Screen Controller               | Enables full control of up to 512 indoor units, ML                                                                        | VRF, network 1:1 model connection interface required for DI/SDI (Excluding high-wall type) |
| BMS-TP0641PWE   | Touch Screen Controller               | Enables full control of up to 64 indoor units with electric billing, ML                                                   | VRF, network 1:1 model connection interface required for DI/SDI (Excluding high-wall type) |
| BMS-TP5121PWE   | Touch Screen Controller               | Enables full control of up to 512 indoor units with electric billing, ML                                                  | VRF, network 1:1 model connection interface required for DI/SDI (Excluding high-wall type) |
| BMS-IFLSV3E     | TCS-Net Relay Interface               | Relay for integration to TCS-Net                                                                                          | Bacnet gateway, Touch-screens & web server                                                 |
| BMS-IFWH4E2     | Energy monitoring relay interface     | Energy monitoring relay interface                                                                                         | Touch screen controller                                                                    |
| BMS-IFDD02E2    | Digital I/O relay interface           | Digital I/O relay interface                                                                                               | Touch screen controller                                                                    |
| BMS-LSV6E       | Intelligent Server                    | Bacnet Gateway                                                                                                            | Requires software BMS-STBN08E & Interface BMS-IFLSV3E                                      |
| BMS-STBN08E     | BACnet                                | Server Software                                                                                                           | Enables integration with BACnet                                                            |
| BMS-STCC06E     | Intelligent Server Software           | Software package for the intelligent server                                                                               |                                                                                            |
| TCB-IFLN640TLE  | Lonworks® Gateway                     | Allows control of 64 indoor units from a Lonworks based BMS                                                               | VRF, network 1:1 model connection interface required for DI/SDI (Excluding high-wall type) |
| TCB-IFMB640TLE  | Modbus interface                      | Modbus interface                                                                                                          | SMMS interface to modbus BMS                                                               |
| TCB-IFCG1TLE    | General purpose interface             | enables control of A/C by the DI/DO and AI/AO                                                                             | Daiseikai, DI, SDI, VRF, Combination with TCB-IFCB640TLE                                   |
| TCB-IFCB640TLE  | Analog interface                      | Control & monitoring up to 64 IU on TCC-link                                                                              | Combination with TCB-IFCG1TLE                                                              |
| TCB-IFGSM1E     | GSM control interface                 | Allows ON/OFF control, operation status monitoring & alarm monitoring of A/C                                              | VRF, DI, SDI (CN61) & Daiseikai (CN08 or 09)                                               |
| TCB-PCNT20E     | Network adapter                       | to connect LC units to TCC-link network                                                                                   | DI, SDI                                                                                    |
| TCB-PCNT30TLE2  | 1:1 model connection interface        | Integration with DI, SDI                                                                                                  | Allows DI/SDI indoor units to be connected to TCC link network                             |
| TCB-PX30MUE     | Terminal box                          | Terminal box to connect to                                                                                                | TCB-PCNT30TLE2                                                                             |
| TCB-PCOS1E2     | Application control kit               | Enables night operation control, demand control, operation monitoring                                                     | All DI units                                                                               |
| TCB-KBOS1E      | Optional connector kit                | Connector kit                                                                                                             | SDI 4 outdoor units                                                                        |
| TCB-KBCN32VEE   | Connectors                            | For CN32                                                                                                                  | DI, SDI & VRF, except Flexi DI                                                             |
| TCB-KBCN60OPE   |                                       | For CN60                                                                                                                  | DI, SDI & VRF, except Flexi DI                                                             |
| TCB-KBCN61HAE   |                                       | For CN61                                                                                                                  | DI, SDI & VRF, except Flexi DI                                                             |
| TCB-KBCN70AOE   |                                       | For CN70                                                                                                                  | DI, SDI & VRF, except HW & Flexi DI                                                        |
| TCB-KBCN73DEE   |                                       | For CN73                                                                                                                  | DI, SDI & VRF, except Flexi DI                                                             |
| TCB-KBCN80EXE   |                                       | For CN80                                                                                                                  | DI, SDI & VRF, except Flexi DI                                                             |

**LonWorks Gateway**

TCB-IFLN640TLE

The LonWorks Gateway uses all standard network variables to control the individual functions of the indoor units. The gateway can also be used with Toshiba Interactive Intelligence.

**Compliant Manager**

BMS-CM1280TLE / BMS-CM1280FTLE\*

WEB function (Intranet)\*

Energy monitoring\*

Easy Settings and operations (10 times per day/ 32 times per week: on/off, operation mode, setting temp, remote ctrl restriction)

TCC-Link direct connection

Password available

Schedule timer

**Intelligent Server**

BMS-LSV6E

The intelligent server is the connection point between the PC station for BACnet system and TCS.NET relay interfaces of the air conditioner network.

**Touch Screen**

BMS-TP0641ACE - up to 64 indoor units

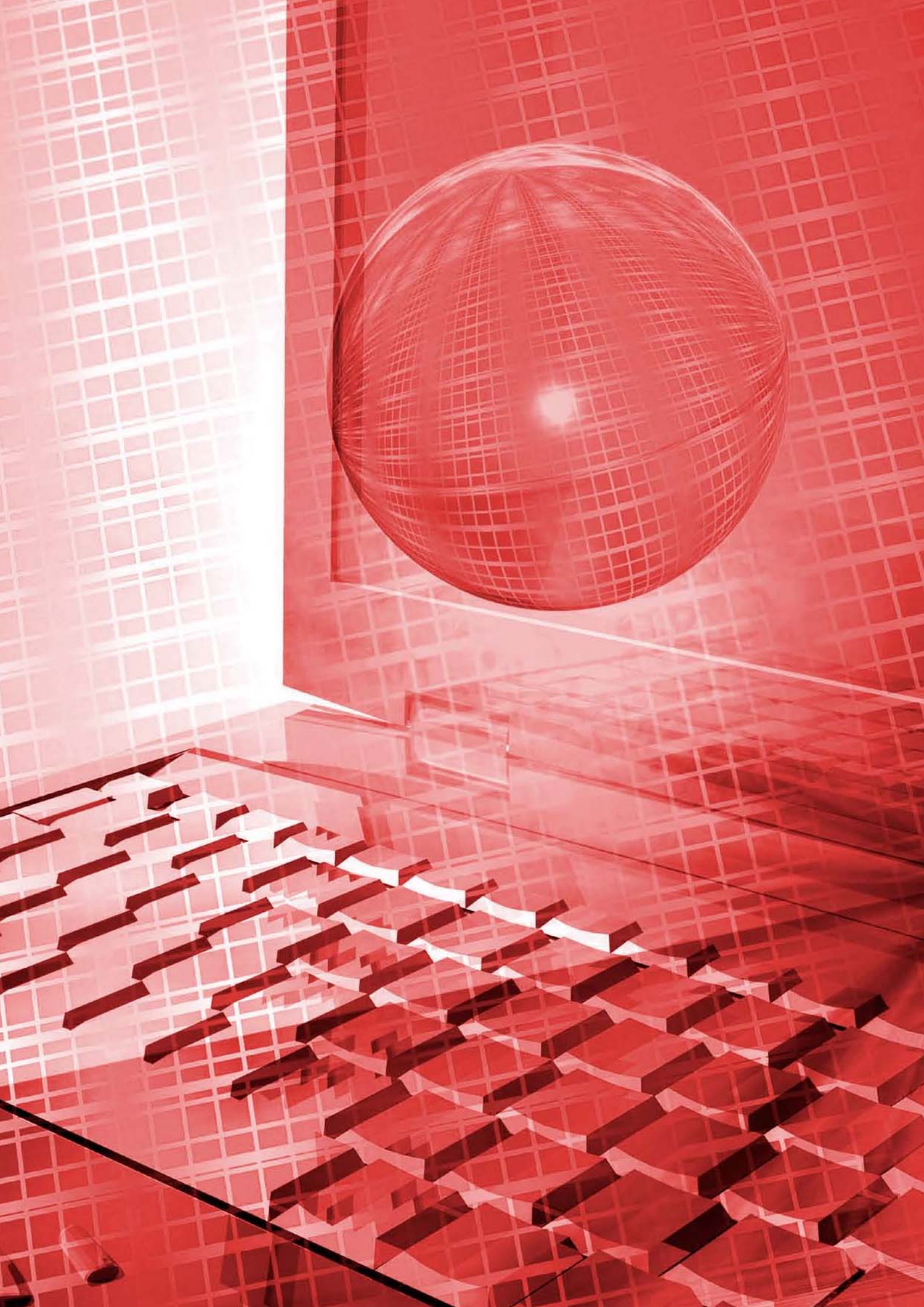
BMS-TP05121ACE - up to 512 indoor units

BMS-TP0641PWE - up to 64 indoor units + electrical bill calculation

BMS-TP5121PWE - up to 512 indoor units + electrical bill calculation

The new Touch Screen Controller provides a graphical interface with the air conditioning system in multilanguage version (EN, FR, IT, DE, ES) It can control each of the individual indoor units and is capable of providing information from the indoor unit settings and malfunction check codes. It is also used to calculate the energy consumption for the selected indoor units. This controller runs time schedules for the indoor units and allows you to set up data logs. The Touch Screen is connected to the air conditioner control network directly by relay interfaces. Password function available.





# Accessories

Residential, Light Commercial and Business.

## Residential and Light Commercial Accessories

| Code         | Description      |                                     | Compatible with                             |                   |
|--------------|------------------|-------------------------------------|---------------------------------------------|-------------------|
|              | Indoor unit type | Parts name                          | Residential units                           | DI, SDI           |
| RB-A607DE    | High wall        | Zeolite filter                      | Super Daisekai                              | RAV-SM**2KRT-E    |
| RB-A608DE    |                  | Zeolite 3G filter                   | Super Daisekai                              |                   |
| RB-A614DE    |                  | Super sterilizer filter             | 7SKV-E, 7SKV-E2                             |                   |
| RB-A615DE    |                  | Super Oxi deo filter (Sasa-Zeolite) | SKVR-E, SKV-E, SKV-E2, M-SKV-E              | RAV-SM**2KRT-E    |
|              |                  |                                     | M-SKCV-E, 13SKVR-E2, SKV(R)-ND              |                   |
|              |                  |                                     | SKP-ES, SKHP-ES, 18/24SK(H)P-EG, GK(H)P-ES2 |                   |
|              | Flexi type       |                                     | GFHP-ES2, GFP-ES2                           | RAV-SM**2XT-E     |
| RB-A616DE    | High wall        | Super sterilizer filter             | SKV-E, SKV-E2, M-SKCV-E                     | RAV-SM**2KRT-E    |
|              |                  |                                     | SKP-ES, SKHP-ES, 18/24SK(H)P-EG, GK(H)P-ES2 |                   |
|              | Flexi type       |                                     |                                             | GFHP-ES2, GFP-ES2 |
| RBC-SH-A1LE2 | Duct type        | Wired remote controller kit         | M-GDV-E                                     |                   |
| RB-F81E2     |                  | Drain pump                          |                                             |                   |

## Light Commercial Accessories

| Model Name  | Description                  | Capacities                    |
|-------------|------------------------------|-------------------------------|
| RBC-TWP30E  | Twin branch kit for DI & SDI | 1,5 HP + 1,5 HP / 2 HP + 2 HP |
| RBC-TWP50E  | Twin branch kit for DI & SDI | 3 HP + 3 HP                   |
| RBC-TRP100E | Triple branch kit for DI     | 1,5 HP + 1,5 HP + 1,5 HP      |

## VRF Accessories

| Model Name   |           | Description                              | Total capacity codes            |
|--------------|-----------|------------------------------------------|---------------------------------|
| RBM-BY54E    |           | Y-shape branching for SMMS and MiNi-SMMS | < 6,4 HP                        |
| RBM-BY104E   |           | Y-shape branching for SMMS               | > 6,4 - 14,2 HP                 |
| RBM-BY204E   |           |                                          | > 14,2 - 25,2 HP                |
| RBM-BY304E   |           |                                          | > 25,2 HP                       |
| RBM-BY54FE   |           |                                          | < 6,4HP                         |
| RBM-BY104FE  |           |                                          | > 6,4 - 14,2 HP                 |
| RBM-BY204FE  |           | Y-shape branching for SHRM               | > 14,2 - 25,2 HP                |
| RBM-BY304FE  |           |                                          | > 25,2 HP                       |
| RBM-HY1043E  | four-way  | Headers branching                        | < 14,2 HP                       |
| RBM-HY1083E  | eight-way |                                          | < 14,2 HP                       |
| RBM-HY2043E  | four-way  |                                          | > 14,2 - 25,2 HP                |
| RBM-HY2083E  | eight-way |                                          | > 14,2 - 25,2 HP                |
| RBM-HY1043FE | four-way  | Headers branching (three pipe SHRM)      | < 14,2 HP                       |
| RBM-HY1083FE | eight-way |                                          | < 14,2 HP                       |
| RBM-HY2043FE | four-way  |                                          | > 14,2 - 25,2 HP                |
| RBM-HY2083FE | eight-way |                                          | > 14,2 - 25,2 HP                |
| RBM-BT13E    |           | T-joint – Two pipe range (SMMS)          | For connection of outdoor units |
| RBM-BT13FE   |           | T-joint – Three pipe range (SHRM)        |                                 |
| RBM-Y1122FE  |           | Flow switch selector (three pipe SHRM)   | < 11,2 Kw indoor units          |
| RBM-Y1802FE  |           |                                          | > 11,2 - 18 Kw indoor units     |
| RBM-Y2802FE  |           |                                          | > 18 - 28 Kw indoor units       |

## Light Commercial and VRF Accessories

| Code             | Description                       |                                 | Compatible with   |                |
|------------------|-----------------------------------|---------------------------------|-------------------|----------------|
|                  | Indoor unit type                  | Parts name                      | SMMS, SHRM        | DI, SDI        |
| RBC-UM11PG(W)E   | Compact 4-way cassette type       | Decoration panel                | MMU-AP***1MH      | RAV-SM**2MUT-E |
| RBC-U31PG(W)-E   | 4-way cassette type               | Standard panel                  | MMU-AP***2H       | RAV-SM**4UT-E  |
| RBC-U31PGS(W)-E  |                                   | MTO straight, white color panel |                   |                |
| RBC-U31PGS(WS)-E |                                   | MTO straight, grey panel        |                   |                |
| RBC-UW136PG      | 2-way Air Discharge cassette type | Decoration panel                | MMU-AP0071-0121WH |                |
| RBC-UW266PG      |                                   |                                 | MMU-AP0151-0301WH |                |
| RBC-UY136PG      | 1-way Air Discharge cassette type | Decoration panel                | MMU-AP0071-0121YH |                |
| RBC-US21PGE      |                                   |                                 | MMU-AP0152-0242SH |                |



## Light Commercial and VRF Accessories

| Code            | Description                       |                                                                                               | Compatible with                           |                                                                                      |                                           |
|-----------------|-----------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------|
|                 | Indoor unit type                  | Parts name                                                                                    | SMMS, SHRM                                | DI, SDI                                                                              |                                           |
| TCB-GFC1602UE2  | 4-way Air Discharge cassette type | Fresh air and filter chamber                                                                  | MMU-AP***2H                               | RAV-SM564UT-E<br>RAV-SM804UT-E<br>RAV-SM1104UT-E<br>RAV-SM1404UT-E<br>RAV-SM1604UT-E |                                           |
| TCB-GB1602UE2   |                                   | Fresh air inlet box                                                                           |                                           |                                                                                      |                                           |
| TCB-FF101URE2   |                                   | Auxiliary fresh air flange                                                                    | MMU-AP***2H, MH, SH, SPH                  |                                                                                      |                                           |
| TCB-SP1602UE    |                                   | Spacer for height adjustment                                                                  | MMU-AP***2H                               |                                                                                      |                                           |
| TCB-BC1602UE    |                                   | Air discharge direction kit                                                                   |                                           |                                                                                      |                                           |
| TCB-BUS21WHE    | 1-way cassette type               | Front air discharge unit                                                                      | MMU-AP0152/0182/0242SH                    |                                                                                      |                                           |
| TCB-UFM11BFCE   | Concealed duct type               | High efficiency filter 65<br>(for rear suction)                                               | MMD-AP0071/0091/0121H                     |                                                                                      |                                           |
|                 |                                   |                                                                                               | MMD-AP0241/0271/0301H (2 pcs.)            |                                                                                      |                                           |
|                 |                                   |                                                                                               | MMD-AP0151/0181H                          |                                                                                      |                                           |
| TCB-UFM21BFCE   |                                   | High efficiency filter 90<br>(for rear suction)                                               | MMD-AP0361/0481/0561H (2 pcs.)            |                                                                                      |                                           |
|                 |                                   |                                                                                               | MMD-AP0071/0091/0121H                     |                                                                                      |                                           |
|                 |                                   |                                                                                               | MMD-AP0241/0271/0301H (2 pcs.)            |                                                                                      |                                           |
| TCB-UFH51BFCE   |                                   | High efficiency filter 90<br>(for rear suction)                                               | MMD-AP0151/0181H                          |                                                                                      |                                           |
|                 |                                   |                                                                                               | MMD-AP0361/0481/0561H (2 pcs.)            |                                                                                      |                                           |
|                 |                                   |                                                                                               | MMD-AP0071/0091/0121H                     |                                                                                      |                                           |
| TCB-FC281BE     |                                   | Filter chamber<br>(for rear suction)                                                          | MMD-AP0151/0181H                          |                                                                                      |                                           |
| TCB-FC501BE     |                                   |                                                                                               | MMD-AP0151/0181H                          |                                                                                      |                                           |
| TCB-FC801BE     |                                   |                                                                                               | MMD-AP0241/0271/0301H                     |                                                                                      |                                           |
| TCB-FC1401BE    |                                   | MMD-AP0361/0481/0561H                                                                         |                                           |                                                                                      |                                           |
| TCB-UFM11BE     |                                   | High efficiency filter 65<br>(For underside suction)                                          | MMD-AP0071/0091/0121BH                    |                                                                                      |                                           |
| TCB-UFM21BE     |                                   |                                                                                               | MMD-AP0151/0181BH                         |                                                                                      |                                           |
| TCB-UFM31BE     |                                   |                                                                                               | MMD-AP0241/0271/0301BH                    |                                                                                      |                                           |
| TCB-UFM41BE     |                                   | MMD-AP0361/0481/0561BH                                                                        |                                           |                                                                                      |                                           |
| TCB-UFH51BE     |                                   | High efficiency filter 90<br>(For underside suction)                                          | MMD-AP0071/0091/0121BH                    |                                                                                      |                                           |
| TCB-UFH61BE     |                                   |                                                                                               | MMD-AP0151/0181BH                         |                                                                                      |                                           |
| TCB-UFH71BE     |                                   |                                                                                               | MMD-AP0241/0271/0301BH                    |                                                                                      |                                           |
| TCB-UFH81BE     |                                   | MMD-AP0361/0481/0561BH                                                                        |                                           |                                                                                      |                                           |
| RBC-UD281PE(W)  |                                   | Ceiling panel<br>(Half panel for underside suction)                                           | MMD-AP0071/0091/0121BH                    |                                                                                      |                                           |
| RBC-UD501PE(W)  |                                   |                                                                                               | MMD-AP0151/0181BH                         |                                                                                      |                                           |
| RBC-UD801PE(W)  |                                   |                                                                                               | MMD-AP0241/0271/0301BH                    |                                                                                      |                                           |
| RBC-UD1401PE(W) |                                   | MMD-AP0361/0481/0561BH                                                                        |                                           |                                                                                      |                                           |
| TCB-CA281BE     |                                   | Suction canvas<br>(For underside suction)                                                     | MMD-AP0071/0091/0121BH                    |                                                                                      |                                           |
| TCB-CA501BE     |                                   |                                                                                               | MMD-AP0151/0181BH                         |                                                                                      |                                           |
| TCB-CA801BE     |                                   |                                                                                               | MMD-AP0241/0271/0301BH                    |                                                                                      |                                           |
| TCB-CA1401BE    |                                   | MMD-AP0361/0481/0561BH                                                                        |                                           |                                                                                      |                                           |
| TCB-FK281BE     |                                   | Filter kit for underside<br>(Kit of underside prefilter & shielding<br>plate of rear suction) | MMD-AP0071/0091/0121BH                    |                                                                                      |                                           |
| TCB-FK501BE     |                                   |                                                                                               | MMD-AP0151/0181BH                         |                                                                                      |                                           |
| TCB-FK801BE     |                                   |                                                                                               | MMD-AP0241/0271/0301BH                    |                                                                                      |                                           |
| TCB-FK1401BE    |                                   | MMD-AP0361/0481/0561BH                                                                        |                                           |                                                                                      |                                           |
| TCB-UFM1D-1E    |                                   | Concealed Duct<br>high static pressure type and fresh air<br>intake unit type                 | High efficiency filter 65                 |                                                                                      | MMD-AP0181H                               |
|                 |                                   |                                                                                               |                                           |                                                                                      | MMD-AP0481H (2 pcs.)                      |
| TCB-UFM2D-1E    |                                   |                                                                                               |                                           |                                                                                      | MMD-AP0241/0271/0361H (2 pcs.)            |
| TCB-UFM3DE      |                                   |                                                                                               | High efficiency filter 90                 |                                                                                      | MMD-AP0721/0961H & MMD-<br>AP0721/0961HFE |
|                 |                                   |                                                                                               |                                           |                                                                                      | MMD-AP0181H                               |
|                 |                                   |                                                                                               |                                           |                                                                                      | MMD-AP0481H (2 pcs.)                      |
| TCB-UFH5D-1E    |                                   |                                                                                               | Long life pre-filter                      |                                                                                      | MMD-AP0241/0271/0361H (2 pcs.)            |
| TCB-UFH6D-1E    |                                   |                                                                                               |                                           |                                                                                      | MMD-AP0721/0961H & MMD-<br>AP0721/0961HFE |
| TCB-UFH7DE      |                                   |                                                                                               |                                           |                                                                                      | MMD-AP0181H                               |
| TCB-PF1D-1E     |                                   |                                                                                               | Filter chamber                            |                                                                                      | MMD-AP0481H (2 pcs.)                      |
| TCB-PF2D-1E     |                                   |                                                                                               |                                           |                                                                                      | MMD-AP0241/0271/0361H (2 pcs.)            |
| TCB-PF3DE       |                                   |                                                                                               |                                           |                                                                                      | MMD-AP0721/0961H & MMD-<br>AP0721/0961HFE |
| TCB-FCY21DE     | Drain pump kit                    |                                                                                               | MMD-AP0181H                               |                                                                                      |                                           |
| TCB-FCY31DE     |                                   |                                                                                               | MMD-AP0241/0271/0361H                     |                                                                                      |                                           |
| TCB-FCY51DE     |                                   |                                                                                               | MMD-AP0481H                               |                                                                                      |                                           |
| TCB-FCY100DE    | High efficiency filter 65         |                                                                                               | MMD-AP0721/0961H & MMD-<br>AP0721/0961HFE |                                                                                      |                                           |
| TCB-DP31DE      |                                   |                                                                                               | MMD-AP0181H to AP0481H                    |                                                                                      |                                           |
| TCB-DP32DE      |                                   |                                                                                               | MMD-AP0721/0961H                          |                                                                                      |                                           |
| TCB-UFM4D-1E    | Fresh air intake type             |                                                                                               | High efficiency filter 65                 |                                                                                      | MMD-AP0481HFE                             |
| TCB-UFH8D-1E    |                                   |                                                                                               | High efficiency filter 90                 |                                                                                      | MMD-AP0481HFE                             |
| TCB-PF4D-1E     |                                   |                                                                                               | Long life filter                          |                                                                                      | MMD-AP0481HFE                             |
| TCB-FCY51DFE    |                                   |                                                                                               | Filter chamber                            |                                                                                      | MMD-AP0481HFE                             |
| TCB-DP32DFE     |                                   |                                                                                               | Drain pump kit                            |                                                                                      | MMD-AP0481/0721/0961HFE                   |
| TCB-DP22CE2     | Ceiling-suspended type            |                                                                                               | Drain pump kit                            |                                                                                      | MMC-AP***1H                               |
| TCB-KP12CE2     |                                   | Elbow Piping Kit                                                                              | MMC-AP0151/0181H                          |                                                                                      |                                           |
|                 |                                   |                                                                                               | MMC-AP0241/0271H                          |                                                                                      |                                           |
| TCB-KP22CE2     | MMC-AP0361/0481H                  |                                                                                               |                                           |                                                                                      |                                           |

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